

From 2D TQFT to Gromov-Witten Theory

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April 23, 2026

Abstract

In this explanatory article, we begin with a brief introduction of 2-dimensional topological quantum field theories (2D TQFTs) and then introduce the Gromov-Witten Theory as a Cohomological Field Theory (CohFT).

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1 Introduction

Topological ideas entered quantum field theory in a particularly concrete way in the late 1980s, when Witten introduced *topological sigma models* and related constructions that produce invariants insensitive to metric deformations [Wit88]. In parallel, Atiyah (building on the functorial viewpoint advocated by Segal) distilled the essential structural axiomatics into what is now called a *topological quantum field theory* (TQFT): a symmetric monoidal functor from a bordism category to vector spaces [Ati88]. This framework elevated “cutting and gluing” from a heuristic principle to a precise algebraic constraint: the value of a theory on a composite cobordism must factor through the state space associated with the cut.

In two dimensions, the functorial axioms become remarkably rigid. If one restricts to oriented bordisms, the entire theory is determined by the single vector space

$$A := Z(S^1),$$

together with the linear maps assigned to a small collection of elementary surfaces (pair-of-pants, cap, cup). By the mid-1990s, it was understood (and is now standard folklore with many clean expositions) that *2D oriented TQFTs are classified by commutative Frobenius algebras* [Abr96; Koc04]. Concretely, the pair-of-pants cobordism induces a multiplication $\mu : A \otimes A \rightarrow A$, a disk induces a unit $\eta : \mathbb{k} \rightarrow A$, and a second disk (viewed oppositely) induces a trace $\epsilon : A \rightarrow \mathbb{k}$, whose associated bilinear form

$$\langle a, b \rangle := \epsilon(\mu(a, b))$$

is nondegenerate and satisfies the Frobenius compatibility. From this perspective, the slogan “2D TQFT $\iff$ Frobenius algebra” is not an analogy but a theorem: any amplitude of a closed surface is computable purely from the algebraic data, via pants decompositions and invariance under the bordism relations [Abr96; Koc04].

However, the same decade also made it clear that many of the most interesting “topological” theories in geometry are not exhausted by this purely topological dependence on the underlying surface. In enumerative geometry and in the sigma-model picture, correlation functions typically vary with the complex structure of the source curve. This naturally forces one to work over the moduli space of curves. The foundational construction of the Deligne–Mumford compactification $\overline{\mathcal{M}}_{g,n}$ [DM69] (see also the textbook treatment [HM98]) provides exactly the geometric arena in which cutting and gluing become boundary phenomena: degenerations to nodal curves define boundary strata, and the corresponding *gluing morphisms* (both separating and non-separating) encode factorization in cohomology rather than merely in numbers.

At roughly the same time, Gromov’s theory of J -holomorphic curves catalyzed a modern definition of *Gromov–Witten (GW) invariants* as intersection numbers on moduli spaces of stable maps. On the algebraic side, Kontsevich introduced stable maps as a compactification amenable to intersection theory and localization, leading to effective computations of rational curve counts [Kon95]. On the symplectic side, Ruan–Tian formulated a rigorous theory of quantum cohomology and GW invariants in a differential-geometric setting [RT95]. A central technical obstacle is that moduli spaces of maps are seldom smooth of the expected dimension. The solution is to replace the fundamental class by a *virtual* fundamental class (or cycle), constructed via different but ultimately compatible analytic and algebro-geometric frameworks [BF97a; FO99; LT98]. This virtual class is precisely what makes the gluing and deformation principles true at the level needed to build field-theoretic structures.

The conceptual synthesis of these ideas appears prominently in the work of Kontsevich–Manin, who introduced the notion of a *cohomological field theory* (CohFT) as a refinement of 2D TQFT in which correlation functions become cohomology classes on $\overline{\mathcal{M}}_{g,n}$ and satisfy functorial gluing axioms [KM94]. In this language, the state space is typically

$$V := H^*(X, \mathbb{Q})$$

for a target space X , equipped with the Poincaré pairing η and unit $\mathbf{1} \in V$. The field theory is a system of multilinear maps

$$\Omega_{g,n} : V^{\otimes n} \rightarrow H^*(\overline{\mathcal{M}}_{g,n})$$

that is S_n -equivariant and compatible with the boundary gluing morphisms by contraction with η^{-1} ; a distinguished unit axiom controls compatibility with forgetting marked points [KM94; Man99]. Gromov–Witten theory furnishes a canonical example: using the stabilization map $st : \overline{\mathcal{M}}_{g,n}(X, \beta) \rightarrow \overline{\mathcal{M}}_{g,n}$ and evaluation maps ev_i , one pushes forward the virtual class with insertions to obtain $\Omega_{g,n}$, and the virtual gluing theorems imply the CohFT axioms [BM96a; FP97; KM94]. In genus zero, the resulting structure recovers quantum cohomology: three-point invariants define a deformation of the cup product, and the gluing axioms along boundary divisors encode associativity [Dub96; Man99; RT95].

Beyond existence, the subsequent history emphasizes structure and classification. Dubrovin and Manin recast genus-zero GW theory in the language of *Frobenius manifolds*, linking it to integrable systems and providing a systematic organizational principle for quantum cohomology [Dub96; Man99]. Givental developed a powerful formalism expressing higher-genus potentials via quantization procedures on symplectic loop spaces, with far-reaching consequences for computation and comparison theorems (e.g. “quantum Riemann–Roch”) [CG07; Giv01]. Teleman later proved a classification theorem for *semisimple* CohFTs, showing that (under semisimplicity) the higher-genus theory is determined by genus-zero data, in precise agreement with the predictions of Givental’s framework [Tel12].

This paper follows this historical and conceptual progression. We begin with the functorial definition of 2D oriented TQFT and its classification by commutative Frobenius algebras [Abr96; Ati88; Koc04]. We then explain why moduli of curves and the Deligne–Mumford compactification are the natural geometric enhancement needed to capture families of theories [DM69; HM98]. Next, we introduce CohFTs in the sense of Kontsevich–Manin [KM94; Man99] and describe how Gromov–Witten theory produces a CohFT via virtual fundamental classes and stabilization maps [BF97a; BM96a; FP97; LT98]. Finally, we illustrate the genus-zero consequences in quantum cohomology and briefly indicate modern structural results (Givental–Teleman) that clarify how much of the higher-genus theory is controlled by genus zero [Giv01; Tel12].

2 TQFT as a Symmetric Monoidal Functor

Michael Atiyah, in the 1980s, had defined the topological quantum field theory (TQFT) in the following way as a functor [Ati88]:

$$\text{A TQFT is a symmetric monoidal functor } Z : \text{Bord}_{\langle n-1, n \rangle} \rightarrow \text{Vect}_{\mathbb{k}}.$$

and we shall follow the notation in the lecture note *Bordism: Old and New* [Fre12], denote symmetric monoidal category of n -dimensional TQFT’s as

$$\text{TQFT}_n = \text{Hom}^{\otimes}(\text{Bord}_{\langle n-1, n \rangle}, \text{Vect}_{\mathbb{k}}) \quad (:= \text{Fun}^{\otimes}(\text{Bord}_{\langle n-1, n \rangle}, \text{Vect}_{\mathbb{k}}))$$

where $\text{Vect}_{\mathbb{k}}$ is the category of vector spaces over the field $\mathbb{k}$.

2.1 Cobordisms and Category $\text{Bord}_{\langle n-1, n \rangle}^{\mathfrak{X}(n)}$

To illustrate the definition of TQFT, recall that a cobordism is a generalization of homotopy, more precisely:

Definition 2.1 (n -Cobordism). Let M_0, M_1 be $(n-1)$ -dimensional closed manifolds, a cobordism $(W, p, \theta_-, \theta_+)$ between M_0 and M_1 is given by a n -dimensional compact manifold W with boundary, and the partition $p : \partial W \rightarrow \{-, +\}$ on the boundary that gives embeddings

$$\begin{aligned} \theta_- &: [0, +1] \times M_0 \rightarrow W \\ \theta_+ &: (-1, 0] \times M_1 \rightarrow W \end{aligned}$$

and also $\theta_{\pm}(0, M_i) = (\partial W)_{\pm} := p^{-1}(\pm)$ for $i = 0, 1$.

The standard way to think about the configuration above is to take $(\partial W)_\pm$ to be the disjoint union of certain boundary components, and $[0, 1] \times M_i \cong (-1, 0] \times (\partial W)_\pm$ can be identified as a collar neighborhood of the boundary component. We connect given $(n-1)$ -dimensional manifolds with the (diffeomorphism class of) n -dimensional manifolds W with the boundary identification defined above. We shall also write $W : M_0 \rightarrow M_1$ refers to W being a cobordism from M_1 to M_2 . The idea can be shown as the following diagram: In this

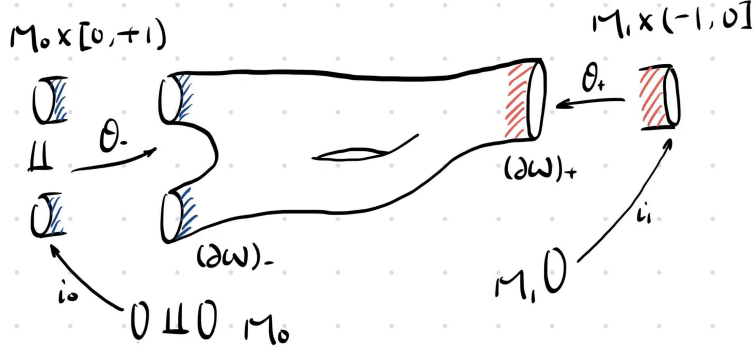


Figure 1: W as a cobordism from M_1 to M_2 .

article, we shall only consider the orientable case where M_0 , M_1 , and W are all orientable manifolds and the boundary of W satisfies

$$\partial W = (\partial W)_- \sqcup (\partial W)_+ \cong \overline{M_0} \sqcup M_1$$

where $\overline{M_1}$ represents M_1 with orientation reversed, and we shall choose $\theta_\pm$ to be orientation compatible embedding (that will ensure the further definition of the gluing operation to generate a well-defined orientation).

Definition 2.2 (Dual Cobordism). Let $(W, p, \theta_-, \theta_+)$ be a cobordism from M_0 to M_1 , the dual cobordism $(W^\vee, p^\vee, \theta_-^\vee, \theta_+^\vee)$ is a cobordism from M_1 to M_0 given by $p^\vee := 1 - p$ and

$$\begin{aligned} \theta_-^\vee(t, x) &:= \theta_+(-t, x) \quad \forall t \in [0, +1] \quad \forall x \in M_1 \\ \theta_+^\vee(t, y) &:= \theta_-(-t, y) \quad \forall t \in (-1, 0] \quad \forall y \in M_0 \end{aligned}$$

The dual cobordism W is just $W^\vee$ "turned around".

With the additional orientation information, the dual should include orientation reversing on the boundary manifold M_0 and M_1 that take the n -manifold $W^\vee$ with

$$\partial W^\vee = (\partial W^\vee)_- \sqcup (\partial W^\vee)_+ \cong \overline{M_1} \sqcup M_0$$

As the generalization of homotopy, we shall define the "gluing" operator between two cobordisms with the boundary identification that coincide with each other.

Definition 2.3 (Gluing of Cobordisms). Suppose $W_0 : M_0 \rightarrow M_1$ and $W_1 : M_1 \rightarrow M_2$ are oriented cobordisms, the gluing operation gives a cobordism $W : M_0 \rightarrow M_2$ which topologically be

$$W := (W_0 \sqcup W_1) / \sim$$

with the equivalence relation given by a diffeomorphism $\varphi := \theta_-^1 \circ (\theta_+^0)^{-1} : (\partial W_0)_+ \xrightarrow{\sim} (\partial W_1)_-$ and

$$\forall x \in (\partial W_0)_+ : \forall y \in (\partial W_1)_- : (x \sim y \iff y = \varphi(x))$$

We denote the gluing cobordism $W = W_0 * W_1 : M_0 \rightarrow M_2$.

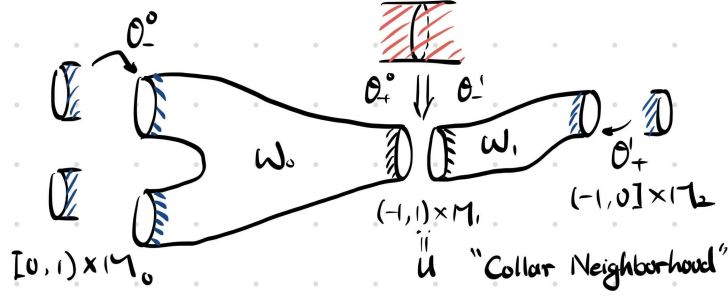


Figure 2: Gluing cobordisms.

The "cobordism" generated by the gluing obviously has a boundary

$$\partial W = (\partial W_0)_- \sqcup (\partial W_1)_+$$

However, we have to show that it admits a smooth structure that is consistent with the smooth structure on W_0 and W_1 , i.e., by restricting the smooth structure onto the components from W_0 and W_1 , we shall have the original smooth structure on the two cobordisms. The result (and actually, all previous definition of cobordisms) rely on the following theorem stating that the boundary inclusion can always be extended to an open neighborhood of the boundary on the manifold.

The detailed proof of the following theorem can be found in any standard smooth manifold textbook, for example, Theorem 9.25 in [Lee03]. We will only give a sketch of the proof, which skips most technical details of constructing a certain smooth map using the partition of unity.

Theorem 2.4 (Collar Neighborhood Theorem). Let W be a smooth manifold with boundary, then there exists a smooth embedding (collar)

$$c : [0, 1) \times \partial W \hookrightarrow W$$

i.e., the boundary ∂W always has a open neighborhood that diffeomorphic to $[0, 1) \times \partial W$

Sketch of Proof. The proof is through three steps:

1. Construct a smooth vector field V that "points strictly inwards" at the boundary ∂W , and everywhere non-zero on the boundary, i.e., a vector field that everywhere is transverse to the boundary.
2. Consider the flow Φ_t of V that gives a local collar neighborhood on an open boundary chart.
3. Reparametrize the "time" of the flow to fit all local collar neighborhoods into $[0, 1) \times \partial W$.

To construct the vector field, $\forall p \in \partial W$, we shall take the boundary chart (U_p, ϕ_p) with $\phi_p : U_p \rightarrow \mathbb{H}^n$ such that $\phi_p(p) = 0$, note that $\mathbb{H}^n = \mathbb{R}^{n-1} \times [0, +\infty)$, we shall let $\phi_p := (u, r)$ which $u(q) \in \mathbb{R}^{n-1}$ and $r(q) > 0$ for any $q \in U_p$. A simple choice of the inward vector field is just to consider $\partial/\partial r$ on $\mathbb{H}^n$ and use the coordinate map to pushforward the vector field onto the manifold:

$$V|_{U_p} := (\phi_p^{-1})_* \frac{\partial}{\partial r} \in \Gamma^\infty(TU_p)$$

and then by applying the partition of unity $\{\rho_i\}$ of an open cover $\{U_{p_i}\}$ of W that covers ∂W , and consider all boundary charts in the open cover (i.e., take $p_i \in \partial W$)

$$V := \sum_i \rho_i V|_{U_{p_i}}$$

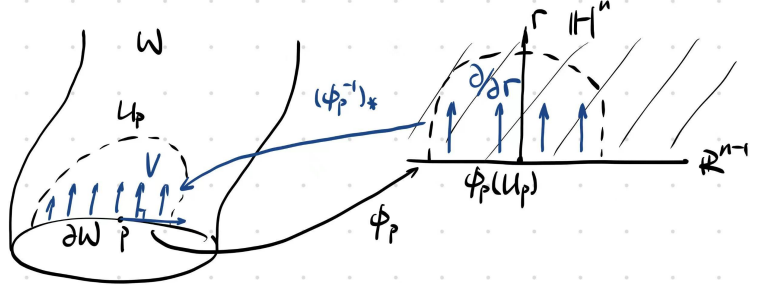


Figure 3: The inward pointing vector field.

By the existence and uniqueness of the ODE solution, one shall claim that $\exists \epsilon(p) > 0$ such that the flow $\Phi_t(p)$ is well-defined on W when $t \in [0, \epsilon(p))$. Let $c_{\text{loc}}(t, p) := \Phi_t(p)$, the pushforward

$$(c_{\text{loc}})_*, (0, p) : \mathbb{R} \oplus T_p \partial W \xrightarrow{\sim} T_p W$$

is a linear isomorphism. By the inverse function theorem, there exists $V := [0, \delta(p)) \times O_p$ such that $c_{\text{loc}}|_V$ is diffeomorphism. Finally, one shall use the partition of unity to show the global existence of the C^∞ -function $\tau : \partial W \rightarrow (0, +\infty)$ such that $\forall p \in \partial W$, $\Phi_t(p)$ exists for $[0, 2\tau(p))$. The global collar neighborhood is given by

$$c : [0, 1) \times \partial W \rightarrow W, \quad c(s, p) := \Phi_{s\tau(p)}(p)$$

since $\Phi_t(p)$ is well-defined for $[0, 2\tau(p))$, $c(s, t)$ exists $\forall s \in [0, 1)$. $\square$

With the collar neighborhood being well-defined, we shall prove the existence of a smooth structure on W :

Theorem 2.5. W can be equipped with a smooth structure and an orientation that coincides with the boundary identification as an orientable smooth cobordism.

Sketch Proof. We shall first construct the smooth structure, and then check the orientation.

Let the map $q : W_0 \sqcup W_1 \rightarrow W$ be the quotient map; the gluing happens on the set

$$S := q((\partial W_0)_+) = q((\partial W_1)_-) \subseteq W$$

which, after gluing, is supposed to be in the interior of the cobordism W . To work in a (open) neighborhood of S , defined $\Psi : (-\epsilon, \epsilon) \times M_1 \rightarrow W$ be

$$\Psi(t, x) := \begin{cases} q \circ \theta_+^0(t, x), & t \in (-\epsilon, 0] \\ q \circ \theta_-^1(t, x), & t \in [0, \epsilon) \end{cases}$$

since at $t = 0$, $q \circ \theta_+^0(0, x) = q \circ \theta_-^1(0, x)$, Ψ is well-defined. By the universal property of quotient, let $A := (-\epsilon, 0] \times M_1$ and $B := [0, \epsilon) \times M_1$, since both q and $\theta_\pm^i$ are continuous, $\Psi|_{C_i} = q \circ \theta_\pm^i$ is continuous. Thus, $\exists ! : \Psi^\sqcup : A \sqcup B \rightarrow W$ such that the following left diagram commutes

$$\begin{array}{ccc} A & \xleftarrow{i_A} & A \sqcup B \xleftarrow{i_B} B \\ & \searrow \Psi|_A & \downarrow \Psi^\sqcup \swarrow \Psi|_B \\ & & W \end{array} \quad \text{and} \quad \begin{array}{ccc} A \sqcup B & & \\ \tilde{q} \downarrow & \searrow \Psi^\sqcup & \\ (-\epsilon, \epsilon) \times M_1 & \xrightarrow{\Psi} & W \end{array}$$

Since $(-\epsilon, \epsilon) \times M_1 = (A \sqcup B) / \sim$ with $0 \in A \sim 0 \in B$, and

$$\Psi^\sqcup|_A(0, x) = \Psi|_A(0, x) = \Psi|_B(0, x) = \Psi^\sqcup|_B(0, x)$$

is a well-defined map with $\Psi \circ \tilde{q} = \Psi^\sqcup$, we know that Ψ is the unique continuous map such that the right diagram above commutes, which shows the continuity of Ψ . Then, by the fact that $\theta_\pm^i$ are embeddings, $A \cong C_0 := \theta_+^0(A)$ and $A \cong C_1 := \theta_-^1(B)$, the following diagram commutes:

$$\begin{array}{ccccc} A \sqcup B & \xrightleftharpoons{\cong} & C_0 \sqcup C_1 & \xrightarrow{\iota} & W_0 \sqcup W_1 \\ \downarrow \tilde{q} & \searrow & \downarrow q|_{C_0 \sqcup C_1} & & \downarrow q \\ (-\epsilon, \epsilon) \times M_1 & \xrightarrow[\cong]{\tilde{q}} & q(C_0 \sqcup C_1) & \xrightarrow{\iota} & W = (W_0 \sqcup W_1)/\sim \end{array}$$

the existence of continuous Ψ^{-1} is just by apply the universal property on the map $f : C_0 \sqcup C_1 \rightarrow (-\epsilon, \epsilon) \times M$ which $q : A \sqcup B \xrightarrow{\sim} C_0 \sqcup C_1 \xrightarrow{f} (-\epsilon, \epsilon) \times M$. Thus, Ψ is an homeomorphism onto its image $(C_0 \sqcup C_1)/\sim$. Let

$$U := q(C_0 \sqcup C_1) \subseteq W$$

The, by the homeomorphism, we consider three types of charts:

- (A) For (V, κ) on W_0 such that $V \cap (\partial W_0)_+ = \emptyset$, then $q|_V : V \rightarrow \text{id}_V$ and thus, we may defined the corresponding chart $(q(V), \kappa \circ q^{-1}|_V)$.
- (B) Similarly, for (V', κ') on W_1 such that $V' \cap (\partial W_1)_- = \emptyset$, take chart $(q(V'), \kappa' \circ (q|_{V'})^{-1})$
- (C) Since $(-\epsilon, \epsilon) \times M_1$ is a smooth manifold, take any chart (O, α) on M_1 with $\alpha : O \rightarrow \mathbb{R}^{n-1}$, define the local chart (U_O, χ_O) as

$$U_O := \Psi((-\epsilon, \epsilon) \times O) \subseteq W, \quad \chi_O(\Psi(t, x)) := (t, \alpha(x)) \in \mathbb{R}^n$$

This is a well-defined chat since Ψ is a homeomorphism onto its image.

The smooth structure on W is simply given by the maximal atlas that is compatible with the following smooth atlas

$$\mathcal{A}_W := \{\text{Type (A) chart}\} \cup \{\text{Type (B) chart}\} \cup \{\text{Type (C) chart}\}$$

It is easy to check that transition maps are all smooth by the compatibility on W_i and M_1 and the smoothness of every map.

Since θ_-^0 and θ_-^1 are assumed to be orientation-compatible collars (with the fixed boundary orientation convention), the induced boundary identification $\varphi := \theta_-^1 \circ (\theta_+^0)^{-1} : (\partial W_0)_+ \rightarrow (\partial W_1)_-$ is automatically orientation-reversing. Hence, the orientations pushed forward from W_0 and W_1 agree on the overlap of the glued collar neighborhood, and define an orientation on W . Thus, we complete the proof of the proposition. $\square$

The gluing and duality of cobordisms allow us to define a relation among smooth manifolds (or submanifolds of some space, in many cases) with certain dimensions. More precisely, we have the following results:

Corollary. *Cobordism defines an equivalence relation. M and N are called cobordant if there exists a cobordism $W : M \rightarrow N$ with inward boundary $(\partial W)_- \cong M$ and outward boundary $(\partial W)_+ \cong N$.*

Also, suppose $f : M \rightarrow N$ gives a diffeomorphism between closed n -manifolds M and N . Consider the cobordism $W := [0, 1] \times N$ with the inward boundary identification be $f : M \rightarrow N \cong (\partial W)_-$ and the outward identification be just id_N , we know that (we just ignor the collar neighborhood here for simplicity)

Corollary. *Diffeomorphic manifolds are cobordant.*

and the cobordism is explicitly given in the construction above.

Since the cobordism, similar to homotopy, have certain freedom of "deform" in "shape", we are supposed to be able to consider the cobordism up to certain equivalence, named isomorphism/diffeomorphism of cobordisms:

Definition 2.6 (Diffeomorphism of Cobordisms). Cobordisms $W_0, W_1 : M_- \rightarrow M_+$ are said to be diffeomorphic if there exists a diffeomorphism $f : W_0 \rightarrow W_1$ of smooth manifolds that commutes with p_i and $\theta_\pm^i$, i.e., let $I_+ := [0, 1)$ and $I_- := (-1, 0]$ the following diagrams commutes

$$\begin{array}{ccc} W_0 & \xrightarrow{f} & W_1 \\ p_0 \searrow & & \swarrow p_1 \\ & \{\pm\} & \end{array} \quad \begin{array}{ccc} W_0 & \xrightarrow{f} & W_1 \\ \theta_\pm^0 \searrow & & \swarrow \theta_\pm^1 \\ & I_\pm \times M_\pm & \end{array}$$

With all results above, the class of $(n-1)$ -manifolds and the n -cobordism between them then gives the following symmetrical monoidal category:

Definition 2.7 (Cobordism Category). The n -dimensional cobordism category $\mathbf{Bord}_{\langle n-1, n \rangle}$ is a symmetrical monoidal category that consists of the following data:

1. The objects are closed $n-1$ manifolds.
2. The morphism $\mathbf{Bord}_{\langle n-1, n \rangle}(M_0, M_1) := \{\text{Diffeomorphism class of cobordisms } W : M_0 \rightarrow M_1\}$.
3. Composition of cobordisms $(W_1, W_2) \mapsto W_2 \circ W_1$ is given by gluing with the identity morphism $\text{id}_M \in \mathbf{Bord}_{\langle n-1, n \rangle}(M, M)$ given by the cobordism $W := I \times M$.
4. The monoidal bifunctor is given by the disjoint union with the unit object given by the empty manifold $\emptyset^{n-1}$.

For any $A, B, C \in \text{ob } \mathbf{Bord}_{\langle n-1, n \rangle}$ the disjoint union, as the monoidal product also been required to satisfies the following natural isomorphisms: $\alpha_{A,B,C} : (A \sqcup B) \sqcup C \cong A \sqcup (B \sqcup C)$ and $s_{A,B} : A \sqcup B \cong B \sqcup A$ such that the following diagram commutes: The inverse law (left) and the unit coherence (right)

$$\begin{array}{ccc} & B \sqcup A & \\ s_{A,B} \nearrow & & \searrow s_{B,A} \\ A \sqcup B & \xrightarrow{\text{id}_{A \sqcup B}} & A \sqcup B \end{array} \quad \begin{array}{ccc} \emptyset^{n-1} \sqcup A & \xrightarrow{\cong} & A \sqcup \emptyset^{n-1} \\ l_A \searrow & & \swarrow r_A \\ & A & \end{array}$$

And finally, the associative coherence

$$\begin{array}{ccc} (A \sqcup B) \sqcup C & \xrightarrow{s_{A,B} \sqcup \text{id}_C} & (B \sqcup A) \sqcup C \\ \downarrow \alpha_{A,B,C} & & \downarrow \alpha_{B,A,C} \\ A \sqcup (B \sqcup C) & & B \sqcup (A \sqcup C) \\ \downarrow s_{A,B \sqcup C} & & \downarrow \text{id}_B \sqcup s_{A,C} \\ (B \sqcup C) \sqcup A & \xrightarrow{\alpha_{B,C,A}} & B \sqcup (C \sqcup A) \end{array}$$

Often, one wants to equip a cobordism, regarded simply as a smooth manifold, with additional structure encoding finer topological information; one basic example is a tangential structure. We have already seen the case of oriented n -cobordisms $\mathbf{Bord}_{\langle n-1, n \rangle}^{\text{SO}(n)}$, where orientation is expressed as a reduction of the structure group of the tangent bundle from $\text{O}(n)$ to $\text{SO}(n)$. The following definition gives a general classifying-space formulation of this idea.

Recall that if X is a Hausdorff paracompact space, then the classifying space BG of principal G -bundles satisfies

$$[X, BG] \cong \{\text{isomorphism classes of principal } G\text{-bundles over } X\},$$

where $[X, BG]$ denotes the set of homotopy classes of continuous maps $X \rightarrow BG$. Here two principal G -bundles $p_i : P_i \rightarrow X$ are identified if there exists a G -equivariant homeomorphism $\phi : P_1 \rightarrow P_2$ such that the following diagram commutes:

$$\begin{array}{ccc} P_1 & \xrightarrow{\phi} & P_2 \\ & \searrow p_1 & \swarrow p_2 \\ & X & \end{array}$$

Moreover, there exists a principal G -bundle

$$G \longrightarrow EG \xrightarrow{\gamma} BG$$

such that

1. EG is contractible.
2. G acts freely on EG .
3. For every principal G -bundle $P \rightarrow X$, there exists a classifying map $f : X \rightarrow BG$ such that

$$P \cong f^*(EG).$$

4. The classifying map f is unique up to homotopy.

Such a principal G -bundle is called the universal bundle.

An n -dimensional tangential structure is encoded by a space lying over $BO(n)$.

Definition 2.8 (Tangential Structure). An n -dimensional tangential structure is a topological space $\mathfrak{X}(n)$ together with a fibration

$$\pi(n) : \mathfrak{X}(n) \rightarrow BO(n).$$

Let $\gamma^n \rightarrow BO(n)$ denote the universal rank- n vector bundle. We write

$$S(n) := \pi(n)^* \gamma^n \rightarrow \mathfrak{X}(n)$$

for its pullback to $\mathfrak{X}(n)$.

Similarly, let M be a smooth manifold of dimension $k \leq n$. Write

$$\tau_M : M \rightarrow BO(n)$$

for the classifying map of the stabilized tangent bundle $\mathbb{R}^{n-k} \oplus TM$. Then an $\mathfrak{X}(n)$ -structure on M is a lift

$$\tilde{\tau}_M : M \rightarrow \mathfrak{X}(n)$$

of τ_M , that is, a map satisfying

$$\pi(n) \circ \tilde{\tau}_M = \tau_M.$$

Equivalently, it is a pullback square

$$\begin{array}{ccc} \mathbb{R}^{n-k} \oplus TM & \longrightarrow & S(n) \\ \downarrow & & \downarrow \\ M & \xrightarrow{\tilde{\tau}_M} & \mathfrak{X}(n). \end{array}$$

In particular,

$$\mathbb{R}^{n-k} \oplus TM \cong \tilde{\tau}_M^* S(n) \cong (\pi(n) \circ \tilde{\tau}_M)^* \gamma^n.$$

We stabilize by $\mathbb{R}^{n-k}$ so that the resulting bundle has rank n .

Example (Orientation as a Tangential Structure). *Consider the map induced on classifying spaces by the natural inclusion $SO(n) \hookrightarrow O(n)$:*

$$\pi(n) : BSO(n) \rightarrow BO(n).$$

Let $E \rightarrow M$ be a rank- n vector bundle, and let

$$c_E : M \rightarrow BO(n)$$

be its classifying map. Since the frame bundle $\text{Fr}(E) \rightarrow M$ is a principal $O(n)$ -bundle, an orientation on E is precisely a reduction of structure group from $O(n)$ to $SO(n)$, and therefore is equivalent to a lift

$$\begin{array}{ccc} & & BSO(n) \\ & \nearrow \tilde{c}_E & \downarrow \pi(n) \\ M & \xrightarrow{c_E} & BO(n). \end{array}$$

From the obstruction-theoretic point of view, the determinant homomorphism

$$\det : O(n) \rightarrow \mathbb{Z}_2 \cong \{\pm 1\}$$

gives a short exact sequence

$$1 \longrightarrow SO(n) \hookrightarrow O(n) \xrightarrow{\det} \mathbb{Z}_2 \longrightarrow 1.$$

We first prove the following proposition.

Proposition 2.9. Let

$$w := B(\det) : BO(n) \rightarrow B\mathbb{Z}_2 \simeq \mathbb{RP}^\infty.$$

Then the short exact sequence above induces a homotopy fiber sequence

$$BSO(n) \longrightarrow BO(n) \xrightarrow{w} B\mathbb{Z}_2 \simeq \mathbb{RP}^\infty.$$

Equivalently,

$$BSO(n) \simeq \text{hofib}(w : BO(n) \rightarrow \mathbb{RP}^\infty).$$

To prove the proposition, we first record a general lemma.

Lemma. *Let $p : P \rightarrow X$ be a principal G -bundle, and let $f : X \rightarrow BG$ be its classifying map. Then there is a homotopy fiber sequence*

$$P \longrightarrow X \xrightarrow{f} BG.$$

Proof. By the definition of the classifying map, there is a pullback square

$$\begin{array}{ccc} P & \longrightarrow & EG \\ p \downarrow & & \downarrow \gamma \\ X & \xrightarrow{f} & BG. \end{array}$$

Since EG is contractible and $\gamma : EG \rightarrow BG$ is a universal principal G -bundle, it follows that P is a model for the homotopy fiber of f . Therefore

$$P \longrightarrow X \xrightarrow{f} BG$$

is a homotopy fiber sequence. □

We now apply this lemma to the determinant map.

Proof of Proposition 2.9. Let

$$EO(n) \rightarrow BO(n)$$

be the universal principal $O(n)$ -bundle. Restricting the $O(n)$ -action to the subgroup $SO(n) \subset O(n)$, the action remains free and the total space $EO(n)$ remains contractible. Hence

$$BSO(n) \simeq EO(n)/SO(n).$$

Since $SO(n) \trianglelefteq O(n)$, the quotient group $O(n)/SO(n) \cong \mathbb{Z}_2$ acts freely on $EO(n)/SO(n)$, and its orbit space is

$$(EO(n)/SO(n)) / (O(n)/SO(n)) \cong EO(n)/O(n) = BO(n).$$

Therefore, the quotient map $p : EO(n)/SO(n) \rightarrow BO(n)$ is a principal $\mathbb{Z}_2$ -bundle.

It remains to identify its classifying map. Consider the principal $\mathbb{Z}_2$ -bundle associated to the universal principal $O(n)$ -bundle by the homomorphism $\det : O(n) \rightarrow \mathbb{Z}_2$. Explicitly, this associated bundle is

$$EO(n) \times_{O(n)} \mathbb{Z}_2 \rightarrow BO(n),$$

where $O(n)$ acts on $\mathbb{Z}_2$ by $A \cdot \varepsilon := \det(A)\varepsilon$. By functoriality of classifying spaces, this principal $\mathbb{Z}_2$ -bundle is classified by

$$B(\det) : BO(n) \rightarrow B\mathbb{Z}_2.$$

We claim that $EO(n)/SO(n) \rightarrow BO(n)$ is isomorphic to $EO(n) \times_{O(n)} \mathbb{Z}_2 \rightarrow BO(n)$. Define

$$\Phi : EO(n)/SO(n) \rightarrow EO(n) \times_{O(n)} \mathbb{Z}_2, \quad [x] \mapsto [x, 1].$$

This is well-defined: if $x' = xA$ for some $A \in SO(n)$, then

$$[x', 1] = [xA, 1] = [x, \det(A) \cdot 1] = [x, 1],$$

since $\det(A) = 1$.

To construct the inverse, choose some $R \in O(n)$ with $\det(R) = -1$, and define

$$\Psi([x, \varepsilon]) := \begin{cases} [x], & \varepsilon = 1, \\ [xR], & \varepsilon = -1. \end{cases}$$

We check that Ψ is well-defined. If $[x, \varepsilon] = [xA, \varepsilon']$ in $EO(n) \times_{O(n)} \mathbb{Z}_2$, then by definition of the quotient relation, $\varepsilon = \det(A)\varepsilon'$. There are four cases:

1. If $\det(A) = 1$ and $\varepsilon = \varepsilon' = 1$, then $\Psi([xA, 1]) = [xA] = [x] = \Psi([x, 1])$.
2. If $\det(A) = 1$ and $\varepsilon = \varepsilon' = -1$, then $\Psi([xA, -1]) = [xAR] = [xR] = \Psi([x, -1])$, since $A \in SO(n)$.
3. If $\det(A) = -1$, $\varepsilon = 1$, and $\varepsilon' = -1$, then $\Psi([xA, -1]) = [xAR] = [x] = \Psi([x, 1])$, because $AR \in SO(n)$.
4. If $\det(A) = -1$, $\varepsilon = -1$, and $\varepsilon' = 1$, then $\Psi([xA, 1]) = [xA] = [xR] = \Psi([x, -1])$, because $R^{-1}A \in SO(n)$.

Thus Ψ is well-defined. It is immediate from the definitions that $\Psi \circ \Phi = \text{id}_{EO(n)/SO(n)}$ and $\Phi \circ \Psi = \text{id}_{EO(n) \times_{O(n)} \mathbb{Z}_2}$. Hence Φ is an isomorphism of principal $\mathbb{Z}_2$ -bundles.

Therefore, the principal $\mathbb{Z}_2$ -bundle $p : EO(n)/SO(n) \rightarrow BO(n)$ is classified by $B(\det)$. By the lemma, its total space is the homotopy fiber of $B(\det)$. Since $EO(n)/SO(n) \simeq BSO(n)$, we obtain the homotopy fiber sequence

$$BSO(n) \longrightarrow BO(n) \xrightarrow{w} B\mathbb{Z}_2 \simeq \mathbb{RP}^\infty.$$

This proves the proposition. □

With Proposition 2.9, for any rank- n vector bundle $E \rightarrow M$ with classifying map

$$c_E : M \rightarrow BO(n),$$

the problem of orienting E is equivalent to the lifting problem

$$\begin{array}{ccc} & BSO(n) & \\ \tilde{c}_E \nearrow & \downarrow \pi(n) & \\ M & \xrightarrow{c_E} BO(n) & \xrightarrow{w} \mathbb{RP}^\infty. \end{array}$$

On the other hand, since $\mathbb{RP}^\infty \simeq K(\mathbb{Z}_2, 1)$, we have

$$H^1(\mathbb{RP}^\infty, \mathbb{Z}_2) \cong \mathbb{Z}_2.$$

Let $u \in H^1(\mathbb{RP}^\infty, \mathbb{Z}_2)$ be the nonzero class. Pulling u back along $w \circ c_E$, we obtain

$$w_1(E) := (w \circ c_E)^* u \in H^1(M, \mathbb{Z}_2),$$

which is the first Stiefel–Whitney class of E .

Moreover,

$$c_E \text{ admits a lift to } BSO(n) \iff w \circ c_E \simeq * \iff (w \circ c_E)^* u = 0 \iff w_1(E) = 0.$$

Therefore $E \rightarrow M$ is orientable if and only if its first Stiefel–Whitney class vanishes.

2.2 Topological Quantum Field Theory

With the cobordism and the category of cobordism, we shall now illustrate the definition of a TQFT of $\mathfrak{X}(n)$ -manifold ($\mathfrak{X}(n)$ is a triangulation structure):

Definition 2.10 (Topological Quantum Field Theory). A n -dimensional TQFT of $\mathfrak{X}(n)$ -manifolds is a (strong) symmetrical monoidal functor (i.e., a functor between symmetrical monoidal categories that preserves the monoidal product)

$$Z : \mathbf{Bord}_{\langle n-1, n \rangle}^{\mathfrak{X}(n)} \rightarrow \mathbf{Vect}_{\mathbb{k}}$$

The cobordisms are allowed to exist with certain tangential structure, for example, orientations.

The definition of an orientable TQFT functor $Z : \mathbf{Bord}_{\langle n-1, n \rangle}^{\mathrm{SO}(n)} \rightarrow \mathbf{Vect}_{\mathbb{k}}$, in an explicit way, is just:

1. On object level, each $n - 1$ manifold M corresponds to a $\mathbb{k}$ -vector space $Z(M)$ such that the monoidal unit is preserved, i.e., $Z(\emptyset^{n-1}) = \mathbb{k}$.
2. As morphisms, each diffeomorphism class of cobordisms $W : M_0 \rightarrow M_1$ is corresponding to a $\mathbb{k}$ -linear map $Z(W) : Z(M_0) \rightarrow Z(M_1)$ such that if $W = I \times M : M \rightarrow M$, then $Z(W) = \mathrm{id}_{Z(M)}$.
3. Let $M := \coprod_{i=1}^n M_i$, then the corresponding vector space is given by

$$Z(M) := \bigotimes_{i=1}^n Z(M_i)$$

and for $W_i : M_i \rightarrow N_i$ be cobordisms, $W := \coprod_{i=1}^n W_i$

$$Z(W) := \bigotimes_{i=1}^n Z(W_i) : \bigotimes_{i=1}^n Z(M_i) \rightarrow \bigotimes_{i=1}^n Z(N_i)$$

and Z preserves composition.

4. The functor should preserves duality in $\text{Bord}_{\langle n-1, n \rangle}^{\mathfrak{X}(n)}$, i.e., on object level, $Z(\overline{M}) \cong Z(M)^*$; and on morphism level,

$$\begin{aligned} W_0 : M_0 &\rightarrow M_1, & Z(W_0) &\in \text{Hom}_{\text{Vect}_{\mathbb{K}}}(Z(M_0), Z(M_1)) \cong Z(M_0)^* \otimes Z(M_1) \\ W_1 : \emptyset^{n-1} &\rightarrow \overline{M_0} \sqcup M_1, & Z(W_1) &\in \text{Hom}_{\text{Vect}_{\mathbb{K}}}(\mathbb{K}, Z(\overline{M_0} \sqcup M_1)) \cong Z(\overline{M_0}) \otimes Z(M_1) \\ W_2 : M_0 \sqcup \overline{M_1} &\rightarrow \emptyset^{n-1}, & Z(W_2) &\in \text{Hom}_{\text{Vect}_{\mathbb{K}}}(Z(M_0 \sqcup \overline{M_1}), \mathbb{K}) \cong Z(M_0) \otimes Z(\overline{M_1}) \end{aligned}$$

the three different morphisms in $\text{Bord}_{\langle n-1, n \rangle}^{\mathfrak{X}(n)}$ can be viewed as somehow the same linear map. Which we shall illustrate in the following diagram

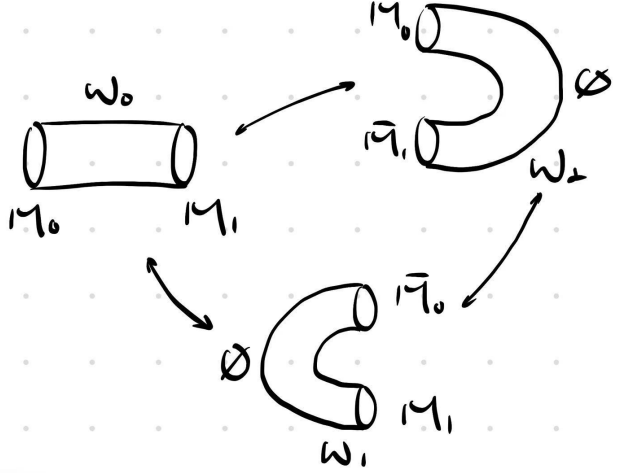


Figure 4: Involutivity of the cobordisms.

It is easy to check that the category of n -dimensional $\mathfrak{X}(n)$ -TQFTs, denoted as $\text{TQFT}_{\langle n-1, n \rangle}^{\mathfrak{X}(n)}$ with the morphism being monoidal natural transformations, is also a symmetric monoidal category.

Before we really come into the 2D TQFT, there are some propositions that we need for the fitness of TQFT:

Theorem 2.11. Let $(Z : \text{Bord}_{\langle n-1, n \rangle}^{\text{SO}(n)} \rightarrow \text{Vect}_{\mathbb{C}}) \in \text{TQFT}_n^{\text{SO}(n)}$ be a n -dimensional orientable TQFT, then $\forall M \in \text{Bord}_{\langle n-1, n \rangle}$, the corresponding vector space $Z(M) \in \text{Vect}_{\mathbb{C}}$ is finite dimensional.

Proof. Fix $M \in \text{Bord}_{\langle n-1, n \rangle}$, $V := Z(M) \in \text{Vect}_{\mathbb{C}}$, and $W := V^* = Z(\overline{M})$. Consider cobordisms given in the following (left) diagram of the "rainbow shape" that $c : \emptyset^{n-1} \rightarrow \overline{M} \sqcup M$ and $e : M \sqcup \overline{M} \rightarrow \emptyset^{n-1}$, then, consider the cobordism $(e \sqcup \text{id}_M) * (\text{id}_M \sqcup c)$ in the following (right) diagram with the "S" shape (We can call it the "zigzag identity"):

The "S" shape cobordism is diffeomorphic to the identity $M \times I$, and the TQFT functor on it gives the following diagram:

$$\begin{array}{ccccc} & & \text{id}_M & & \\ & \nearrow & & \searrow & \\ M \sqcup \emptyset^{n-1} \cong M & \xrightarrow{\text{id}_M \sqcup c} & M \sqcup \overline{M} \sqcup M & \xrightarrow{e \sqcup \text{id}_M} & \emptyset^{n-1} \sqcup M \cong M \\ \downarrow & & \downarrow & & \downarrow \\ V & \xrightarrow{\text{id}_V \otimes \text{coev}} & V \otimes W \otimes V & \xrightarrow{\text{ev} \otimes \text{id}_V} & V \\ \xi \longmapsto & \longrightarrow & \xi \otimes \left(\sum_i w_i \otimes v_i \right) & \longmapsto & \sum_i \text{ev}(\xi \otimes w_i) v_i \end{array}$$

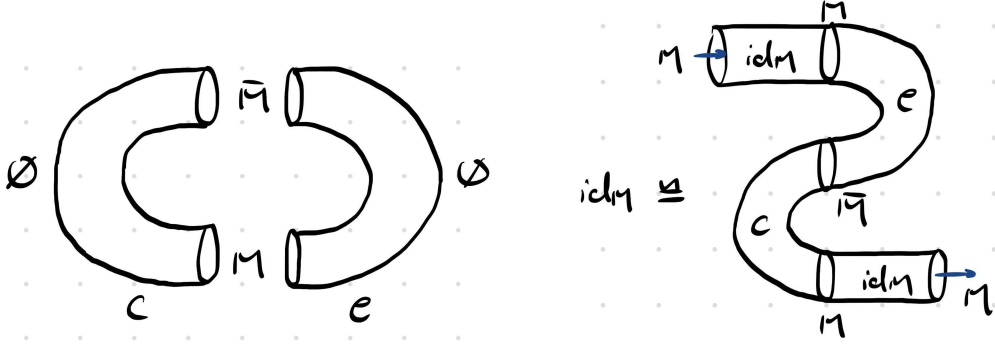


Figure 5: (Left) The "rainbow" shape cobordism. (Right) The "S" shape cobordism.

Where $\text{ev} := Z(e) : V \otimes W \rightarrow \mathbb{C}$ and $\text{coev} := Z(c) : \mathbb{C} \rightarrow V \otimes W$, since

$$\text{coev}(1) = \sum_{i=1}^n w_i \otimes v_i$$

are only allowed to be a finite sum, the commutative diagram gives $\forall \xi \in V$

$$\xi = Z((e \sqcup \text{id}_M) * (\text{id}_M \sqcup c)) = \sum_{i=1}^n \text{ev}(\xi \otimes w_i) \otimes v_i$$

i.e., $V = \text{Span}\{v_i \mid i = 1, \dots, n\}$ is finite dimensional. We shall also call the identity $(\text{ev} \otimes \text{id}_V) \circ (\text{id}_V \otimes \text{coev}) = \text{id}_V$ the "zigzag" identity due to the shape of the corresponding cobordism. $\square$

We shall end this section with the classification theorem of 1-dimensional oriented TQFT:

Theorem 2.12. Every finite-dimensional vector space $V \in \mathbf{Vect}_{\mathbb{k}}$ gives a unique (up to natural isomorphism) 1-dimensional oriented TQFT functor $Z \in \mathbf{TQFT}_1^{\text{SO}(1)}$ with $Z(\text{pt}_+) = V$, where pt_+ is a point with positive orientation.

Proof. (Existence) By classification of 1-dimensional manifolds, the compact 1-dimensional manifold can be either a closed interval $[0, 1]$ or a circle S^1 . Given a n -dimensional $\mathbb{k}$ -vector space V , we have the functor Z

1. On objects: $Z(\emptyset) = \mathbb{k}$, $Z(\text{pt}_+) = V$, $Z(\text{pt}_-) = V^*$ and for disjoint union, $Z(X \sqcup Y) = Z(X) \otimes Z(Y)$
2. On morphisms: Consider the 1-cobordisms "c" and "e" defined as in Figure 5. For "c", we defined the corresponding linear map

$$\text{coev} := Z(c) : \mathbb{k} \rightarrow V^* \otimes V, \quad 1 \mapsto \text{coev}(1) := \sum_i \epsilon_i \otimes e_i$$

where ϵ_i and e_i are (dual) basis for V^* and V ; and for "e", the linear map to be the evaluation map of a linear functional in V^*

$$\text{ev} := Z(e) : V \otimes V^* \rightarrow \mathbb{k}, \quad \phi \otimes v \mapsto \text{ev}(\phi \otimes v) := \phi(v)$$

Also, for the cobordism W that exchanges the order as shown in Figure 6, we defined the linear map

$$Z(W) : V \otimes V^* \rightarrow V^* \otimes V, \quad v \otimes \phi \mapsto \phi \otimes v$$

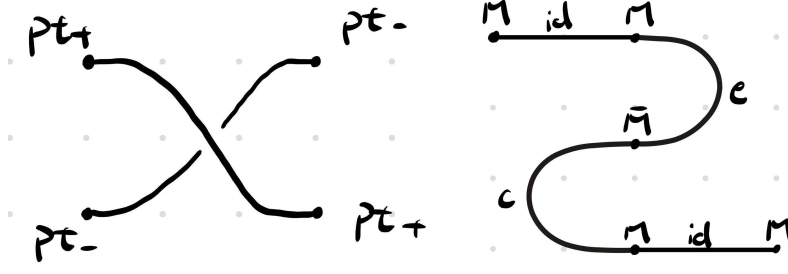


Figure 6: (Left) The "swap order" cobordism. (Right) The "S" shape cobordism.

And we shall check the consistency of these definitions of morphism given by the "S" shape cobordism in (Right) Figure 6, which, in $\text{Vect}_{\mathbb{k}}$, is just the "zigzag" identity given by the following diagram:

$$\begin{array}{ccc}
 V & \xrightarrow{\text{id}_V} & V \\
 \text{id}_V \otimes \text{coev} \swarrow & & \searrow \text{ev} \otimes \text{id}_V \\
 V \otimes V^* \otimes V & \xrightarrow{\text{ev} \otimes \text{id}_V} & V \\
 v \longmapsto v \otimes \left(\sum_i \epsilon_i \otimes e_i \right) & \longmapsto & \sum_i \epsilon_i(v) e_i
 \end{array}$$

or just $(\text{ev} \otimes \text{id}_V) \circ (\text{id}_V \otimes \text{coev}) = \text{id}_V$. Take the vector $v \in V$ such that the definition of the dual basis gives

$$v = \sum_{i=1}^n v_i e_i, \quad \epsilon_j(v) = \sum_{i=1}^n v_i \delta_{ij} = v_j$$

which proves the "zigzag" identity since

$$(\text{ev} \otimes \text{id}_V) \circ (\text{id}_V \otimes \text{coev})v = \sum_{i=1}^n \text{ev}(v \otimes \epsilon_i) \otimes \text{id}_V(e_i) = \sum_{i=1}^n \epsilon_i(v) e_i = v$$

This shows the existence of TQFT.

(Uniqueness) Let Z be any oriented TQFT with $V = Z(\text{pt}_+)$ and $W = Z(\text{pt}_-)$, we claim that the linear map $Z_e : V \otimes W \rightarrow \mathbb{k}$ corresponding to the cobordism e is a perfect pairing, which gives a linear isomorphism $\Phi : W \rightarrow V^*$ by $\Phi(w)v = Z_e(w \otimes v) \forall w \in W$. The linearity is by definition, and we shall write the inverse as

$$\Phi^{-1}(\alpha) := (\alpha \otimes \text{id}_W)(Z_c(1)) \forall \alpha \in V^*$$

with $Z_c : \mathbb{k} \rightarrow V \otimes W$ be the linear map corresponding to c . We shall check that it is indeed the inverse:

$$\begin{aligned}
 [(\Phi \circ \Phi^{-1})\alpha]v &:= Z_e(\Phi^{-1}(\alpha) \otimes v) = Z_e((\alpha \otimes \text{id}_W)(Z_c(1)) \otimes v), \quad Z_c(1) := \sum_i v_i \otimes w_i \\
 &= \sum_i \alpha(v_i) Z_e(w_i \otimes v) = \alpha((\text{id}_V \otimes Z_e) \circ (Z_c \otimes \text{id}_V)v) \quad \forall \alpha \in V^*
 \end{aligned}$$

But by the "S" shape figure, we know that $v = (\text{id}_V \otimes Z_e) \circ (Z_c \otimes \text{id}_V)v$. Thus, $(\Phi \circ \Phi^{-1})\alpha(v) = \alpha(v) \forall v \in V$, i.e., $\Phi \circ \Phi^{-1} = \text{id}_{V^*}$. Also, the other composition is given by

$$(\Phi^{-1} \circ \Phi)(w) = [(\Phi(w) \otimes \text{id}_W) \circ Z_c](1) = (Z_e \otimes \text{id}_W) \circ (\text{id}_W \otimes Z_c)w = w$$

The final step is still due to the "S" shape cobordism. Thus, given $V = Z(\text{pt}_+)$, we can automatically get $W = Z(\text{pt}_-) \cong V^*$. By setting the functor preserves the monoidal product, i.e., for $X = \text{pt}_\pm \sqcup \dots \sqcup \text{pt}_\pm$, given a natural isomorphism

$$\theta_X : Z(X) \rightarrow Z_V(X)$$

which take id_V on all positive points pt_+ and Φ on all negative points pt_- , we have the unique symmetry monoidal functor $Z \in \text{TQFT}_{(0,1)}^{\text{SO}(1)}$ up to isomorphisms. This completes the proof. $\square$

3 Frobenius Algebras and 2D TQFT

In the last section, we have defined TQFT to be the analogy of path homotopy, with the "boundary points" being closed manifolds and "path class" being the diffeomorphism class of cobordisms. In the following passages, we shall denote the n -dimensional TQFT with tangential structure $\mathfrak{X}(n)$ and take values in symmetrical monoidal category C as

$$\text{TQFT}_n^{\mathfrak{X}(n)}[C] := \text{Fun}^\otimes(\text{Bord}_{(n-1,n)}^{\mathfrak{X}(n)}, C)$$

In this section, we shall revisit the famous result of [Abr96] that there is an equivalence of categories between oriented 2D TQFT and commutative Frobenius algebra:

$$\text{TQFT}_2^{\text{SO}(2)}[\text{Vect}_{\mathbb{k}}] \cong \text{cFrob}_{\mathbb{k}}$$

and for simplicity of notations, we denote $\mathbb{k}\text{TQFT}_n^{\mathfrak{X}(n)} := \text{TQFT}_n^{\mathfrak{X}(n)}[\text{Vect}_{\mathbb{k}}]$.

In the following passages, we shall first introduce the definition of a commutative Frobenius algebra. Without further explanation, by TQFT, we refer to oriented TQFT, i.e., objects in $\text{TQFT}_n^{\text{SO}(n)}$.

3.1 Frobenius Algebra

Fix a field $\mathbb{k}$; all the algebras we mentioned in this section will be assumed to be finite-dimensional and commutative. The (commutative) Frobenius algebra over $\mathbb{k}$ is defined as follows:

Definition 3.1 (Commutative Frobenius Algebra). A commutative Frobenius algebra is an $\mathbb{k}$ -algebra (A, m, Θ) such that

1. $m : A \otimes A \rightarrow A$ is an associative and commutative product with unit 1_A .
2. A $\mathbb{k}$ -linear form $\Theta : A \rightarrow \mathbb{k}$ (called "Frobenius trace") such that $\ker \Theta$ contains no non-zero ideal.

We shall name the second condition the "Frobenius structure".

Remark. Here are some remarks on notations. We shall always write $ab := m(a \otimes b)$ be the multiplication of $a, b \in A$, and write $\langle a, b \rangle$ is the perfect pairing given by $\langle a, b \rangle := \Theta \circ m(a \otimes b) \forall a, b \in A$.

To simplify our following discussion of the algebraic structure of 2D TQFT, we shall remark that there are several different forms of Frobenius structure:

Proposition 3.2. The following structures are equivalent as Frobenius structures on (A, m) :

1. A $\mathbb{k}$ -linear form $\Theta : A \rightarrow \mathbb{k}$ (called "Frobenius form") such that $\ker \Theta$ contains no non-zero ideal.
2. An associative and commutative non-degenerate pairing $\beta : A \otimes A \rightarrow \mathbb{k}$.
3. Coassociative and cocommutative comultiplication (coalgebra structure) $\Delta : A \rightarrow A \otimes A$ with a counit, such that $\Delta \circ m = (m \otimes \text{id}_A) \circ (\text{id}_A \otimes \Delta)$ (we may call this "Frobenius relation").

For the equivalence of the first two conditions, it is clear that we can simply take the non-degenerate pairing to be $\beta := \Theta \circ m$, or, conversely, take $\Theta := \beta(1_A, a)$, where 1_A is the multiplication unit. The proof is mostly by computation, and we may assume the equivalence of these three definitions for a commutative Frobenius algebra without proof.

3.2 Algebraic Structure of 2D TQFT

We shall now discuss the algebraic structure of 2D TQFTs. Note that the objects in $\mathbf{Bord}_{\langle 1,2 \rangle}$ are closed 1-manifolds, and we have the following classification theorem:

Theorem 3.3 (Classification of 1-Manifolds). Let M be a connected 1-manifold, then M is either

1. Compact: S^1 (without boundary) or closed interval $[0, 1]$ (with boundary).
2. Noncompact: Open interval $(0, 1)$ (without boundary) or half open interval $(0, 1]$ (with boundary).

and any other 1-manifold would be the disjoint union of the case listed above.

Every connected closed 1-manifold is diffeomorphic to S^1 , hence every object is a finite disjoint union of circles. Thus,

Proposition 3.4. To defined $Z \in \mathbb{k}\text{TQFT}_2$ on object level, we only need to specify $Z(S^1) := A \in \text{Vect}_{\mathbb{k}}$.

With $A := Z(S^1)$, we shall claim that a TQFT gives a commutative Frobenius $\mathbb{k}$ -algebra (A, m, Δ) where

$$m := Z(W_1 : S^1 \sqcup S^1 \rightarrow S^1), \quad \Delta := Z(W_1^\vee : S^1 \rightarrow S^1 \sqcup S^1)$$

We may intuitively give the correspondence between 2D TQFT and a Frobenius algebra as shown in the following figure: where 1_A and Θ are the unit/counit (trace) and the $\Delta : A \rightarrow A \otimes A$ is the comultiplication

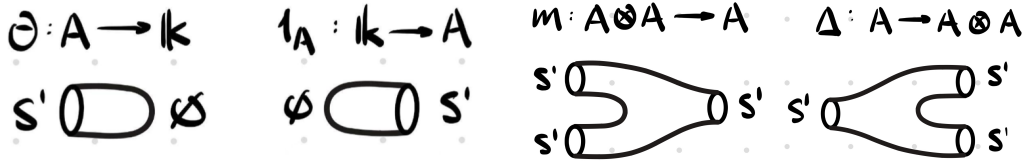


Figure 7: The correspondence between commutative Frobenius algebra and 2D TQFT.

given in the Frobenius structures in Proposition 3.2.

The Frobenius relation is motivated by the following cobordism: Note that we can also write the algebraic

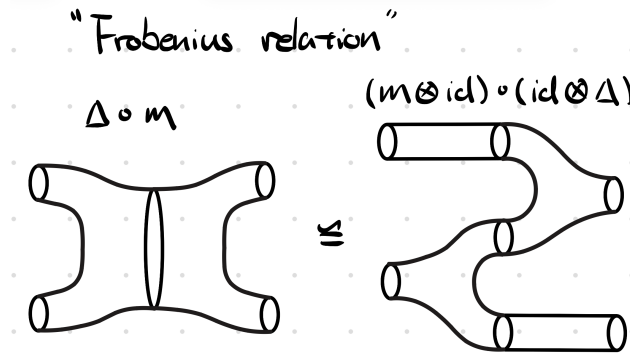


Figure 8: The cobordism corresponding to the "Frobenius relation".

structure above using indices: By the fact that we choose $\mathbb{k}$ to be a field, the commutative Frobenius $\mathbb{k}$ -algebra we are considering is a finite-dimensional (by Theorem 2.11) $\mathbb{k}$ -vector space. We may take basis $\{e_i \mid i = 1, \dots, n\}$, and then the multiplication is given by (we apply Einstein notation of summation)

$$m(e_i \otimes e_j) := c^k_{ij} e_k, \quad g_{ij} := \langle e_i, e_j \rangle := \Theta \circ m(e_i \otimes e_j) = c_{ij}^k \Theta(e_k)$$

and then, by the nondegeneracy of the pairing $\Theta \circ m$, matrix $[g_{ij}]$ is invertible, and we shall write the inverse as g^{ij} (thus, $g^{ij}g_{jk} = \delta^i_k$). Then, the multiplication unit is given by the $\mathbb{k}$ -linear map $1_A : \mathbb{k} \rightarrow A$, which only depends on its value at $1 \in \mathbb{k}$, $1_A(1) = \eta = \eta^i e_i$ and the multiplication gives $c^k_{ij} \eta^i = \delta^k_j$.

Recall that the involutivity of the cobordisms gives an equivalence between the identity cobordism, the "c" cobordism, and the e cobordism. We shall make a further illustration on that: The perfect pairing $\langle -, - \rangle : A \otimes A \rightarrow \mathbb{k}$ gives a natural isomorphism (as an inner product) $\varphi : A \rightarrow A^*$ between A and its dual space. Then, in finite-dimensional linear algebra, we have the isomorphism

$$A \otimes A \cong \text{Hom}_{\text{Vect}_{\mathbb{k}}}(A^*, A), \text{ given by } \sum_i a_i \otimes b_i \mapsto \left(\sum_i f(a_i) b_i \right)$$

which, in this case, we shall find the element $\Gamma \in A \otimes A$ (or, equivalently, the map $\mathbb{k} \ni 1 \mapsto \Gamma \in A \otimes A$) gives the corresponding element of $\varphi^{-1} : \langle a, - \rangle \rightarrow a$

$$\Gamma := g^{ij} e_i \otimes e_j, \quad a = \varphi^{-1}(\langle a, - \rangle) = (\langle a, - \rangle \otimes \text{id})(\Gamma) = g^{ij} \langle a, e_i \rangle e_j = g^{ij} a_i e_j = a^j e_j$$

and thus, we have an explicit formula of the coproduct:

Proposition 3.5. The coproduct $\Delta : A \rightarrow A \otimes A$ is given by $\Delta := (m \otimes \text{id}) \circ (\text{id} \otimes \Gamma)$, which can be explicitly given by

$$\Delta(e_i) = g^{kl} c^j_{ik} e_j \otimes e_l$$

The proposition above can be shown from the following cobordism: As a summary of all the statements we

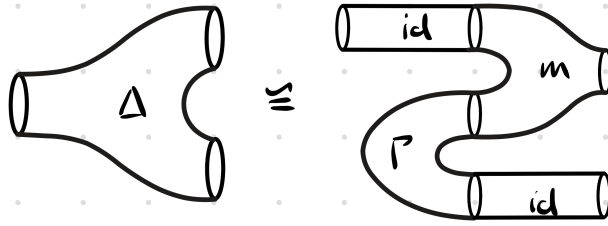


Figure 9: Expression of coproduct

have made above, we have the classification theorem of 2D-TQFT:

Theorem 3.6 (Classification of 2D Oriented TQFT). Let $Z \in \mathbb{k}\text{TQFT}_2^{\text{SO}(2)}$ be an oriented 2D TQFT with $Z(S^1) = A$ and $A \in \text{Vect}_{\mathbb{k}}$ be a vector space over field $\mathbb{k}$. Then, A admits the structure of a finite-dimensional commutative Frobenius algebra given by the 2D TQFT.

Conversely, every finite-dimensional commutative Frobenius algebra A , there is a unique (up to isomorphism) $Z_A \in \mathbb{k}\text{TQFT}_2^{\text{SO}(2)}$ such that $Z(S^1) = A$.

The previous argument in this section already showed that every 2D TQFT gives a commutative Frobenius algebra. In the following proof, we shall show the converse, that every commutative Frobenius algebra over $\mathbb{k}$ gives a unique 2D TQFT.

The idea of proof follows from the fact that any closed 1-manifold (as the boundary of cobordisms) is S^1 up to diffeomorphism, and any cobordism can be properly cut into the basic components of 2D TQFT that correspond to the algebraic structure in Frobenius algebra, as we have intuitively known from Fig. 7. An example of this canonical decomposition (without any cup/cap) can be shown as the following figure:

Proof of the Classification Theorem 3.6. We consider the commutative unital associative $\mathbb{k}$ -algebra $(A, m, 1_A)$ and Frobenius trace $\Theta : A \rightarrow \mathbb{k}$. Given basis $\{e_i \mid i = 1, \dots, n\}$ of A and $e^i = g^{ij} e_j$ as defined. The coproduct

$$\Delta(a) = a e_i \otimes e^i = a^i c_{ij}^k g^{jl} e_k \otimes e_l, \quad \forall a = a^i e_i \in A$$

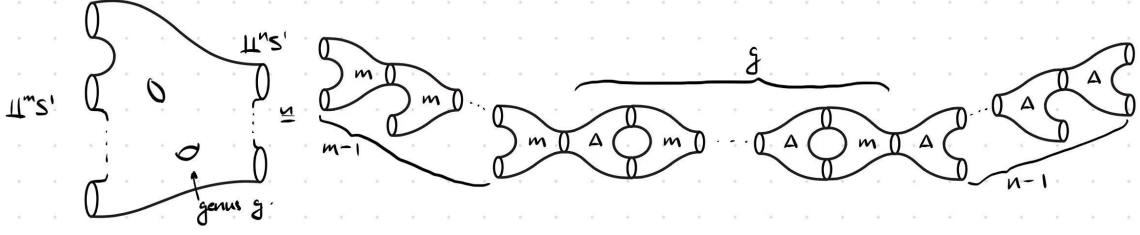


Figure 10: The decomposition of cobordism $W : \coprod^m S^1 \rightarrow \coprod^n S^1$ with out any "cup" or "cap"

and $(A, m, 1_A, \Delta, \Theta)$ satisfies the Frobenius relation

$$(m \otimes \text{id}) \circ (\text{id} \otimes \Delta) = \Delta \otimes m = (m \otimes \text{id}) \circ (\Delta \otimes \text{id})$$

Since $A \in \text{Vect}_{\mathbb{k}}$, we shall defined $Z_A \in \mathbb{k}\text{TQFT}_2^{\text{SO}(2)}$ such that $Z_A(S^1) = A$. Then, we shall consider the following "generating cobordisms" shown in Figure 7 using the fundamental structures on A as a Frobenius algebra:

1. Cup ($\emptyset \rightarrow S^1$): $Z_A(\text{cup}) = 1_A : \mathbb{k} \rightarrow A$
2. Cap ($S^1 \rightarrow \emptyset$): $Z_A(\text{cap}) = \Theta : A \rightarrow \mathbb{k}$
3. Pair-of-pants ($S^1 \sqcup S^1 \rightarrow S^1$): $Z_A(\text{pants}) = m : A \otimes A \rightarrow A$
4. Copair-of-pants ($S^1 \rightarrow S^1 \sqcup S^1$): $Z_A(\text{copants}) = \Delta : A \rightarrow A \otimes A$

And to ensure the functor is symmetrical, we need to make the conversion that each time a specific calculation is performed, a homeomorphism is selected to number/sort the boundary branches; changing the sorting only introduces commutative isomorphisms of tensor factors (given by symmetry).

Then, let $W : M_- \rightarrow M_+$ be any 2-cobordism, where

$$M_- \cong \coprod^m S^1, \quad M_+ \cong \coprod^n S^1$$

We shall construct the corresponding linear map due to the TQFT by the following decomposition: Take a Morse function $f : W \rightarrow [0, 1]$ such that $f^{-1}(0) = M_-$ and $f^{-1}(1) = M_+$, with all critical points in the interior of W . Suppose the regular values are

$$0 < c_1 < \dots < c_r < 1$$

and regular values $t_i \in (c_i, c_{i+1}) \forall i \in \{0, \dots, r+1\}$ such that $t_0 = 0$ and $t_{r+1} = 1$. Let the "time slice" be $W_i := f^{-1}([t_i, t_{i+1}])$ and each time slice contains at most 1 critical point with index either 0, 1, or 2.

For each regular value t_i , the preimage is given by

$$f^{-1}(t_i) \cong \coprod^{k_i} S^1$$

and thus, we shall choose an identification (up to permutation) $\phi_i : f^{-1}(t_i) \xrightarrow{\cong} \coprod^{k_i} S^1$. For any permutation $\sigma \in S_k$, there is canonical isomorphism

$$\tau_\sigma : A^{\otimes k} \rightarrow A^{\otimes k}, \quad \tau_\sigma(a_1 \otimes \dots \otimes a_k) = a_{\sigma(1)} \otimes \dots \otimes a_{\sigma(k)}$$

which is the symmetrical structure of $\text{Vect}_{\mathbb{k}}$. We shall claim that each slice is a cobordism $W_i : f^{-1}(t_i) \rightarrow f^{-1}(t_{i+1})$, which is corresponding to a $\mathbb{k}$ -linear map under the TQFT, given by

$$Z_i : A^{\otimes k_i} \rightarrow A^{\otimes k_{i+1}}$$

We deal with the Morse index of each critical point of the Morse function to get the linear map. The relation between Morse indexes and the decomposition law can be shown as the following figure

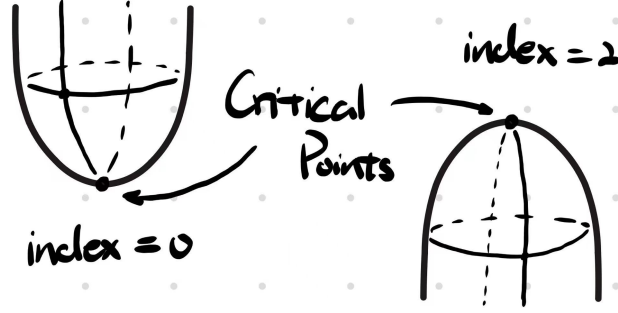


Figure 11: The corresponding cobordism segment of each interval between regular values.

1. Index 0 (birth): This implies that along the direction of cobordism (from input to output), there is a local minimum, i.e., we shall insert a cup $1_A : \mathbb{k} \rightarrow A$. More precisely, suppose the newly generated circle has index p in the disjoint union, then the linear map corresponds to a cobordism segment is

$$Z_i := \text{id}^{\otimes(p-1)} \otimes 1_A \otimes \text{id}^{\otimes(k_i-p+1)} : A^{\otimes k_i} \rightarrow A^{\otimes(k_i+1)}$$

2. Index 2 (death): Similarly, index 2 gives a local maximum, which corresponds to a cap $\Theta : A \rightarrow \mathbb{k}$ or the "Frobenius trace" as a linear map. The linear map corresponding to the cobordism segment is given by (assume the trace kills the q -th component)

$$Z_i := \text{id}^{\otimes(q-1)} \otimes \Theta \otimes \text{id}^{\otimes(k_i-q)} : A^{\otimes k_i} \rightarrow A^{\otimes(k_i-1)}$$

3. Index 1 (merge/split): In this case, we have a saddle of the Morse function, which can be either multiplication $m : A \otimes A \rightarrow A$ or comultiplication $\Delta : A \rightarrow A \otimes A$. Thus, we may check the number of connected components of $f^{-1}(t_i)$ and $f^{-1}(t_{i+1})$.

If $|f^{-1}(t_i)| > |f^{-1}(t_{i+1})|$, then we have

$$Z_i := \text{id}^{\otimes(q-1)} \otimes m \otimes \text{id}^{\otimes(k_i-q-1)} : A^{\otimes k_i} \rightarrow A^{\otimes(k_i-1)}$$

and if $|f^{-1}(t_i)| < |f^{-1}(t_{i+1})|$, then we have

$$Z_i := \text{id}^{\otimes(q-1)} \otimes \Delta \otimes \text{id}^{\otimes(k_i-q)} : A^{\otimes k_i} \rightarrow A^{\otimes(k_i+1)}$$

Finally, we shall composite all time slices as

$$Z_A(W) := Z_{r+1} \circ Z_r \circ \cdots \circ Z_1 : A^{\otimes m} \rightarrow A^{\otimes n}$$

which gives the corresponding $\mathbb{k}$ -linear map for any cobordism.

It remains to show that the above definition of $Z_A(W)$ is independent of the choice of Morse data. Let $f_0, f_1 : W \rightarrow [0, 1]$ be two relative Morse functions with $f_i^{-1}(0) = M_-$, $f_i^{-1}(1) = M_+$, and all critical points in the interior of W . By relative Cerf theory, there exists a generic homotopy $\{f_s\}_{s \in [0,1]}$ from f_0 to f_1 such that the induced handle decomposition changes only at finitely many parameters $s = s^*$, and at each such parameter one encounters only the following local Cerf moves:

1. Birth–death: creation/annihilation of a cancelable pair of critical points of indices $(0, 1)$ or $(1, 2)$;
2. Critical value crossing: two critical values interchange, equivalently swapping the order of two adjacent elementary cobordisms.

On the algebraic side, Move (1) corresponds to inserting/removing one of the composites

$$m \circ (1_A \otimes \text{id}), \quad m \circ (\text{id} \otimes 1_A), \quad (\Theta \otimes \text{id}) \circ \Delta, \quad (\text{id} \otimes \Theta) \circ \Delta$$

which equal id by the unit/counit axioms. Move (2) corresponds to changing parentheses and/or permuting adjacent tensor factors; invariance under such changes follows from associativity/coassociativity/commutativity together with the Frobenius relation

$$\Delta \circ m = (m \otimes \text{id}) \circ (\text{id} \otimes \Delta) = (\text{id} \otimes m) \circ (\Delta \otimes \text{id})$$

Therefore, $Z_A(W)$ is unchanged under each Cerf move, hence is independent of the chosen Morse function, regular values, and boundary orderings (which only contribute canonical symmetry isomorphisms in $\text{Vect}_{\mathbb{k}}$). Consequently, $Z_A(W)$ depends only on the diffeomorphism class of W relative to the boundary, so the assignment is well-defined. Compatibility with gluing and disjoint union is immediate from the construction, hence Z_A defines a symmetric monoidal functor $Z_A \in \mathbb{k}\text{TQFT}_2^{\text{SO}(2)}$. $\square$

Moreover, this one-to-one correspondence gives rise to an equivalence of categories:

Theorem 3.7. The functor $F : \mathbb{k}\text{TQFT}_2^{\text{SO}(2)} \rightarrow \text{cFrob}_{\mathbb{k}}$ gives a equivalence of category.

Recall that equivalence between categories $\mathcal{C}$ and $\mathcal{D}$ is a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ which exists another functor $G : \mathcal{D} \rightarrow \mathcal{C}$ such that there exists natural isomorphisms $\alpha_x : x \rightarrow GFx$ and $\beta_y : y \rightarrow FGY$ for any $x \in \text{ob } \mathcal{C}$ and any $y \in \text{ob } \mathcal{D}$. To prove Theorem 3.7, we just need to show the relation above.

Sketch Proof of Theorem 3.7. Consider the functor $F : \mathbb{k}\text{TQFT}_2^{\text{SO}(2)} \rightarrow \text{cFrob}_{\mathbb{k}}$ defined as follows. On objects, for an oriented 2-dimensional $\mathbb{k}$ -TQFT $Z : \text{Bord}_2^{\text{SO}(2)} \rightarrow \text{Vect}_{\mathbb{k}}$, let $F(Z) := Z(S^1)$, where the commutative Frobenius algebra structure on $Z(S^1)$ is induced by the four standard generating cobordisms: the cup $\emptyset \rightarrow S^1$, the cap $S^1 \rightarrow \emptyset$, the pair-of-pants $S^1 \sqcup S^1 \rightarrow S^1$, and the co-pair-of-pants $S^1 \rightarrow S^1 \sqcup S^1$, exactly as in Theorem 3.6. On morphisms, if $\theta : Z \Rightarrow Z'$ is a symmetric monoidal natural transformation, define

$$F(\theta) := \theta_{S^1} : Z(S^1) \rightarrow Z'(S^1).$$

We first check that $F(\theta)$ is indeed a morphism in $\text{cFrob}_{\mathbb{k}}$. Since θ is natural with respect to the cup cobordism, one has

$$\theta_{S^1} \circ \eta_Z = \eta_{Z'},$$

where $\eta_Z : \mathbb{k} \rightarrow Z(S^1)$ and $\eta_{Z'} : \mathbb{k} \rightarrow Z'(S^1)$ are the unit maps induced by the cup. Likewise, naturality with respect to the cap cobordism gives

$$\varepsilon_{Z'} \circ \theta_{S^1} = \varepsilon_Z,$$

where $\varepsilon_Z : Z(S^1) \rightarrow \mathbb{k}$ and $\varepsilon_{Z'} : Z'(S^1) \rightarrow \mathbb{k}$ are the counit maps. Next, naturality with respect to the pair-of-pants cobordism, together with the monoidal structure of θ , gives

$$\theta_{S^1} \circ \mu_Z = \mu_{Z'} \circ (\theta_{S^1} \otimes \theta_{S^1}),$$

where $\mu_Z : Z(S^1) \otimes Z(S^1) \rightarrow Z(S^1)$ and $\mu_{Z'} : Z'(S^1) \otimes Z'(S^1) \rightarrow Z'(S^1)$ are the multiplication maps. Similarly, naturality with respect to the co-pair-of-pants cobordism gives

$$(\theta_{S^1} \otimes \theta_{S^1}) \circ \Delta_Z = \Delta_{Z'} \circ \theta_{S^1},$$

where $\Delta_Z : Z(S^1) \rightarrow Z(S^1) \otimes Z(S^1)$ and $\Delta_{Z'} : Z'(S^1) \rightarrow Z'(S^1) \otimes Z'(S^1)$ are the comultiplication maps. Therefore θ_{S^1} preserves the unit, counit, multiplication, and comultiplication, so $F(\theta)$ is a morphism of commutative Frobenius algebras. Hence F is a well-defined functor.

Now define $G : \text{cFrob}_{\mathbb{k}} \rightarrow \mathbb{k}\text{TQFT}_2^{\text{SO}(2)}$ as follows. On objects, for a finite-dimensional commutative Frobenius algebra A , let $G(A)$ be the oriented 2-dimensional $\mathbb{k}$ -TQFT constructed from A in the previous proof of

Theorem 3.6. In particular, $G(A)(S^1) = A$, and the structure maps assigned by $G(A)$ to the cup, cap, pair-of-pants, and co-pair-of-pants cobordisms are precisely the unit, counit, multiplication, and comultiplication of A .

On morphisms, let $f : A \rightarrow B$ be a morphism in $\mathbf{cFrob}_k$. We define a symmetric monoidal natural transformation

$$G(f) : G(A) \Rightarrow G(B)$$

by specifying its component on each object M . Since every object M in $\mathbf{Bord}_2^{\mathrm{SO}(2)}$ is a finite disjoint union of circles, say

$$M = \coprod_{i=1}^n S^1,$$

the symmetric monoidal structures of $G(A)$ and $G(B)$ yield canonical isomorphisms

$$\phi_{A,M} : G(A)(M) \xrightarrow{\sim} A^{\otimes n}, \quad \phi_{B,M} : G(B)(M) \xrightarrow{\sim} B^{\otimes n}.$$

We then set

$$G(f)_M := \phi_{B,M}^{-1} \circ f^{\otimes n} \circ \phi_{A,M} : G(A)(M) \rightarrow G(B)(M).$$

In the case $M = \emptyset$, this simply gives $G(f)_\emptyset = \mathrm{id}_k$.

We claim that the family $\{G(f)_M\}_M$ is natural in M . Indeed, it is enough to check naturality for the generating cobordisms, since every oriented 2-dimensional cobordism is generated under composition and disjoint union by the cup, cap, pair-of-pants, and co-pair-of-pants cobordisms. For the cup cobordism, the naturality condition is exactly

$$G(f)_{S^1} \circ \eta_A = \eta_B$$

which holds because f preserves the unit. For the cap cobordism, the naturality condition is

$$\varepsilon_B \circ G(f)_{S^1} = \varepsilon_A$$

which holds because f preserves the counit. For the pair-of-pants cobordism, naturality becomes

$$G(f)_{S^1} \circ \mu_A = \mu_B \circ (G(f)_{S^1} \otimes G(f)_{S^1}),$$

which is exactly the condition that f preserve multiplication. For the co-pair-of-pants cobordism, naturality becomes

$$(G(f)_{S^1} \otimes G(f)_{S^1}) \circ \Delta_A = \Delta_B \circ G(f)_{S^1},$$

which is exactly the condition that f preserve comultiplication. Hence $G(f)$ is a natural transformation. Since its components are defined by tensor powers of f through the monoidal structure isomorphisms, it is automatically compatible with the symmetric monoidal structures. Therefore G is a well-defined functor.

We now construct natural isomorphisms

$$\beta : \mathrm{id}_{\mathbf{cFrob}_k} \Rightarrow FG \quad \text{and} \quad \alpha : GF \Rightarrow \mathrm{id}_{\mathbf{kTQFT}_2^{\mathrm{SO}(2)}}.$$

First, let $A \in \mathbf{cFrob}_k$. By construction of $G(A)$, one has

$$FG(A) = G(A)(S^1) = A$$

as commutative Frobenius algebras. Thus, we may define $\beta_A := \mathrm{id}_A$. This is clearly natural in A , so β is a natural isomorphism.

Next, let $Z \in \mathbf{kTQFT}_2^{\mathrm{SO}(2)}$, and set $A := F(Z) = Z(S^1)$. Then $GF(Z) = G(A)$ is the TQFT reconstructed from the Frobenius algebra A . For each object $M = \coprod_{i=1}^n S^1$, define

$$\alpha_{Z,M} : GF(Z)(M) = A^{\otimes n} = Z(S^1)^{\otimes n} \xrightarrow{\sim} Z(M)$$

to be the inverse of the iterated monoidal structure isomorphism of Z . In particular, for $M = S^1$, this is just $\alpha_{Z, S^1} = \text{id}_{Z(S^1)}$. We claim that $\alpha_Z : GF(Z) \Rightarrow Z$ is a natural transformation. Let $W : M_0 \rightarrow M_1$ be any cobordism. We must show that

$$\alpha_{Z, M_1} \circ GF(Z)(W) = Z(W) \circ \alpha_{Z, M_0}.$$

Again, it suffices to verify this equality for the generating cobordisms. For the cup, cap, pair-of-pants, and co-pair-of-pants cobordisms, the map assigned by $GF(Z)$ is, by construction, exactly the unit, counit, multiplication, and comultiplication extracted from $Z(S^1)$. But these are precisely the maps that Z assigns to those same generating cobordisms. Therefore, the above identity holds for the generators, and hence for all cobordisms by functoriality and monoidality. Since each $\alpha_{Z, M}$ is an isomorphism and is defined using the monoidal structure of Z , the family α_Z is a symmetric monoidal natural isomorphism.

Consequently, F and G are quasi-inverse functors. Therefore, we have proved that there is an equivalence of categories $\mathbb{k}\text{TQFT}_2^{\text{SO}(2)} \cong \text{cFrob}_{\mathbb{k}}$. $\square$

Note that $\text{TQFT}_n^{\mathfrak{X}(n)}[C]$ is a groupoid (category where every morphism is an isomorphism). We shall end our discussion of the details of TQFT by pointing out the central problem of TQFT.

Central Problem: Consider the n -dimensional TQFT with tangential structure $\mathfrak{X}(n)$ and codomain category C . The goal of studying TQFT is to "compute" the groupoid $\text{TQFT}_n^{\mathfrak{X}(n)}[C]$. More precisely, we want to find the equivalence of groupoids between $\text{TQFT}_n^{\mathfrak{X}(n)}[C]$ and some other groupoid that we are more familiar with. Theorem 2.12 and 3.6 show that we have a nice answer for $n = 1$ and the orientable cobordism case when $n = 2$.

At the end of this section, we may give a remark on the motivation of TQFT and an overview of further steps, which explain how to use the idea of TQFT to give more detailed information about complex structures on manifolds and counting curves on algebraic varieties.

3.3 Limitations of 2D TQFT: Why the Atiyah-Segal axioms are not enough?

Before moving on to moduli spaces of (stable) curves, it is helpful to clarify what the Atiyah–Segal notion of a 2-dimensional TQFT does extremely well, and what it cannot do. The axioms are optimized for cut-and-paste topology, but the geometric problems in symplectic/algebraic geometry force us to remember extra structure (complex structure, marked points, degenerations) that bordisms forget.

A guiding principle in topology is that meaningful invariants should be (i) functorial and (ii) computable by decomposition. Euler’s polyhedron formula is the classical prototype:

$$\chi = V - E + F.$$

In modern language, χ is a homotopy invariant expressible via homology:

$$\chi(X) := \sum_{k \geq 0} (-1)^k \dim H_k(X, \mathbb{Q}),$$

and the basic “decompose-and-reconstruct” mechanism is excision encoded by Mayer–Vietoris.

Theorem (Mayer–Vietoris sequence). Let $X = U \cup V$ with $U, V \subset X$ open (or suitably nice subspaces). Then there is a natural long exact sequence in homology

$$\cdots \longrightarrow H_k(U \cap V) \xrightarrow{(i_*, -j_*)} H_k(U) \oplus H_k(V) \xrightarrow{+} H_k(X) \xrightarrow{\partial} H_{k-1}(U \cap V) \longrightarrow \cdots$$

Philosophically: global invariants are constrained—and often determined—by how local pieces glue.

Atiyah–Segal TQFT as “cut-and-paste linear algebra”. A d -dimensional TQFT in the Atiyah–Segal sense is a symmetric monoidal functor

$$Z : \text{Bord}_d^{\text{or}} \rightarrow \text{Vect}_{\mathbb{k}}$$

sending a closed oriented $(d-1)$ -manifold to a vector space (“states”), and a d -dimensional cobordism to a linear map (“time evolution”), such that:

- disjoint union corresponds to tensor product;
- gluing cobordisms along matching boundaries corresponds to the composition of linear maps;
- orientation reversal corresponds (morally) to dualization.

In $d = 2$, objects are disjoint unions of circles and morphisms are compact oriented surfaces. The axioms make surface-cutting into algebra: decompose into pairs of pants/caps/cylinders, translate to operations, then compose.

Concretely, for a 2D TQFT one sets $A = Z(S^1)$, and the values of Z on basic cobordisms encode

$$m : A \otimes A \rightarrow A \quad (\text{pair-of-pants}), \quad 1_A : \mathbb{k} \rightarrow A \quad (\text{cap}), \quad \Theta : A \rightarrow \mathbb{k} \quad (\text{cup}),$$

with nondegenerate pairing $\langle a, b \rangle = \Theta(m(a, b))$. Equivalently, a 2D TQFT is a (commutative) Frobenius algebra package: once $(A, m, 1_A, \Theta)$ is fixed, $Z(\Sigma)$ is computed from any pants decomposition.

So what is missing? From the point of view of symplectic/algebraic geometry, the Atiyah–Segal framework is too coarse in three related ways.

(1) No dependence on complex structure, hence no room for moduli. A bordism is defined up to diffeomorphism, so $Z(\Sigma)$ depends only on genus. But GW theory needs marked Riemann surfaces varying in families; the natural objects live over $\overline{\mathcal{M}}_{g,n}$, not as a single number for a topological type.

(2) No place to store cohomological data (“gravitational descendants”). GW theory inserts cohomology classes and characteristic classes (e.g. ψ -classes) and naturally produces cohomology classes on $\overline{\mathcal{M}}_{g,n}$ (numbers arise only after integration). A 2D TQFT outputs linear maps/scalars, not classes in $H^*(\overline{\mathcal{M}}_{g,n})$.

(3) The relevant “gluing” is degeneration to nodal curves, not cutting along circles. The key identities in GW theory come from nodal degenerations forming boundary strata of $\overline{\mathcal{M}}_{g,n}$, with gluing morphisms such as

$$\overline{\mathcal{M}}_{g_1, n_1+1} \times \overline{\mathcal{M}}_{g_2, n_2+1} \rightarrow \overline{\mathcal{M}}_{g_1+g_2, n_1+n_2}, \quad \overline{\mathcal{M}}_{g-1, n+2} \rightarrow \overline{\mathcal{M}}_{g,n}.$$

This is sewing along the boundary of moduli, not along geometric boundary circles of a bordism. This is exactly why the next step must be the geometry/topology of $\overline{\mathcal{M}}_{g,n}$ and its compactification.

4 Moduli of Stable Curves and Deligne–Mumford Compactification

The purpose of this section is to introduce the necessary information about the moduli space of stable curves and its compactification, since this will be the working space of the CohFT. In many cases (especially in symplectic geometry), we study the moduli space $\mathcal{M}_{g,n}$ of genus- g curves with n -marked points; however, the moduli space of curves is often not compact.

The noncompactness of the moduli space is due to the “nodes” that may be generated from the “limit” on a series of algebraic curves (pinching or bubbling). For example, consider curve C , we shall take the limit of a simple closed curve γ on C such that the geodesic length of the curve converges to zero in the limit, as shown in Fig. 12.

We give an example here to show the first taste of the idea on the node by a pinching process:

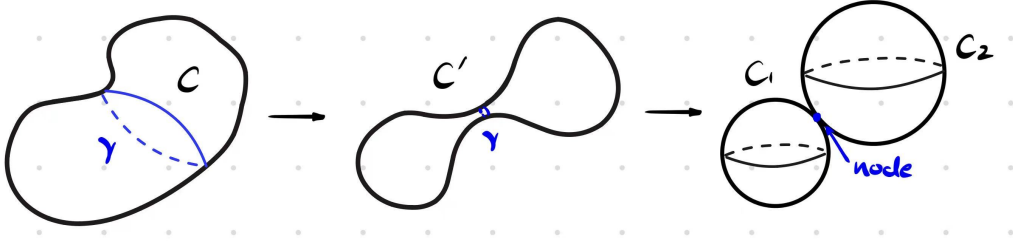


Figure 12: The nodal curve, where the node is locally isomorphic to $zw = 0$ in $\mathbb{C}^2$.

Example (Node of two disks). Take $D_z := \{z \in \mathbb{C} \mid |z| < \epsilon\}$ and $D_w := \{w \in \mathbb{C} \mid |w| < \epsilon\}$, consider a small parameter $q \in \mathbb{C}$ with $0 \neq |q| < \epsilon^2 \ll 1$. The equation $zw = q$ defines the curve

$$C_q := \{(z, w) \in D_z \times D_w \mid zw = q\}$$

on $D_z \times D_w \subseteq \mathbb{C}^2$.

Claim. $zw = q$ is a smooth curve if $q \neq 0$.

The proof of the proposition is easy since this is the level set of the holomorphic function $F(z, w) = zw - q$. Then, defined the annulus

$$A_q = \{z \in \mathbb{C} \mid |q|/\epsilon < |z| < \epsilon\}$$

A clear observation is the following proposition:

Claim. For $q \neq 0$, $\phi : A_q \rightarrow C_q$, $\phi(z) = (z, q/z)$ is biholomorphism.

Then, we shall define the gluing operation. Let $D_z^* := D_z \setminus \{0\}$ and $D_w^* := D_w \setminus \{0\}$, consider the small annulus $U_z(q) := \{z \in \mathbb{C} \mid |q|/\epsilon < |z| < \epsilon\} \subseteq D_z^*$ and similar for $U_w(q)$. The gluing is given by the pushout of the diagram

$$D_z^* \xleftarrow{i_z} U_z(q) \cong U_w(q) \xleftarrow{i_w} D_w^*$$

where $U_z(q) \cong U_w(q)$ is given by $g(z) = q/z$. g is indeed a biholomorphism since multiplication are holomorphic and $g^{-1}(w) = q/w$. We shall denote the gluing as $X_q := D_z^* \sqcup_{U(q)} D_w^*$. It is easy to check that X_q is a smooth curve.

For now, what we construct is nothing but a connected sum $D^2 \# D^2$. However, if we consider the limit of this construction (since the construction depends on q) as $q \rightarrow 0$,

$$C := C_0 = \{(z, w) \in D_z \times D_w \mid zw = 0\} = (D_z \times \{0\}) \cup (\{0\} \times D_w)$$

The nodal curve C_0 is smooth everywhere except $(0, 0)$, and the conformal modulus of the annulus

$$\text{mod}(A_q) = \frac{1}{2\pi} \ln \frac{\epsilon^2}{|q|} \longrightarrow +\infty$$

which shows that the length of the annulus "neck" $\rightarrow \infty$ as $q \rightarrow 0$, this is exactly a node from pinching.

In more algebraic geometrical language, the analytic local ring $\mathcal{O}_{C,p}^{\text{an}}$ (analytic/holomorphic function germs) near $p \in C$ has the completion:

$$\hat{\mathcal{O}}_{C,p} = \mathbb{C}[[z, w]]/(zw)$$

which means this "pinching a neck" limit makes the curve locally isomorphic to $zw = 0$ analytically near p .

The key idea for solving this problem goes back to Deligne and Mumford [DM69], who introduced the notion of stable curves and the Deligne–Mumford compactification. In what follows, the author recasts their method from a more geometric viewpoint, which the author finds helpful for clarifying the objects involved.

To study this method, we first need to understand what a node is and the properties of curves with nodes. We still use Fig. 12 as an example. In such a limit process, the remaining parts C_1 and C_2 independently are both well-defined smooth curves, and thus, a limit can be viewed as the one-point union of two curves, in this case, topologically being $S^2 \vee S^2$. To rigidify the automorphism and generate nodes, we shall replace the elements of the moduli space from an algebraic curve C_g with genus g to a curve $(C_g; p_1, \dots, p_n)$ with genus g and n marked points, which by the homogeneity of the manifolds can be any points on the curve. And we may pick random points on both C_1 and C_2 and glue them together to generate the limit elements in the moduli space $\overline{\mathcal{M}}_{g,n}$ and thus, make the compactification of the moduli space. An example of such a gluing operation is shown in Fig. 13.

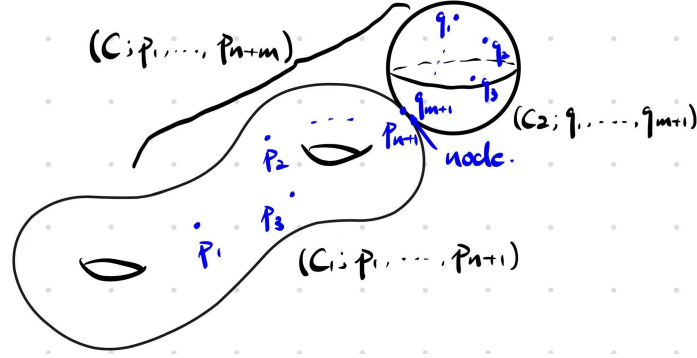


Figure 13: The gluing of curves with marked points $(C_1; p_1, \dots, p_{m+1})$ and $(C_2; q_1, \dots, q_{m+1})$ and obtain $(C; p_1, \dots, p_{m+n})$ with a node p_{m+n+1} .

4.1 Prestable curves and stability

To formalize the idea we have talked about above, we need to clarify the definition of "a curve with n marked points" as the elements of the moduli space. More precisely, we need a stabilization condition on the moduli space.

Fix integers g, n in the stable range $2g - 2 + n > 0$. A smooth n -pointed curve of genus g means a compact Riemann surface C of genus g with n ordered, distinct marked points $(p_1, \dots, p_n) \in C^n$. Isomorphisms are biholomorphisms preserving the labels.

Definition 4.1 (Nodal Curve). A (complex) curve C is nodal if it is a compact connected 1-dimensional complex analytic space whose only singularities are ordinary double points (nodes), i.e., every singular point admits local coordinates in which C is analytically isomorphic to $\{zw = 0\} \subset \mathbb{C}^2$.

Definition 4.2 (Prestable Curve). A prestable n -pointed curve is a nodal curve C together with n pairwise distinct marked smooth points $p_1, \dots, p_n \in C$.

However, recall that we defined the marked points on curves to define the gluing operation shown in Fig. 13. When counting special points, we should not only consider the marked points but also take the node as a special point.

Note that each node is locally $zw = 0$, then we shall consider the following normalization: Note that locally biholomorphic to $zw = 0$ means that we can take two branches of the small neighborhood $(z, 0)$ and $(0, w)$, which intersect transversely at $(0, 0)$. We shall denote the set of nodes as N , consider a compact (may not be connected) Riemann surface $\tilde{C}$:

$$\tilde{C} := \coprod_k \tilde{C}_k$$

and consider surjection $\nu : \tilde{C} \rightarrow C$ such that:

1. $\nu|_{\tilde{C} \setminus \nu^{-1}(N)} : \tilde{C} \setminus \nu^{-1}(N) \rightarrow C \setminus N$ is biholomorphism.
2. For any node $p \in N$, there are exactly two preimage $\nu^{-1}(p) = \{p_+, p_-\}$.
3. For small neighborhood U_+ and U_- of p_+ and p_- , ν locally behave as

$$U_+ \sqcup U_- \subseteq D_z \sqcup D_w \rightarrow C, \quad z \mapsto (z, 0), \quad w \mapsto (0, w)$$

which sends both center $0 \mapsto (0, 0)$.

Also, since the normalization map is biholomorphic, for all marked points $p_i \notin N$, $\exists! \tilde{p}_i \in \tilde{C}$ that $\nu(\tilde{p}_i) = p_i$. Thus, we shall define the set of special points to be

$$S := \nu^{-1}(N) \cup \{\tilde{p}_1, \dots, \tilde{p}_n\} \subseteq \tilde{C}$$

and for each connected component of $\tilde{C}$, consider $S_k := S \cap \tilde{C}_k$ and $n_k = |S_k|$. Recall that the Euler characteristic of a closed curve with genus g is given by

$$\chi(\Sigma_g) = 2 - 2g$$

Then, since after we remove all special points, C_k is then a well-defined smooth surface that is holomorphic to $\tilde{C}_k \setminus S_k$, we shall compute its Euler characteristic:

Proposition 4.3. Let C_k be a irreducible component of C with genus g (well-defined by the genus of $\tilde{C}_k = \nu^{-1}(C_k)$), then let C_k^o be C_k with all nodes and marked points been removed

$$\chi(C_k^o) = \chi(\tilde{C}_k \setminus S_k) = 2 - 2g_k - n_k$$

Proof. The first equal sign $\chi(C_k^o) = \chi(\tilde{C}_k \setminus S_k)$ is obvious since $\nu : \tilde{C}_k \rightarrow C_k$ is biholomorphism. Then, homotopically, we may consider

$$C_k^o \cong \tilde{C}_k \setminus S_k \simeq C_k \setminus \bigcup_{x \in S_k} \text{int}(D_x)$$

And then Euler's characteristic is just

$$\chi(C_k^o) = 2 - 2g - \# \text{ of boundary components} = 2 - 2g - n_k$$

which completes the proof. $\square$

Following the description of the special points on curves, we consider the following stability condition:

Definition 4.4 (Stability). A prestable n -pointed curve $(C; p_1, \dots, p_n)$ is stable if for every component $\tilde{C}_k$ one has $2g_k - 2 + n_k > 0$. Equivalently, $\chi(C_k^o) < 0$.

The idea of the definition of the stable curve above is to reduce the automorphism of the curve to 0-dimensional (discrete): We achieve this goal by requiring special points to be the fixed points of automorphisms. For each smooth component $(\tilde{C}_k, S_k)$ of the nodal curve, we define the automorphism to be

$$\text{Aut}(\tilde{C}_k, S_k) := \{\text{Biholomorphism } h : \tilde{C}_k \rightarrow \tilde{C}_k \text{ that } \forall p_i \in S_k : h(p_i) = p_i\}$$

and the automorphism is just the map $h : (C, S) \rightarrow (C, S)$ that is an automorphism when we restrict it to any smooth component.

Theorem 4.5. Let C be a nodal curve with genus g and set of special points S ($n = |S|$), then (C, S) stable implies $\text{Aut}(C, S)$ finite.

Proof. Note that the stability condition $\chi(\tilde{C}_k) < 0$ ($\tilde{C}_k := \tilde{C} \setminus S$) means $\tilde{C}_k$ is hyperbolic. By the uniformization theorem, there exists a unique complete Riemannian metric with constant curvature -1 on a given conformal structure of X , denoted as $\rho_{\tilde{C}_k}$, and $(\tilde{C}_k, \rho_{\tilde{C}_k})$ has finite area. More precisely, by the Gauss-Bonnet Theorem,

$$\text{Area}(\tilde{C}_k, \rho_{\tilde{C}_k}) = -2\pi\chi(\tilde{C}_k) = 2\pi(2g - 2 + n) < \infty$$

Then, $\forall \varphi \in \text{Aut}(C, S)$, the restriction on $\tilde{C}_k$ gives biholomorphism $\varphi|_{\tilde{C}_k} : \tilde{C}_k \rightarrow \tilde{C}_k$. Since $\rho_{\tilde{C}_k}$ is compatible with the complex structure, the pullback $\varphi^*\rho_{\tilde{C}_k}$ is also a complete metric with curvature -1 . Thus, by uniqueness, $\rho_{\tilde{C}_k} = \varphi^*\rho_{\tilde{C}_k}$, i.e., we have the inclusion

$$\text{Aut}(\tilde{C}_k, S_k) \hookrightarrow \text{Isom}^+(\tilde{C}_k, \rho_{\tilde{C}_k})$$

and thus, it is enough to claim that $|\text{Aut}(C_k, S_k)| < \infty$ by the classical fact that the isometric group of a complete hyperbolic surface with finite area is finite. The proof of this fact is in Appendix A. Since the automorphism of (C, S) is just combining the automorphism on each stable component with the permutation of the components

$$\text{Aut}(C, S) = \left(\prod_{v=1}^k \text{Aut}(C_k, S_k) \right) \rtimes S_k$$

where k is the number of irreducible components on the curve, the finiteness of automorphisms of each irreducible component $\text{Aut}(C_k, S_k)$ implies the finiteness of the automorphism of the entire stable curve. Thus, we proved $|\text{Aut}(C, S)| < \infty$. $\square$

4.2 Deligne-Mumford Stack and Moduli of Stable Maps

We begin with the formal definition of Deligne-Mumford (DM) stack:

Definition 4.6 (Deligne-Mumford Stack). The moduli space $\overline{\mathcal{M}}_{g,n}$ is the space of genus g stable curves with n marked points up to biholomorphism.

For expository purposes, we often view $\overline{\mathcal{M}}_{g,n}$ as an orbifold-like moduli space; strictly speaking, it is a proper smooth Deligne–Mumford stack.

Note that by Theorem 4.5, there is a heuristic characterization that defines the moduli space as

$$\overline{\mathcal{M}}_{g,n} := \{(\Sigma_g, S) \mid \text{Aut}(\Sigma_g, S) \text{ finite, } |S| = n\}$$

which replaces the condition on the number of marked points with just requiring the automorphism to be finite (Σ_g means curves with genus g).

There is an obvious forgetful morphism $p : \overline{\mathcal{M}}_{g,n+1} \rightarrow \overline{\mathcal{M}}_{g,n}$ given by dropping the last marked point: $p(C; p_1, \dots, p_{n+1}) = (C; p_1, \dots, p_n)$, where C is a nodal curve and p_i are marked points. However, dropping a marked point may cause the curve component to be unstable when the component has $g = 0, 1$. We shall discuss the stability case by case on each irreducible component:

1. Genus 0 ($\widehat{\mathbb{C}}$ component): The stable component C_k need at least 3 special points.
2. Genus 1 (complex torus component): The stable component C_k needs at least 1 special point.
3. Genus ≥ 2 : Automatically stable.

The method that deals with this problem is to contract the unstable components after removing the marked points, an intuitive example is shown in Fig. 14, where the forgetful map makes the $\widehat{\mathbb{C}}$ component remain only two marked points, which cause instability. Thus, under the stabilization, we shall forget this component, and the other part remains stable.

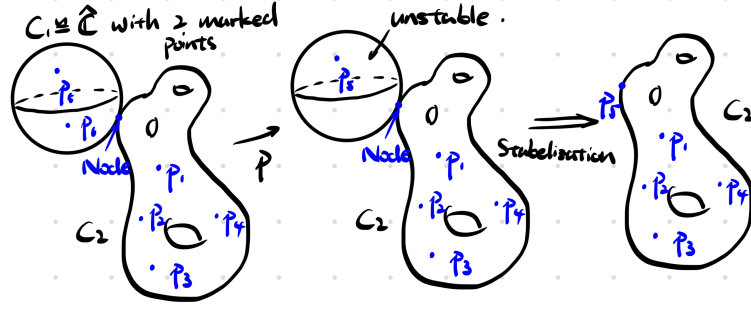


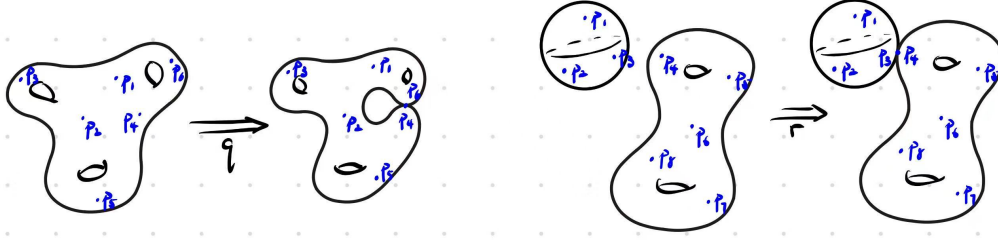
Figure 14: The forgetful map and stabilization

Another natural morphism is given by the gluing of stable curves, which is already intuitively given in Fig 13, which is similar to the structure of TQFT:

Definition 4.7 (Gluing). The gluing operation on the DM stack is to glue two nodes and form a node on the curve, which can be formally classified by the property of generated nodes:

1. Non-separating node/Self gluing: $q : \overline{\mathcal{M}}_{g-1, n+2} \rightarrow \overline{\mathcal{M}}_{g, n}$, which glue two marked points on a single curve.
2. Separating node: $r : \overline{\mathcal{M}}_{g_1, n_1+1} \times \overline{\mathcal{M}}_{g_2, n_2+1} \rightarrow \overline{\mathcal{M}}_{g, n}$ where $g = g_1 + g_2$ and $n = n_1 + n_2$.

The idea of gluing is shown in Fig. 15


 Figure 15: The gluing map r and q .

Since we are counting special points, the preimage of a node under normalization maps is a pair of special points; we can argue that the gluing operation does not break the stability condition. However, it remains an important problem: What is the topology on the moduli spaces? Without any topology, there is no way to talk about any of the following steps and the compactification of the DM stack. To give DM stack a topology naturally, we need to introduce the Teichmüller space: Let C be a closed orientable smooth surface, then we define the *space of complex structure* as

$$\mathcal{J}(C) := \{J \in \Gamma^\infty(\text{End}(TC)) \mid J^2 = -\text{id}\}$$

An observation is that the (almost) complex structure¹ with an induced orientation on the manifold. Thus, we shall consider the space of the complex structure that is compatible with the given orientation:

$$\mathcal{J}^+(C) := \{J \in \mathcal{J}(C) \mid J \text{ compatible with the given orientation on } C\}$$

Note that $\mathcal{J}^+(C)$ is a contractable space; a proof is in Appendix C. We may give $\mathcal{J}^+(C) \subseteq \Gamma^\infty(\text{End}(TC))$ as a set of C^∞ -tensor field on C and equip with the C^∞ -topology.

¹This definition gives an almost complex structure in general. However, any almost complex structure gives rise to a complex structure in dimension 2 (the Nijenhuis tensor is constant zero, and thus, every almost complex structure is integrable).

Consider $\text{Diff}^+(C) \trianglelefteq \text{Diff}(C)$ consists of all orientation-preserving diffeomorphisms, and $\text{Diff}^0(C) \trianglelefteq \text{Diff}^+(C)$ be the diffeomorphisms that are isotopic to the identity on C . All diffeomorphism groups above act naturally on $\mathcal{J}^+(C)$. Then, the Teichmüller space is given by:

Definition 4.8 (Teichmüller Space). The Teichmüller space contains the class of conformal structures under the identity components of the diffeomorphism group:

$$\mathcal{T}(C) := \mathcal{J}^+(C)/\text{Diff}^0(C)$$

Instead, the moduli space of complex structures on a surface C with genus g is given by

$$\mathcal{M}_g := \mathcal{J}(C)/\text{Diff}(C) = \mathcal{J}^+(C)/\text{Diff}^+(C)$$

To connect these two concepts, we shall define the mapping class group as follows:

Definition 4.9 (Mapping Class Group). The mapping class group on C is given by

$$\text{MCG}(C) := \text{Diff}^+(C)/\text{Diff}^0(C)$$

and thus, the moduli space of genus g curve (Riemann surface) is just

$$\mathcal{M}_g = \mathcal{T}(C)/\text{MCG}(C)$$

which contains the quotient topology on from $\mathcal{T}(C)$. More details about the C^k -topology and the inherent topology on the moduli space of curves can be found in Appendix B. Finally, as we have stated in 4.6, we shall take the moduli space of genus g curve with n marked points

$$\mathcal{M}_{g,n} := \mathcal{J}(C)/\text{Diff}(C, S) = \mathcal{J}^+(C)/\text{Diff}^+(C, S)$$

where the DM stack $\overline{\mathcal{M}}_{g,n}$ is given by the space $\mathcal{M}_{g,n}$ with all nodal cases been added in it.

We may also list some useful properties of Deligne-Mumford stacks without proof: Recall that an orbifold is a generalization of a manifold, which allows a finite group action on each chart (Thus, instead of locally Euclidean, locally modeled as quotients of $U \subseteq \mathbb{R}$).

1. $\overline{\mathcal{M}}_{g,n}$ is a closed nonsingular, irreducible orbifold of (complex) dimension $3g - 3 + n$.
2. The "boundary"² of the Deligne-Mumford compactification given by the curves with at least one node:

$$\partial\overline{\mathcal{M}}_{g,n} = \overline{\mathcal{M}}_{g,n} \setminus \mathcal{M}_{g,n}$$

This can be illustrated as the "points" that have been added into $\mathcal{M}_{g,n}$, which are obtained by the limit that broke smoothness.

3. $\text{im}(r)$ and $\text{im}(q)$ are both in the boundary $\partial\overline{\mathcal{M}}_{g,n}$ (since gluing generates nodal curves).

Here are some simple examples of a DM stack:

Example. We may take $\overline{\mathcal{M}}_{0,3}$ and $\overline{\mathcal{M}}_{0,4}$ as examples:

1. The moduli space of a genus 0 curve with 3 marked points $\overline{\mathcal{M}}_{0,3} = \{\text{pt}\}$ is just a point, since three given fixed points can fix a Möbius transformation on the Riemann sphere.

²Since the previous statement already states that $\overline{\mathcal{M}}_{g,n}$ is a closed nonsingular orbifold, the boundary here is NOT in the sense of orbifold with boundary, instead, by "boundary", we mean the limit case of nodal curves.

2. In the same method, the only freedom on the moduli space of a genus 0 curve with 4 marked points is the choice of the last marked point p_4 . The limit case gives $p_4 \rightarrow p_i$ for $i = 1, 2, 3$, and the detail is given in the next subsection, which shows that we may use a graph to represent stable curves. In this case, $\overline{\mathcal{M}}_{0,4} = \mathbb{P}^1$.

Recall that at the beginning of this section, we mentioned that we are considering the map from a genus g nodal curve with marked points to some target space X (often a symplectic manifold or projective algebraic manifold) given by

$$f : (C, S) \rightarrow X$$

The automorphism of (C, S) induces the automorphism of stable maps, given by the following diagram

$$\begin{array}{ccc} (C, S) & \xrightarrow{f} & X \\ h \downarrow & \nearrow f & \\ (C, S) & & \end{array}$$

where $h \in \text{Aut}(C, S)$. This allows us to say such a map is stable if and only if the prestable curve (C, S) is stable.

Definition 4.10 (Moduli Space of Stable Maps). The moduli space we consider is given by

$$\overline{\mathcal{M}}_{g,n}(X) := \{f : (C, S) \rightarrow X \mid f \text{ stable holomorphic}\}$$

where (C, S) is a genus g curve with n special points, i.e., $|S| = n$. Furthermore, we can choose a characteristic homology class $\beta \in H_2(X)$ and restrict the map as

$$\overline{\mathcal{M}}_{g,n}(X, \beta) = \{f : (C, S) \rightarrow X \mid f_*[C] = \beta, f \text{ stable holomorphic}\}$$

The stability condition can also be written on irreducible components, which requires the map $f : (C; p_1, \dots, p_n) \rightarrow X$ to satisfy one of the following conditions:

1. $f|_{C_k}$ is nonconstant on irreducible component $C_k \subseteq C$.
2. $f|_{C_k}$ is constant on genus g_k component C_k with n_k special points on it, and $2g_k - 2 + n_k > 0$.

When we talk about the stable maps, there are also natural evaluation and forgetful maps given by:

1. Evaluation on marked point p_i : $\text{ev}_i : \overline{\mathcal{M}}_{g,n}(X, \beta) \rightarrow X$ for $i = 1, \dots, n$ which is defined by

$$\text{ev}_i : (f : (C; p_1, \dots, p_n) \rightarrow X) \mapsto f(p_i)$$

2. Forgetful map: $\pi : \overline{\mathcal{M}}_{g,n}(X, \beta) \rightarrow \overline{\mathcal{M}}_{g,n}$ that forget the map to target space.

Example. We may consider a simplest model as an example: Consider the curves in $\mathbb{P}^2$, with $H_2(\mathbb{P}^2) = \mathbb{Z} = \langle 1 \rangle$, then $\overline{\mathcal{M}}_{0,3}(\mathbb{P}^2, 1)$ consists of the map $\mathbb{P}^1 \rightarrow \mathbb{P}^2$ that by requiring the map has degree 1, we claim that the holomorphic map $f : \mathbb{P}^1 \rightarrow \mathbb{P}^2$ must be a embedding of line:

$$f : \mathbb{P}^1 \xrightarrow{\sim} L \subseteq \mathbb{P}^2, \quad x_i := f(p_i)$$

Thus, we shall claim that the moduli space of the stable map can be described as

$$\overline{\mathcal{M}}_{g,n}(\mathbb{P}^2, 1) \cong \{(L; x_1, x_2, x_3) \mid L \subseteq \mathbb{P}^2 \text{ is a line, } x_i \in L\}$$

where the isomorphism is explicitly given by taking $L = f(\mathbb{P}^1)$ and $x_i = f(p_i) \forall i$. To further illustrate this description, consider the parametrized space of lines in $\mathbb{P}^2$, which can be written as a Grassmannian

$$B := \text{Gr}_2(\mathbb{C}^3) \cong \{\text{All lines in } \mathbb{P}^2\} \cong \mathbb{P}^2$$

The marked points induce the space $U := \{(L, x) \in B \times \mathbb{P}^2 \mid x \in L \subseteq \mathbb{P}^2\}$ with natural projection given by

$$\rho : U \rightarrow B, (L, x) \mapsto L; \quad \pi : U \rightarrow \mathbb{P}^2, (L, x) \mapsto x$$

which $\forall L \in B : \rho^{-1}(L) = \{(L, x) \mid x \in L\} \cong L \cong \mathbb{P}^1$. Thus, $U \rightarrow B$ is a $\mathbb{P}^1$ -bundle, and the moduli space is given by the fiber product

$$\overline{\mathcal{M}}_{0,3}(\mathbb{P}^2, 1) \cong U \times_B U \times_B U$$

and $\dim_{\mathbb{C}} B = 2$, $\dim_{\mathbb{C}} \mathbb{P}^1 = 1$, this directly gives the dimension of the moduli space:

$$\dim_{\mathbb{C}} \overline{\mathcal{M}}_{0,3}(\mathbb{P}^2, 1) = \dim_{\mathbb{C}}(U \times_B U \times_B U) = 5$$

4.3 Stable Graphs and Stratification of $\overline{\mathcal{M}}_{g,n}$

The topological type of stable curves can be represented by a stable graph given by

$$\Gamma = (V, H, L, g : V \rightarrow \mathbb{Z}_{\geq 0}, v : H \rightarrow V, i : H \rightarrow H)$$

consists of the following elements:

1. A set of vertices V , which corresponds to the irreducible components of the curve with a genus function $g : V \rightarrow \mathbb{Z}_{\geq 0}$ that marks the genus of each irreducible component.
2. A set of half edges H with vertex assignment $v : H \rightarrow V$ and involution³ $i : H \rightarrow H$ (multiple edges and loops are allowed).
3. The set of edges E is defined by the 2-cycles of i in H , corresponding to nodes. The set of external edges (or "legs") $L := \{h \in H \mid i(h) = h\}$ is given by the self-edge of i in H , which corresponds to marked points (on components of the curve). A special case of the edge is that for $h, h' \in H$, if $v(h) = v(h')$, then there is a loop on the corresponding vertex. The loop is allowed in this setting.
4. The graph (V, E) is a connected graph, and the stability condition is given by for any vertices $v \in V$

$$2g(v) - 2 + n(v) > 0$$

where $n(v)$ is the valence (number of half-edges) on v .

A simple example of stable graphs of the boundary case in $\overline{\mathcal{M}}_{0,4}$ is given in Fig. 16. The genus of the stable graph is given by

$$g(\Gamma) := h^1(\Gamma) + \sum_{v \in V} g(v)$$

where $h^1(\Gamma) = |E| - |V| + 1$ is the first Betti number of Γ . Let $G_{g,n}$ be the set of graphs of genus g with n legs up to isomorphism.

Recall that a stratification often means writing the space as the disjoint union (on set level!) of some locally closed subsets. There is a natural construction of $\overline{\mathcal{M}}_{g,n}$ using the stable graph that for any $\Gamma \in G_{g,n}$, we can find a corresponding strata

$$\mathcal{S}_{\Gamma} := \{(C; p_1, \dots, p_n) \in \overline{\mathcal{M}}_{g,n} \mid \text{dual graph of } (C; p_1, \dots, p_n) \text{ is } \Gamma\}$$

³involution means $i \circ i = \text{id}$

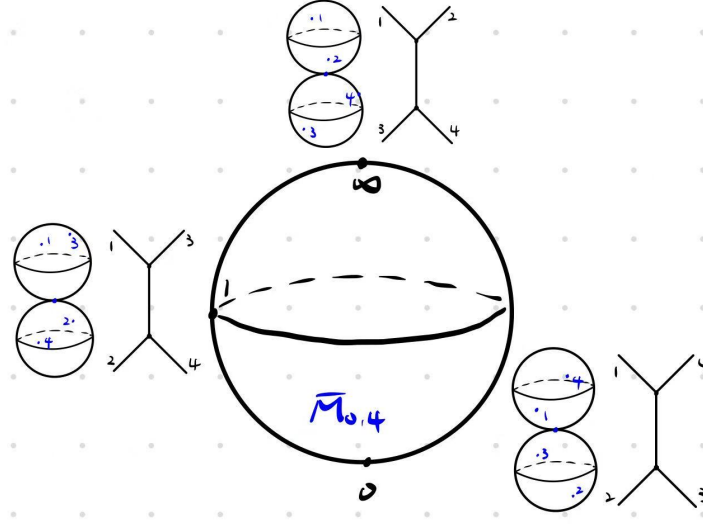


Figure 16: The boundary case of moduli space in $\overline{\mathcal{M}}_{0,4}$, where $\mathcal{M}_{0,4} \cong \mathbb{P}^1 \setminus \{0, 1, \infty\}$ and its compactification is just $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ with the boundary case at $0, 1, \infty$ listed in the figure.

and the stratification of $\overline{\mathcal{M}}_{g,n}$ is given by the disjoint union (again, on set level!) of strata

$$\overline{\mathcal{M}}_{g,n} = \coprod_{\Gamma \in G_{g,n}} \mathcal{S}_{\Gamma}$$

For each stable graph Γ , we associate the following moduli space:

$$\mathcal{M}_{\Gamma} := \prod_{v \in V} \mathcal{M}_{g(v), n(v)}, \quad \overline{\mathcal{M}}_{\Gamma} := \prod_{v \in V} \overline{\mathcal{M}}_{g(v), n(v)}$$

which contains the curves that each irreducible components corresponding to the information on the vertices. Here, we need to remark about the moduli space $\mathcal{M}_{\Gamma}$ and its compactification $\overline{\mathcal{M}}_{\Gamma}$.

Remark. The moduli space $\mathcal{M}_{\Gamma}$ associated to the stable graph Γ parametrizes the smooth vertex-wise normalization components together with the marked points corresponding to legs and preimages of nodes; note that it does not directly parametrize the already glued stable curves. Its DM compactification $\overline{\mathcal{M}}_{\Gamma}$ allows each such vertex-wise factor to degenerate further to a stable nodal curve. Correspondingly, $\mathcal{S}_{\Gamma}$ denotes the exact strata, whereas $\overline{\mathcal{S}}_{\Gamma}$ is its closure in $\overline{\mathcal{M}}_{g,n}$.

The closure relation of $\mathcal{S}_{\Gamma}$ is that

$$\mathcal{S}_{\Gamma_1} \subseteq \overline{\mathcal{S}}_{\Gamma_2} \iff \Gamma_1 \text{ is obtained from } \Gamma_2 \text{ by contracting a}$$

On the moduli space of stable curves with the same stable graph type $\overline{\mathcal{M}}_{\Gamma}$, one can define the natural projection $\pi_v : \overline{\mathcal{M}}_{\Gamma} \rightarrow \overline{\mathcal{M}}_{g(v), n(v)}$ for vertex v and gluing morphism, which "glues" the vertices with half-edges on it to the graph:

$$i_{\Gamma} : \overline{\mathcal{M}}_{\Gamma} \rightarrow \overline{\mathcal{M}}_{g,n}$$

that has image $i_{\Gamma}(\overline{\mathcal{M}}_{\Gamma}) = \overline{\mathcal{S}}_{\Gamma}$, sends each graph type (with freedom on genus) to the corresponding strata. Note that for any Γ with $|E(\Gamma)| \geq 1$, the strata contains only nodal curve, and $i_{\Gamma}(\overline{\mathcal{M}}_{\Gamma}) \subseteq \partial \overline{\mathcal{M}}_{g,n} = \overline{\mathcal{M}}_{g,n} \setminus \mathcal{M}_{g,n}$.

After we introduce the moduli space of stable curves and stable maps, we may start introducing the axiom of cohomological field theory.

5 Definition of Cohomological Field Theory

After we explained the prerequisites of moduli spaces of stable curves/maps, we shall now state the axiom of Cohomological Field Theory (CohFT). We may start with a small comparison between 2D TQFT and CohFT, which will help us see how CohFT packages the higher-level information related to the intersection theory:

1. 2D TQFT: equipped the boundary with a vector space, and the cobordism with a linear map.
2. CohFT: The "space of states" starts from a finite-dimensional vector space V (over $\mathbb{Q}$ or $\mathbb{C}$) with unit and a nondegenerate symmetric bilinear form $\eta : V \otimes V \rightarrow \mathbb{Q}$. Instead of giving a linear map for each cobordism, for a given stable pair (g, n) , we give a map

$$\Omega_{g,n} : V^{\otimes n} \rightarrow H^*(\overline{\mathcal{M}}_{g,n})$$

with a certain gluing operation, which will be the analogy of gluing cobordisms (composition of linear maps). Note that the coefficient of the cohomology needs to be in the base field of the vector space to ensure the tensor product makes sense.

5.1 Axioms of Cohomological Field Theory

The story of CohFT, as we have mentioned at the start of the section, begins from the triple of state space $(V, \eta, \mathbf{1})$ where

1. V is a finite dimensional vector space over $\mathbb{F} = \mathbb{Q}$ or $\mathbb{C}$.
2. $\eta : V \otimes V \rightarrow \mathbb{F}$ is a nondegenerate symmetric bilinear form, and we denote $\eta_{ij} := \eta(e_i, e_j)$ with the inverse $(\eta^{ij}) := (\eta_{ij})^{-1}$.
3. $\mathbf{1} \in V$ is a specific element in V which will be the unit.

We may assume that V is a $\mathbb{Q}$ -vector space and the cohomology of the moduli space has coefficients in $\mathbb{Q}$ in the following axioms.

A CohFT is a system $\Omega = (\Omega_{g,n})_{2g-2+n>0}$ of tensors

$$\Omega_{g,n} \in H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) \otimes (V^*)^{\otimes n}, \quad 2g - 2 + n > 0$$

more precisely, $\Omega_{g,n} : V^{\otimes n} \rightarrow H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q})$ is a symmetrical multilinear map. Recall that we have three natural map on $\overline{\mathcal{M}}_{g,n}$:

1. Forgetful map: $p : \overline{\mathcal{M}}_{g,n+1} \rightarrow \overline{\mathcal{M}}_{g,n}$ with stabilization in low genus.
2. Gluing map on a single curve: $q : \overline{\mathcal{M}}_{g-1,n+2} \rightarrow \overline{\mathcal{M}}_{g,n}$.
3. Gluing map on two curves: $r : \overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \rightarrow \overline{\mathcal{M}}_{g_1+g_2,n_1+n_2}$.

The system must satisfy the following axioms (at least the first two axioms), which constrain the behavior of the system under the map induced by the natural existence morphisms above:

Axiom 1 (Symmetry). *Each tensor $\Omega_{g,n}$ in the system is S_n -invariant under the natural symmetry action on $H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) \otimes (V^*)^{\otimes n}$, i.e., $\forall v_1, \dots, v_n \in V : \forall \sigma \in S_n :$*

$$\Omega_{g,n}(v_1, \dots, v_n) = \sigma^* \Omega_{g,n}(v_1, \dots, v_n) = \Omega_{g,n}(v_{\sigma(1)}, \dots, v_{\sigma(n)}) \in H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q})$$

Axiom 2 (Gluing). *Start from the given data $\Omega_{g,n} \in H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) \otimes (V^*)^{\otimes n}$. We may pullback the tensor via boundary morphisms q and r , and we require that*

1. The pullback tensor $q^*\Omega_{g,n} \in H^*(\overline{\mathcal{M}}_{g-1,n+2}, \mathbb{Q}) \otimes (V^*)^{\otimes n}$ is obtained by $\Omega_{g-1,n+2}$ contract with then bivector $\sum_{i,j} \eta^{ij} e_i \otimes e_j \in V \otimes V$, i.e., $\forall v_i \in V$

$$q^*\Omega_{g,n}(v_1, \dots, v_{n-1}) = \sum_{i,j} \eta^{ij} \Omega_{g-1,n+2}(v_1, \dots, v_n, e_i, e_j)$$

2. Similarly, the pullback tensor $r^*\Omega_{g,n} \in H^*(\overline{\mathcal{M}}_{g_1,n_1+1}, \mathbb{Q}) \otimes H^*(\overline{\mathcal{M}}_{g_2,n_2+1}, \mathbb{Q}) \otimes (V^*)^{\otimes n}$ ($g = g_1 + g_2$ and $n = n_1 + n_2$) is given by contract $\Omega_{g_1,n_1+1} \otimes \Omega_{g_2,n_2+1}$ with the bivector, i.e., $\forall v_i \in \Omega_{g,n}$

$$r^*\Omega_{g,n}(v_1, \dots, v_n) = \sum_{i,j} \eta^{ij} \Omega_{g_1,n_1+1}(v_1, \dots, v_{n_1}, e_i) \otimes \Omega_{g_2,n_2+1}(v_{n_1+1}, \dots, v_{n_1+n_2}, e_j)$$

Axiom 3 (Unit). Given $\Omega_{g,n} \in H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) \otimes (V^*)^{\otimes n}$, the pullback tensor induced by the forgetful morphism $p^*\Omega_{g,n} \in H^*(\overline{\mathcal{M}}_{g,n-1}, \mathbb{Q}) \otimes (V^*)^{\otimes n}$ is defined by $\forall v_i \in V$

$$p^*\Omega_{g,n}(v_1, \dots, v_{n-1}) := \Omega_{g,n+1}(v_1, \dots, v_n, \mathbf{1})$$

In addition, the in the base case $(g, n) = (0, 3)$, the tensor is required to be $\Omega_{0,3}(v_1, v_2, \mathbf{1}) := \eta(v_1, v_2)$.

The three axioms give rise to the definition of a CohFT

Definition 5.1 (Cohomological Field Theory). A cohomological field theory is a system $\Omega = (\Omega_{b,n})_{2g-2+n>0}$ of tensors $\Omega_{g,n} \in H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) \otimes (V^*)^{\otimes n}$ that satisfies the axion 1 and 2. If a CohFT Ω also satisfies axiom 3, then it is said to be a CohFT with a unit.

Remark. We need to remark here that we can allow the CohFT to be graded S_n -equivariant, i.e.

$$\Omega_{g,n}(v_1, \dots, v_n) = \epsilon(\sigma; \gamma_1, \dots, \gamma_n) \sigma^* \Omega_{g,n}(v_1, \dots, v_n)$$

where $\epsilon(\sigma; \gamma_1, \dots, \gamma_n)$ is a $\mathbb{Z}_2$ -grading. In the following section, we will see that one of the most important example case is that V is an graded vector space (cohomology group, actually). More precisely, when V is a $\mathbb{Z}_2$ -graded supervector space, we may choose the bilinear form that $\forall a, b \in V$

$$\eta(a, b) = (-1)^{|a||b|} \eta(b, a)$$

and the axiom 1 also needs to be modified to $\mathbb{Z}_2$ -graded version.

As an example, we first consider a trivial case:

Example (Trivial CohFT). The trivial CohFT is the theory given by taking $V = \mathbb{Q}$, $\eta(1, 1) = 1$, and $\mathbf{1} = 1$. With the tensor $\Omega_{g,n}(1, \dots, 1) = 1 \in H^0(\overline{\mathcal{M}}_{g,n}, \mathbb{Q})$.

A more interesting example is actually the motivation of CohFT, the Gromov-Witten theory over a nonsingular projective variety X :

Example (Gromov-Witten). We may take $V = H^*(X, \mathbb{Q})$, and fix $\mathbf{1} = 1$ and the pairing on V is given by the Poincaré pairing⁴ that $\forall \alpha, \beta \in H^*(X, \mathbb{Q})$

$$\eta(\alpha, \beta) = \int_X \alpha \wedge \beta$$

This is the theory given by Kontsevich and Manin in [KM94]. More about Gromov-Witten Theory will be in Section 6. The tensor here is defined by the Gromov-Witten classes $\Omega_{g,n,\beta}^X$, which $\forall \gamma_i \in H^*(X, \mathbb{Q})$

$$\Omega_{g,n}(\gamma_1, \dots, \gamma_n) := \sum_{\beta \in H_2(X, \mathbb{Q})} q^\beta \Omega_{g,n,\beta}^X(\gamma_1, \dots, \gamma_n)$$

where q is the Novikov parameter. Note that in this case, the $\mathbb{Q}$ -vector space $V = H^*(X, \mathbb{Q})$, so we need to use the $\mathbb{Z}_2$ -graded version of the CohFT axioms.

⁴To ensure the pairing is symmetric, we need either consider a $\mathbb{Z}_2$ -grading, or take X be some target space with no odd cohomology.

5.2 Topological Part of CohFT and Quantum Product

Similar to what we do to 2D TQFT, we may discuss the algebraic structure behind the CohFT. To begin with, the structure of a CohFT naturally contains a TQFT, which is called the "topological part of CohFT": in the following definition, the cohomological ring

$$H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) := \bigoplus_{k \geq 0} H^k(\overline{\mathcal{M}}_{g,n}, \mathbb{Q})$$

has natural projection $[-]^k : H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) \rightarrow H^k(\overline{\mathcal{M}}_{g,n}, \mathbb{Q})$.

Definition 5.2 (Topological Part of CohFT). Given CohFT $\Omega = (\Omega_{g,n})_{2g-2+n>0}$, the topological part ω of Ω is given by

$$\omega_{g,n} := [\Omega_{g,n}]^0 : (V^*)^{\otimes n} \rightarrow H^0(\overline{\mathcal{M}}_{g,n}, \mathbb{Q})$$

This topological part of CohFT is the coarsest topological information of the system. In the following passage, we may see that the topological part of the CohFT is determined by the "three-points function" of the genus-zero part of the theory.

The most well understood algebraic structure of CohFT is in the the three points function of genus-zero part $\Omega_{0,3}$, which gives the quantum product:

Theorem 5.3 (Quantum Product). Let $\Omega = (\Omega_{g,n})_{2g-2+n>0}$ be a CohFT with unit. Then the genus-zero part $\Omega_{0,3} \in H^*(\overline{\mathcal{M}}_{0,3}, \mathbb{Q}) \otimes (V^*)^{\otimes 3}$ of the CohFT give rise to a commutative, associative, and unital quantum product on V defined by

$$\eta(v_1 \bullet v_2, v_3) = \Omega_{0,3}(v_1, v_2, v_3)$$

Proof. Note that $\overline{\mathcal{M}}_{0,3} = \{\text{pt}\}$ is a point, $H^*(\overline{\mathcal{M}}_{0,3}, \mathbb{Q}) = \mathbb{Q}$. Thus, by the fact that $\eta : V \otimes V \rightarrow \mathbb{Q}$ is nondegenerate, fix $a, b \in V$, the map

$$c \mapsto \eta(a \bullet b, c) = \Omega_{0,3}(a, b, c)$$

gives a linear functional (Riesz representation) $l_{a,b} : V \rightarrow \mathbb{Q}$. But by the nondegeneracy of η

$$\Phi_\eta : V \rightarrow V^*, \quad x \mapsto \eta(x, -)$$

is injective (since $\ker \Phi = \{0\}$) and surjective in finite dimensions (since $\dim V = \dim V^*$), thus, is an isomorphism. Thus, for any linear functional $l_{a,b} : V \rightarrow \mathbb{Q}$ with $l_{a,b}(c) = \Omega_{0,3}(a, b, c)$, we may find unique $x \in V$ such that $l_{a,b} = \Phi_\eta(x)$, and we denote $x := a \bullet b$. This shows the well-definedness of the quantum product.

(Commutative) By Axiom 1 (symmetric axiom), the CohFT tensor $\Omega_{g,n}$ is invariant under the natural action of S_n , thus

$$\eta(a \bullet b, c) = \Omega_{0,3}(a, b, c) = \Omega_{0,3}(b, a, c) = \eta(b \bullet a, c)$$

which gives $a \bullet b = b \bullet a$ and proves the commutativity.

(Unital) By Axiom 3 (unit axiom), we may take $\mathbf{1} \in V$ and

$$\eta(a \bullet \mathbf{1}, b) = \Omega_{0,3}(a, \mathbf{1}, b) = \Omega_{0,3}(a, b, \mathbf{1}) = \eta(a, b)$$

The nondegeneracy gives $a \bullet \mathbf{1} = a$. By commutativity, $a \bullet \mathbf{1} = \mathbf{1} \bullet a = a$.

(Associative) Note that to prove associativity, we consider $a, b, c, d \in V$ and

$$\eta(a \bullet (b \bullet c), d) = \Omega_{0,3}(a, b \bullet c, d), \quad \eta((a \bullet b) \bullet c, d) = \Omega_{0,3}(a \bullet b, c, d)$$

We may take basis $\{e_i\}$ of space V and defined $\eta_{ij} := \eta(e_i, e_j)$, $(\eta^{ij}) = (\eta_{ij})^{-1}$. Then, we may write $a \bullet b$ also in coordinates: Since $\eta(a \bullet b, e_k) = \Omega_{0,3}(a, b, e_k)$, suppose $a \bullet b = \sum_j x^j e_j$, we can write

$$\sum_j x^j \eta_{jk} = \Omega(a, b, e_k)$$

One can solve this equation with the inverse η^{ij}

$$x^i = \sum_j x^j \sum_k \eta_{jk} \eta^{ki} = \sum_k \Omega_{0,3}(a, b, e_k) \eta^{ki}$$

and then we can write $\eta((a \bullet b) \bullet c, d)$ and $\eta(a \bullet (b \bullet c), d)$ both in coordinate as:

$$\begin{aligned} \eta((a \bullet b) \bullet c, d) &= \sum_{i,j} \Omega_{0,3}(a, b, e_i) \eta^{ij} \Omega_{0,3}(c, d, e_j) \\ \eta(a \bullet (b \bullet c), d) &= \sum_{i,j} \Omega_{0,3}(b, c, e_i) \eta^{ij} \Omega_{0,3}(a, d, e_j) \end{aligned}$$

To prove the associativity, we can just prove that the two expressions above are equal, i.e., the WDVV-type (Witten-Dijkgraaf-Verlinde-Verlinde) equation (we make the convention that all basis elements should be written in the last entry, by the symmetric axiom, we are allowed to do this)

$$\sum_{i,j} \Omega_{0,3}(a, b, e_i) \eta^{ij} \Omega_{0,3}(c, d, e_j) = \sum_{i,j} \Omega_{0,3}(b, c, e_i) \eta^{ij} \Omega_{0,3}(a, d, e_j)$$

However, by Axiom 2 (gluing axiom), the equation is just

$$\text{LHS} = r^* \Omega_{0,4}(a, b, c, d), \quad \text{RHS} = r^* \Omega_{0,4}(b, c, a, d) = r^* \sigma^* \Omega_{0,4}(a, b, c, d)$$

By Axiom 1 (symmetry axiom), $\sigma^* \Omega_{0,4}(a, b, c, d) = \Omega_{0,4}(a, b, c, d)$ and the two expressions above are equal, and thus,

$$\eta(a \bullet (b \bullet c), d) = \eta((a \bullet b) \bullet c, d) \quad \forall d \in V$$

which proves the associativity of the quantum product. $\square$

Note that for $\mathbb{Z}_2$ -graded version of CohFT require the bilinear form η to be an even supersymmetric pairing, so we naturally have supercommutative quantum product, i.e.,

$$a \bullet b = (-1)^{|a||b|} b \bullet a$$

where $|a|$ and $|b|$ are the degree of $a, b \in V$.

We may also remark another way to illustrate the associativity of the quantum product using the independence of the point-inclusion pullback on cohomology:

Remark. Another way to understand the associativity relation is to look directly at the geometry of $\overline{M}_{0,4} \cong \mathbb{P}^1$. Its boundary

$$\partial \overline{M}_{0,4} = \overline{M}_{0,4} \setminus M_{0,4} = \{12|34, 13|24, 14|23\}$$

consists of three zero-dimensional boundary strata, each corresponding to one of the three possible degenerations of a four-pointed stable rational curve into two irreducible components. Equivalently, these three boundary points encode the three different ways of breaking a four-point correlator into a product of two three-point correlators, so they represent the geometric origin of the associativity relation.

For each such degeneration, the gluing morphism

$$r : \overline{M}_{0,3} \times \overline{M}_{0,3} = \{\text{pt}\} \longrightarrow \overline{M}_{0,4}$$

is simply the inclusion of the corresponding boundary point. Since $H^*(\text{pt}, \mathbb{Q}) = \mathbb{Q}$ is concentrated in degree 0, the induced pullback

$$r^* : H^*(\overline{M}_{0,4}, \mathbb{Q}) \rightarrow H^*(\text{pt}, \mathbb{Q}) = \mathbb{Q}$$

kills all positive-degree classes and retains only the degree-zero part. In other words, if $\alpha \in H^*(\overline{M}_{0,4}, \mathbb{Q})$, then $r^*(\alpha) = [\alpha]^0$, where $[\alpha]^0 \in \mathbb{Q}$ denotes the degree-0 component of α . Thus, from the viewpoint of the pullback to a boundary point, all higher-degree information on $\overline{M}_{0,4}$ disappears.

Moreover, although the three boundary strata are disconnected as a subspace of $\partial\overline{M}_{0,4}$, the ambient space $\overline{M}_{0,4} \cong \mathbb{P}^1$ is path connected. Hence, any two-point inclusions

$$\text{pt} \hookrightarrow \overline{M}_{0,4}$$

are homotopic, and therefore induce the same map on cohomology. It follows that the projection

$$r^* : H^*(\overline{M}_{0,4}, \mathbb{Q}) \rightarrow \mathbb{Q}$$

is independent of which boundary point, or equivalently, which gluing map, is chosen. This gives another conceptual proof of the identity

$$r^*\Omega_{0,4}(a, b, c, d) = r^*\Omega_{0,4}(b, c, a, d)$$

which is exactly the cohomological expression of the associativity of the quantum product.

This viewpoint is useful for the discussion below because it isolates the mechanism by which a CohFT reduces to its topological part. Once one projects to degree 0, the geometric complexity of the moduli space no longer contributes: what remains is only the combinatorics of how markings are distributed among components together with the contraction determined by the bilinear form. In this sense, the topological part of a CohFT behaves like a purely graph-theoretic shadow of the full theory. For the next result, it is therefore convenient to introduce an operation that records this contraction process explicitly, namely the contraction along a graph.

Definition 5.4 (Contraction along Stable Graph). Let $(C; p_1, \dots, p_n) \in \overline{\mathcal{M}}_{g,n}$ be a stable curve with dual graph $\Gamma = \Gamma_C$, $\Omega = (\Omega_{g,n})_{2g-2+n>0}$ is a CohFT. Then we can define the contraction of Ω along the graph Γ as a tensor Ω_Γ , where, for any $u_1, \dots, u_n \in V$, the tensor $\Omega_\Gamma : V^{\otimes n} \rightarrow H^*(\overline{\mathcal{M}}_\Gamma, \mathbb{Q})$ is given by

$$\begin{aligned} \Omega_\Gamma(u_1, \dots, u_n) &= \text{Cont}_\Gamma \left(\bigotimes_{v \in V(\Gamma)} \Omega_{g(v), n(v)} \right) \in H^*(\overline{\mathcal{M}}_\Gamma, \mathbb{Q}) \\ &= \sum_{\{\mu_e, \nu_e\}_{e \in E(\Gamma)}} \left(\prod_{e \in E(\Gamma)} \eta^{\mu_e \nu_e} \right) \bigotimes_{v \in V(\Gamma)} \Omega_{g(v), n(v)}(w_{v,1}, \dots, w_{v,n(v)}) \end{aligned}$$

where for each vertex v , we assign the corresponding tensor $\Omega_{g(v), n(v)}$ by the entries $w_{v,1}, \dots, w_{v,n(v)} \in V$ of every half-edges on v that defined as follows:

1. If the i -th half-edge on v is a leg with index $j \in \{1, \dots, n\}$, then $w_{v,i} = u_j$ is an input vector.
2. If the corresponding half is a half-edge of an internal edge $e \in E(\Gamma)$, then the entry on the pair of half-edge is a pair of basis vectors e_{μ_e} and e_{ν_e} that contract with $\eta^{\mu_e \nu_e}$.

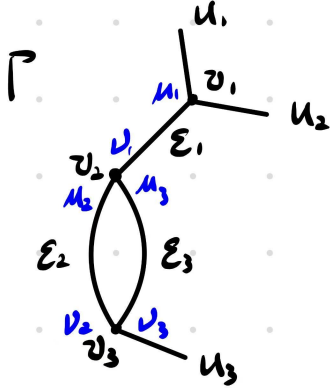
We now give a concrete computation illustrating the topological part of a CohFT. The calculation is a direct application of the stable-graph formalism. Given a stable graph Γ , the marked points are represented by the legs, the irreducible components of the corresponding stable curve are represented by the vertices, and the nodes are represented by the internal edges. To each leg we assign the corresponding input $u_1, \dots, u_n$, and to each vertex $v \in V(\Gamma)$ we assign the tensor $\Omega_{g(v), n(v)}$. The contribution of the graph is then obtained by contracting, along every internal edge, the tensor factors attached to the two corresponding half-edges by means of the inverse bilinear form η^{-1} . In this way, the combinatorial data of Γ encode the degeneration type of the stable curve, while the CohFT gluing axiom turns these data into a cohomology class on the boundary stratum determined by Γ . More precisely, we define

$$\Omega_\Gamma(u_1, \dots, u_n) := \text{Cont}_\Gamma \left(\bigotimes_{v \in V(\Gamma)} \Omega_{g(v), n(v)} \right) \in H^*(\overline{\mathcal{M}}_\Gamma, \mathbb{Q}) \cong \bigotimes_{v \in V(\Gamma)} H^*(\overline{\mathcal{M}}_{g(v), n(v)}, \mathbb{Q})$$

Here Cont_Γ denotes the total contraction prescribed by the internal edges of Γ . Equivalently, after choosing a basis of the state space V , one inserts basis elements at the two branches of each node and sums over all indices

against the coefficients of the inverse metric. Thus each vertex contributes a local tensor, whereas each internal edge performs one contraction between two such local contributions. In the topological sector, this description is especially simple: since only degree-zero classes appear, no higher-degree tautological contribution is involved, and the graph contribution is entirely determined by repeated contractions of copies of $\Omega_{0,3}$ using the Frobenius pairing. The following example makes this procedure completely explicit in the case of genus 1 with three marked points.

Example (Computation of $\omega_{1,3}$). *We shall consider the following stable graph Γ and the corresponding stable curve, in the graph, all vertices has genus 0:*



The left graph is the stable graph of a stable curve with genus $g = 1$, 3 nodes, and 3 marked points. Each vertex corresponds to an irreducible component isomorphic to $\mathbb{P}^1$ with three special points.

Let $\omega_{1,3} := [\Omega_{1,3}]^0$ be the topological part of some CohFT on $(V, \eta, 1)$. Then for all $u_1, u_2, u_3 \in V$, we have

$$\begin{aligned} \Omega_\Gamma(u_1, u_2, u_3) &= \text{Cont}_\Gamma \left(\bigotimes_{i=1}^3 \Omega_{0,3} \right) \in H^*(\overline{\mathcal{M}}_\Gamma, \mathbb{Q}) \\ &= \sum_{\{\mu_i, \nu_i\}} \eta^{\mu_1 \nu_1} \eta^{\mu_2 \nu_2} \eta^{\mu_3 \nu_3} \Omega_{0,3}(u_1, u_2, e_{\mu_1}) \Omega_{0,3}(e_{\nu_1}, e_{\mu_2}, e_{\mu_3}) \Omega_{0,3}(e_{\nu_2}, e_{\nu_3}, u_3) \end{aligned}$$

The dual graph of a stable curve in $\overline{\mathcal{M}}_{1,3}$.

By the next theorem, the topological part of genus 1 with three marked points $\omega_{1,3} = \Omega_\Gamma$.

Theorem 5.5 (Topological Part is TQFT). Let Ω be a CohFT. The topological part ω of Ω is a TQFT that is uniquely and effectively determined by the three-point function $\Omega_{0,3}$ of the genus-zero part.

Proof. The topological part $\omega_{g,n} := [\Omega_{g,n}]^0 : V^{\otimes n} \rightarrow \mathbb{Q}$ is a multilinear form, and $\omega_{0,3} = \Omega_{0,3}$.

(Topological part is a TQFT) We may claim that the $\mathbb{Q}$ -algebra $(V, \bullet, \eta, 1)$ is a commutative Frobenius algebra, and thus, leads to a 2D TQFT. We have already shown in Theorem 5.3 that $(V, \bullet, \eta, 1)$ is a commutative, associative, and unital algebra, with a nondegenerate pairing η . Note that η is associative since $\forall a, b, c \in V$

$$\eta(a \bullet b, c) = \omega_{0,3}(a, b, c) = \omega_{0,3}(b, c, a) = \eta(b \bullet c, a) = \eta(a, b \bullet c)$$

By Proposition 3.2, V admits the structure of a commutative Frobenius algebra.

(Uniquely Determined by $\Omega_{0,3}$) Note that $\forall v_1, \dots, v_n \in V$, then the topological part $\omega_{g,n}(v_1, \dots, v_n) = \lambda \cdot 1 \in \mathbb{Q}$ is the scalar multiplication of the identity class (which is just 1 since the cohomology group is $\mathbb{Q}$). Thus, we may pick a maximally degenerate curve $(C; p_1, \dots, p_n) \in \overline{\mathcal{M}}_{g,n}$, that is, a stable curve all of whose irreducible components are isomorphic to $\mathbb{P}^1$, each carrying exactly three special points (i.e., the point in $\overline{\mathcal{M}}_{0,3}$). We may restrict the cohomology class to the given point (more precisely, pullback the cohomology class by the inclusion map $i : \{(C; p_1, \dots, p_n)\} \hookrightarrow \overline{\mathcal{M}}_{g,n}$, since a point only contains 0-th cohomology

$$\omega_{g,n}(v_1, \dots, v_n) \big|_{(C; p_1, \dots, p_n)} = \Omega_{g,n}(v_1, \dots, v_n) \big|_{(C; p_1, \dots, p_n)}$$

Thus, by repeated application of the gluing axiom along every internal edge of the dual graph Γ , we can write the topological part as

$$\omega_{g,n}(v_1, \dots, v_n) = \text{Cont}_\Gamma \left(\bigotimes_{v \in V(\Gamma)} \omega_{0,3} \right)$$

and by the fact that $\omega_{0,3} = \Omega_{0,3}$, we can show that ω is effectively determined by the genus-zero three-point function $\Omega_{0,3}$.

It remains to show the uniqueness. Suppose we have another TQFT ω' with the same three-point function

$\omega'_{0,3} = \Omega_{0,3}$, the decomposition of the TQFT as genus zero part of CohFT is the same as that of ω (i.e., provides the same class/number). Since both class lie in $H^0(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) \cong \mathbb{Q}$, they must coincide. This completes the proof of the theorem. $\square$

Thus, the CohFT on V leads to a commutative, associative, and unital $\mathbb{Q}$ -algebra $(V, \bullet, \mathbf{1})$. Recall that a finite-dimensional $\mathbb{Q}$ -algebra V is semisimple if it can be written as the product of simple algebras. Equivalently, V is semisimple if and only if there exists a basis $\{e_i\}$ of idempotents in $V_{\mathbb{C}} := V \otimes_{\mathbb{Q}} \mathbb{C}$ that

$$e_i \bullet e_j = \delta_{ij} e_i$$

Definition 5.6 (Semisimple CohFT). A CohFT $\Omega = (\Omega_{g,n})_{2g-2+n>0}$ is semisimple if the corresponding $\mathbb{Q}$ -algebra $(V, \bullet, \mathbf{1})$ is semisimple.

6 Gromov-Witten Invariants

In the last section of the main text, we are going to introduce one of the most important examples of CohFT, the example that motivated the theory, namely the Gromov-Witten (GW) Theory. A famous fact is that the GW theory discusses the moduli space of stable maps $\overline{\mathcal{M}}_{g,n}(X, \beta)$ to "count curves on target space X ". In an ideal case, such a space that parametrizes curves on X would be smooth, compact, and pure-dimensional (namely, every component having the expected dimension), and all of which would allow us to use the ordinary intersection theory to count curves. For the sake of exposition, let us temporarily assume that this is the case.

Remark. In the following passages, we may informally treat the cohomology class as "differential forms", and use the notation $\wedge$ as the cup product (which is usually denoted as $\smile$).

We may fix a target space $X \subseteq \mathbb{P}^n$ as an m -dimensional smooth complex projective variety (which is also a compact complex manifold) and consider the holomorphic map $f : C \rightarrow X$ for some Riemann surface C . Recall that we have defined the moduli space of stable maps with fixed $\beta \in H_2(X, \mathbb{Q})$ as:

$$\overline{\mathcal{M}}_{g,n}(X, \beta) = \{(C, f) \mid C \in \overline{\mathcal{M}}_{g,n}, f : C \rightarrow X \text{ holomorphic}, u_*[C] = \beta\}$$

such that the automorphism of $f : C \rightarrow X$ is finite. The natural evaluation map is given by

$$\text{ev}_i : \overline{\mathcal{M}}_{g,n}(X, \beta) \rightarrow X$$

which evaluate the holomorphic map on the marked points $\forall i \in \{1, \dots, n\}$

$$(f : (C; p_1, \dots, p_n) \rightarrow X) \mapsto f(p_i) \in X$$

To defined the GW invariant, consider subvarieties $V_1, \dots, V_n$ of X with (real) codimension n_i and their homology class are $[V_i] \in H_{2m-n_i}(X, \mathbb{Q})$. We may consider the cohomology class given by the Poincaré duals of these homology classes $\gamma_i := \text{PD}[V_i] \in H^{n_i}(X, \mathbb{Q})$, then, the pullback class $\text{ev}_i^* \gamma_i$ gives the constraint that $f(p_i) \in V_i$.

Remark. To clarify this condition, note that the condition that $f(p_i) \in V_i$ corresponds to the preimage set

$$\text{ev}_i^{-1}(V_i) := \{f \in \overline{\mathcal{M}}_{g,n}(X, \beta) \mid f(p_i) \in V_i \text{ for the fixed } i\}$$

Then, if $\text{ev}_i \pitchfork V_i$, we can say that $\text{ev}_i^{-1}(V_i)$ is a smooth submanifold with codimension n_i (same with V_i), and thus,

$$\text{PD}_{\overline{\mathcal{M}}_{g,n}(X, \beta)}[\text{ev}_i^{-1}(V_i)] = \text{ev}_i^* \text{PD}_X[V_i] = \text{ev}_i^* \gamma_i$$

which gives the condition above.

Moreover, since the Poincaré duality has the following property

$$\mathrm{PD}_X[V_i] \wedge \mathrm{PD}_X[V_j] = \mathrm{PD}[V \cap W]$$

if V_i and V_j intersect transversely with the same orientation. With the condition $\mathrm{ev}_i \pitchfork V_i \forall i$, we may consider the cohomology class given by

$$\bigwedge_{i=1}^n \mathrm{ev}_i^* \gamma_i = \mathrm{PD}_{\overline{\mathcal{M}}_{g,n}(X,\beta)} \left[\bigcap_{i=1}^n \mathrm{ev}_i^{-1}(V_i) \right]$$

This cohomology class represents exactly (in a suitably generic setting) those stable maps $f : C \rightarrow X$ such that $f(p_i) \in V_i \forall i$.

Having interpreted the classes $\mathrm{ev}_i^* \gamma_i$ as incidence conditions, the natural follow-up question is when their product can be integrated over the moduli space to get the curve count. This is governed by a dimension constraint: the total degree of the insertions must equal the dimension of the moduli space. Under suitable transversality assumptions, this condition is equivalent to requiring that the constrained moduli space be zero-dimensional, so that the resulting integral becomes a finite signed count of stable maps.

Let $d := \dim_{\mathbb{R}} \overline{\mathcal{M}}_{g,n}(X,\beta)$, since each cohomology class $\mathrm{ev}_i^* \gamma_i$ has degree n_i , then, let $a = \sum_i n_i$ and the product is given by

$$\bigwedge_{i=1}^n \mathrm{ev}_i^* \gamma_i \in H^a(\overline{\mathcal{M}}_{g,n}(X,\beta), \mathbb{Q})$$

Therefore, in order to integrate this class over the moduli space and obtain a number, we must impose the dimension constraint

$$\sum_{i=1}^n n_i = d$$

Under the transversality assumptions above, the intersection $\bigcap_i \mathrm{ev}_i^{-1}(V_i)$ is a smooth submanifold of codimension d (under the constraint), i.e., the intersection is zero-dimensional, and thus consists of finitely many points for generic choices of the constraints $V_1, \dots, V_n$. Hence, we compute the count of smooth curves by integrate the cohomology class (or pairing with the top homology class) as follows:

$$\begin{aligned} \langle \gamma_1, \dots, \gamma_n \rangle_{g,\beta}^X &:= \int_{\overline{\mathcal{M}}_{g,n}(X,\beta)} \bigwedge_{i=1}^n \mathrm{ev}_i^* \gamma_i = \int_{\overline{\mathcal{M}}_{g,n}(X,\beta)} \mathrm{PD}_{\overline{\mathcal{M}}_{g,n}(X,\beta)} \left[\bigcap_{i=1}^n \mathrm{ev}_i^{-1}(V_i) \right] \\ &= \int_{\bigcap_{i=1}^n \mathrm{ev}_i^{-1}(V_i)} \mathrm{PD}_{\overline{\mathcal{M}}_{g,n}(X,\beta)} [\overline{\mathcal{M}}_{g,n}(X,\beta)] = \int_{\bigcap_{i=1}^n \mathrm{ev}_i^{-1}(V_i)} 1 = \left| \bigcap_{i=1}^n \mathrm{ev}_i^{-1}(V_i) \right|_{\mathrm{alg}} \end{aligned}$$

Strictly speaking, the right-hand side is not merely the cardinality of the constrained moduli space, but its algebraic count: each isolated solution contributes a local sign determined by the orientations. In the orbifold/stack setting, one must further weight each point by the reciprocal of the order of its automorphism group.

However, this setting does not work in general. In general, the moduli space of stable maps is not a smooth manifold of the dimension suggested by the naive transversality picture. It may be singular, may contain strata of different actual dimensions, and the ordinary fundamental class may no longer provide the correct object against which to integrate. What remains well-behaved is instead the infinitesimal deformation problem of a stable map, whose index gives the expected dimension of the moduli space. This number is called the virtual dimension. The virtual fundamental class is then designed to replace the ordinary fundamental class by a homology class of precisely this virtual dimension.

6.1 A Simple Example of Virtual Dimension and Virtual Class

The naive curve-counting construction implicitly assumes that the moduli space $M = \overline{\mathcal{M}}_{g,n}(X,\beta)$ is a smooth compact space of the expected dimension. In practice, this is rarely the case. Even after stable compactifi-

cation, the moduli space may still be singular and may contain components or strata of different dimensions. Consequently, the ordinary fundamental class of M is not the correct object for defining enumerative invariants.

To remedy this, one does not use the actual dimension of the moduli space, but rather the dimension predicted by the deformation theory of stable maps. At a given stable map, the infinitesimal theory is governed by a deformation complex, and its index gives the virtual dimension:

$$\text{vdim} = \dim(\text{deformation space}) - \dim(\text{automorphism space}) - \dim(\text{obstruction space})$$

The crucial point is that this index is deformation-invariant, even when the moduli space itself is singular or fails to have pure dimension. The virtual fundamental class

$$[\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\text{vir}}$$

is then constructed so as to play the role of the fundamental class of a hypothetical smooth space of dimension vdim . Integrating against this virtual class produces the correct curve-counting invariants.

Before we discuss the stable map, we may give a simple finite-dimensional example to illustrate the idea.

Example (Vector Bundle with Rank r). *Let U be a smooth, compact, and oriented manifold with $\dim_{\mathbb{R}} U = n$, $p : E \rightarrow U$ gives a smooth, oriented real vector bundle such that $\text{rank}_{\mathbb{R}} E = r$. Let $s \in \Gamma(U, E)$ be a smooth section of the vector bundle. We want to solve for something in the set of zeros (as the "moduli space") of the section*

$$Z = s^{-1}(0)$$

We first study this problem locally, consider $p \in Z$ with a locally trivialized chart, we may write the section as the map $s = (s_1, \dots, s_r) : W \subseteq U \rightarrow \mathbb{R}^r$, and the condition that

$$s_1(p) = \dots = s_r(p) = 0$$

Thus, we have n variables and r equations; thus, the naive intuition is that the solution space is $n - r$ dimensions, i.e., the expected dimension is $n - r$.

To study the local geometry rigorously, we consider the differential of s near $p \in Z$ given by $ds_p : T_p U \rightarrow E_p$, which is a finite-dimensional linear map. We rewrite this structure as

$$0 \longrightarrow \ker(ds_p) \longrightarrow T_p U \xrightarrow{ds_p} E_p \longrightarrow \text{coker}(ds_p) \longrightarrow 0$$

Here, we may remark the following illustration:

1. $\ker(ds_p)$ is corresponding to the infinitesimal deformations.

Given a (first-order) deformation direction, we may ask the natural question of whether it can be integrated to a family of solutions (as a curve); then we need at least $ds_p(v) = 0$.

2. $\text{coker}(ds_p)$ is the obstruction.

If the map is not surjective, then there is some direction that cannot be reached by any first-order deformation. Thus, those "losing directions" are the obstruction, and all obstructions are contained in

$$\text{coker}(ds_p) = E_p / \text{im}(ds_p)$$

Most importantly,

$$\dim \text{im}(ds_p) = r - \dim \text{coker}(ds_p) = n - \dim \ker(ds_p)$$

and the virtual (expected) dimension is just the dimension of Z , given by

$$\dim Z = n - r = \dim \ker(ds_p) - \dim \text{coker}(ds_p)$$

We may further consider two possibilities:

(When $s \pitchfork 0_E$) In this case, we have no obstruction, i.e., $\dim \text{coker}(ds_p) = 0$, and $\dim \ker(ds_p) = n - r$. By

the implicit function theorem, we claim that Z is a smooth manifold of dimension $n - r$, and we have a short exact sequence

$$0 \longrightarrow T_p Z \longrightarrow T_p U \xrightarrow{ds_p} E_p \longrightarrow 0$$

since both U and E are oriented, we can give Z an inherent orientation. By pullback the Thom class $\tau_E \in H_c^r(E, \mathbb{Q})$ by the zero section

$$e(E) := 0_E^*(\tau_E) \in H^r(U, \mathbb{Q})$$

We can obtain the Euler class. With the inclusion map $i : Z \hookrightarrow U$, then $i_*[Z] = e(E) \frown [U]$, equivalently $\text{PD}_U(i_*[Z]) = e(E)$.

(If s is not transverse with 0_E) Note that Z is compact. We may take a small neighborhood $Z \subseteq N$ such that s has no zero on ∂N . Take a generic perturbation $\nu \in \Gamma(U, E)$ that $\text{supp } \nu \subseteq N$ such that $s + \nu \pitchfork 0_E$. Then, after the perturbation, the preimage

$$Z_\nu := (s + \nu)^{-1}(0)$$

is a smooth manifold with dimension $n - r$, and $Z_\nu \subseteq N$. Since Z_ν for different generic perturbations are cobordant, they define the same cohomology class. Thus, although the space Z is bad, we still get a homology class

$$[Z]^{\text{vir}} = [Z_\nu] \in H_{n-r}(N) \cong H_{n-r}(Z)$$

Here, the isomorphism $H_{n-r}(N) \cong H_{n-r}(Z)$ is because we can choose N such that there is a deformation retract from N to Z . By the same method, the transverse case, we have $i_*[Z]^{\text{vir}} = e(E) \cap [U]$.

In general, deformations are not always finite dimensional. If we want the trick above to work, the deformation operator needs to be at least Fredholm:

Definition 6.1 (Fredholm Operator). A bounded linear operator $D : X \rightarrow Y$ between Banach spaces is Fredholm if $\dim \ker(D)$ and $\dim \text{coker}(D)$ are both finite, and $D(X) \subseteq Y$ is closed. The Fredholm index of D is defined by

$$\text{ind}(D) := \dim \ker(D) - \dim \text{coker}(D)$$

6.2 J -Holomorphic Equation as a Section

In the previous example, we saw that when the moduli space is not a "good" space, we can take a generic approximation using the virtual fundamental class rather than the fundamental class, and this will be consistent with the virtual dimension. The first question is:

What is the equation that determines a stable map in the moduli space $\overline{\mathcal{M}}_{g,n}(X, \beta)$?

The answer is clear. Let (X, J) be (almost) complex manifold, $(C; p_1, \dots, p_n)$ be a stable curve with complex structure j , and $D := p_1 + \dots + p_n$. Let $f : C \rightarrow X$ be a holomorphic map such that $f_*[C] = \beta$ that satisfies the J -holomorphic equation

$$\bar{\partial}_J f = \frac{1}{2}(\text{d}f + J \circ \text{d}f \circ j) = 0 \in \Omega^{0,1}(C, f^*TX)$$

where the pullback tangent bundle is just

$$f^*TX := \{(z, v) \mid z \in C, V \in T_{f(z)}X\}$$

and the f^*TX -value $(0, 1)$ -form is given by

$$\Omega^{0,1}(C, f^*TX) := \{\eta \in \Omega^1(C, f^*TX) \mid \eta \circ j = -J \circ \eta\}$$

we need to check the well-definedness:

Proposition 6.2. The form $\Omega^{0,1}(C, f^*TX)$ we defined above is indeed in $(0, 1)$ -type.

Proof. For any f^*TX valued form $A \in \Omega^1(C, f^*TX)$, the $(0, 1)$ projection is defined for any $c \in TC$ by

$$A^{0,1}(jv) = \frac{1}{2}(A(jv) + JA(j^2v)) = \frac{1}{2}(A(jv) - JA(v))$$

and also,

$$-JA^{0,1}(v) = -\frac{1}{2}(JA(v) + J^2A(jv)) = \frac{1}{2}(A(jv) - JA(v))$$

Thus, the $(0, 1)$ part of A satisfies

$$A^{0,1}(jv) + JA^{0,1}(v) = 0$$

which complete the check of well-definedness. $\square$

Thus, $\forall df \in \Omega^1(C, f^*TX)$, the $(0, 1)$ part is exactly

$$(df)^{0,1} = \bar{\partial}_J f = \frac{1}{2}(df + J \circ df \circ j)$$

and that is the operator of J -holomorphic equation.

The J -holomorphic equation is a nonlinear first-order elliptic PDE. In a more abstract way, we may take the Banach manifold $\mathcal{B} := W^{k,p}(C, X)$ with $k \geq 1$ and $p > 2^5$. We claim that, similar to the example in the last subsection, the J -holomorphic equation is an equation that determines the zeros of a section of the Banach vector bundle $\mathcal{E} \rightarrow \mathcal{B}$, where $\forall f \in \mathcal{B}$, the fiber is defined by

$$\mathcal{E}_f := W^{k-1,p}(\Omega^{0,1}(C, f^*TX))$$

then, the differential operator $\bar{\partial}_J : \mathcal{B} \rightarrow \mathcal{E}$ is a smooth section of the bundle. The J -holomorphic map is just the set of zeros:

$$\mathcal{M}_J(X, \beta; J) = \bar{\partial}_J^{-1}(0_{\mathcal{E}})$$

All structures above are parallel with our finite-dimensional example, so the natural following step is to linearize the section.

Suppose $f : C \rightarrow X$ is J -holomorphic, we may take $\xi \in \Gamma(C, f^*TX)$ as the infinitesimal deformation. Fix a connection ∇^6 , let

$$f_t(z) = \exp_{f(z)}(t\xi(x))$$

When the perturbation $\xi \in \Gamma(C, f^*TX)$ is sufficiently small, the inverse of exponential map

$$\text{Exp}_f(\xi)(z) := \exp_{f(z)}(\xi(z))$$

gives the local coordinate (V_f, ϕ_f) that for neighborhood $V_f \subseteq \mathcal{B}$ that contains f , $\phi_f := \text{Exp}_f^{-1} : V_f \rightarrow W^{k,p}(C, f^*TX)$. This allows us to compute the tangent space of $\mathcal{B}$, given by the following proposition:

Proposition 6.3 (Tangent Space of $\mathcal{B}$). There is an isomorphism $T_f\mathcal{B} = W^{k,p}(C, f^*TX)$.

Proof. Note the the f_t we defined above is a path in $\mathcal{B}$ such that $f_0 = f$. Fix $z \in C$, $t \mapsto f_t(z)$ is a curve on X . Thus, for fixed $z \in C$, since $f_t = \text{Exp}_f(t\xi)$ is given by the local chart on $\mathcal{B}$, and by Sobolev embedding, we may consider the following derivative

$$\xi(z) := \left. \frac{d}{dt} \right|_{t=0} f_t(z) \in T_{f(z)}X$$

⁵This condition ensures that we have the Sobolev embedding $W^{k,p}(C) \hookrightarrow C^0(C)$, which ensure the map to be at least continuous.

⁶When X is a smooth projective variety, we can choose the Chern connection ∇^{Ch} on the holomorphic Hermitian vector bundle $(T^{1,0}X, h)$ equipped with the Hermite metric h , we may assume the connection is torsion free.

and thus $z \mapsto \xi(z)$ is a section in $\Gamma(C, f^*TX)$, where, in our frame work, the allowed section is just $W^{k,p}(C, f^*TX)$, thus, $T_f\mathcal{B} \subseteq W^{k,p}(C, f^*TX)$.

In another side, given any $\xi \in W^{k,p}(C, f^*TX)$, we can construct the map

$$f_t(z) := \exp_{f(z)}(t\xi(z))$$

and the velocity of this curve at $t = 0$ is just $\xi(z)$. Thus, $W^{k,p}(C, f^*TX) \subseteq T_f\mathcal{B}$, which completes the proof of the proposition. $\square$

In the following proposition, we may define the linearization of the J -holomorphic curve equation:

$$D_f : T_f\mathcal{B} \rightarrow \mathcal{E}_f$$

We will show that it is elliptic and then compute the Fredholm index as part of the deformation.

Proposition 6.4 (Linearization). For any J -holomorphic map $f : C \rightarrow X$, define the linearization operator $D_f : W^{k,p}(C, f^*TX) \rightarrow W^{k-1,p}(\Omega^{0,1}(C, f^*TX))$ such that

$$D_f\xi := \frac{d}{d\tau} \Big|_{\tau=0} \bar{\partial}_J(f_\tau)$$

With the choice of connection, we have an explicit formula that

$$D_f\xi = \frac{1}{2}(\nabla\xi + J\nabla\xi \circ j) + \frac{1}{2}(\nabla_\xi J) \circ df \circ j$$

Proof. Given $\xi \in T_f\mathcal{B} = W^{k,p}(C, f^*TX)$ and the variation path f_τ generated by the exponential map. We may treat the variation as the map

$$F : (-\epsilon, \epsilon) \times C \rightarrow X, \quad (\tau, z) \mapsto f_\tau(z)$$

where $F(0, z) = f(z)$ and $\Xi(\tau, z) := \partial_\tau F(\tau, z) \in T_{f_\tau(z)}X$. We know that $\Xi(0, z) = \xi(z)$. We may pullback ∇ to F^*TX , then we may take covariant derivative ∇_{∂_τ} to f^*TX valued form. The linearization is given by

$$D_f\xi = \nabla_{\partial_\tau}|_{\tau=0} \bar{\partial}_J f_\tau = \frac{1}{2} \nabla_{\partial_\tau}|_{\tau=0} (df + J \circ df \circ j)$$

We compute this covariant derivative term by term:

1. For $\nabla_{\partial_\tau}|_{\tau=0} df$, take $v \in \Gamma(TC)$, note that v can be also viewed as the vector field on $(-\epsilon, \epsilon) \times C$ without τ -component and thus

$$df_\tau(v) = dF(v)$$

Since we can always choose ∇ to be torsion-free, we may claim that since v has no τ component, $[\partial_\tau, v] = 0$, and thus,

$$\nabla_{\partial_\tau} dF(v) - \nabla_v dF(\partial_\tau) = [\partial_\tau, v] = 0$$

Thus, we have $\nabla_{\partial_\tau} df_\tau(v) = \nabla_v \Xi$, which can be written as follows

$$\nabla_{\partial_\tau}|_{\tau=0} df = \nabla_v \xi$$

when $\tau = 0$. Thus, the first term is just $\nabla_{\partial_\tau}|_{\tau=0} df = \nabla_v \xi$.

2. To compute the second term, we also take $v \in \Gamma(TC)$ and consider $\nabla_{\partial_\tau}|_{\tau=0} (Jdf(jv))$. By the Leibniz rule

$$\begin{aligned} \nabla_{\partial_\tau}|_{\tau=0} (Jdf(jv)) &= (\nabla_{\partial_\tau} J)|_{\tau=0} df(jv) + J(\nabla_{\partial_\tau} df(jv))|_{\tau=0} \\ &= (\nabla_\xi J) df(jv) + J \nabla_{jv} \xi \end{aligned}$$

where the second step use the fact that ∇ is torsion free, $\nabla_{\partial_\tau}|_{\tau=0} df_\tau(v) = \nabla_v \xi$.

Thus, the linearization of infinitesimal deformation is given by

$$D_f \xi = \frac{1}{2}(\nabla \xi + J \nabla \xi \circ j) + \frac{1}{2}(\nabla_\xi J) \circ df \circ j = (\nabla \xi)^{0,1} + \frac{1}{2}(\nabla_\xi J) \circ df \circ j$$

which completes the proof of the proposition. $\square$

We may also write this in local coordinates. Take holomorphic coordinate $z = s + it$ on C such that

$$j\partial_s = \partial_t, \quad j\partial_t = -\partial_s$$

We use the following notation $f_s := df(\partial_s)$, $f_t := df(\partial_t)$ as the derivative of $f : C \rightarrow X$, and deformation

$$\nabla_s \xi := \nabla_{\partial_s} \xi, \quad \nabla_t \xi := \nabla_{\partial_t} \xi$$

The J -holomorphic equation is just

$$\bar{\partial}_J f = f_t + J_{f(z)} f_s = 0$$

and the linearization

$$D_f \xi(\partial_s) = \frac{1}{2}(\nabla_s \xi + J \nabla_t \xi + (\nabla_\xi J) J f_s) = 0$$

gives the linearized Cauchy-Riemann equation.

Remark. *This linearized equation is indeed a Cauchy-Riemann type equation since the principal part of the equation is given by*

$$(\nabla \xi)^{0,1} = \frac{1}{2}(\nabla \xi + J \circ \nabla \xi \circ j)$$

which is exactly the same as the principal symbol of the Dolbeault operator. More precisely, $\forall \zeta \in T_z^ C \setminus \{0\}$, the principal symbol of the operator is given by*

$$\sigma_\zeta(D_f)(\xi) = \frac{1}{2}(\zeta \otimes \xi + J \circ (\zeta \otimes \xi) \circ j)$$

Moreover, the result does not depend on connection, since for any $\nabla = d + A$, the second term A only provides a lower order term, which does not change the principal symbol.

A famous result is that the Cauchy-Riemann equation is first-order elliptic. Thus, we know that the linearization operator $D_f : W^{k,p}(C, f^*TX) \rightarrow W^{k-1,p}(\Omega^{0,1}(C, f^*TX))$ is Fredholm, and $\ker D_f$ is the infinitesimal deformation direction and $\operatorname{coker} D_f$ is the obstruction. Also,

1. $\dim_{\mathbb{C}} \ker D_f$ and $\dim_{\mathbb{C}} \operatorname{coker} D_f$ is finite.
2. $\operatorname{ind}_{\mathbb{C}} D_f := \dim_{\mathbb{C}} \ker D_f - \dim_{\mathbb{C}} \operatorname{coker} D_f$ can be compute by Riemann-Roch theorem.

We also need to make a remark on the deformation domain, since the example in the last subsection is the deformation with a fixed domain. However, in the case of stable maps, we also have the domain deformation, and the operator linearization itself is not the whole story; we need the deformation complex:

$$[\mathcal{T}_C(-D) \rightarrow f^*TX]$$

We may use the following example to see what the deformation is on the domain, and why we need it:

Example. *Take the target space $X = \{\text{pt}\}$ to be a point. With the fixed domain $(C, j; S)$, the map $f : C \rightarrow \{\text{pt}\}$ does not contain any other information, and the moduli space should be just $\{f\}$. However, we know that*

$$\overline{\mathcal{M}}_{g,n}(\{\text{pt}\}, 0) = \overline{\mathcal{M}}_{g,n}$$

Is clearly not a point if $(g, n) \neq (0, 3)$.

Thus, we should treat the deformation of the domain and the operator separately.

6.3 Deformation and Virtual Dimension

In this subsection, we may complete the calculation of the virtual dimension and define the GW invariant.

Deformation of J -holomorphic equation. We may first consider the deformation with a fixed domain, which completely depends on the Fredholm index of D_f . To compute the Fredholm index of D_f , we may consider first the case that the target space (X, J) is a complex manifold, i.e., J integrable. Then the pullback bundle $E := f^*TX$ is a rank m ($\dim X = m$) holomorphic vector bundle, we may study the pull back bundle locally by holomorphic coordinate $w^1, \dots, w^m$ on X . Let $f = (f^1, \dots, f^m)$, it is holomorphic equivalent to

$$\frac{\partial f^\alpha}{\partial \bar{z}} = 0, \quad \alpha = 1, \dots, m$$

For any section $\xi \in \Gamma(C, f^*TX)$ with local coordinate representation

$$\xi = \sum_{\alpha=1}^m \xi^\alpha(z, \bar{z}) \frac{\partial}{\partial w^\alpha} \Big|_{f(z)}$$

and the induced Dolbeault operator is given by

$$\bar{\partial}_E \xi = \sum_{\alpha=1}^m \frac{\partial \xi^\alpha}{\partial \bar{z}} d\bar{z} \otimes \frac{\partial}{\partial w^\alpha} \Big|_{f(z)}$$

On the other hand, if we take a family f_t of holomorphic maps such that $f_0 = f$, then, let

$$\xi := \frac{d}{dt} \Big|_{t=0} f_t$$

Since for any $t \in [0, 1]$, f_t holomorphic

$$\frac{\partial \xi^\alpha}{\partial \bar{z}} = \frac{\partial}{\partial \bar{z}} \left(\frac{d}{dt} \Big|_{t=0} f_t^\alpha \right) = \frac{d}{dt} \Big|_{t=0} \frac{\partial f_t^\alpha}{\partial \bar{z}} = 0$$

Thus, $\bar{\partial}_E \xi = 0$, which means that

Proposition 6.5. If (X, J) is a complex manifold, the linearization operator D_f is just the Dolbeault operator on the pullback bundle $E = f^*TX$, namely, $\bar{\partial}_E : \Omega^0(C, E) \rightarrow \Omega^{0,1}(C, E)$.

And the direct corollary is that

$$\ker D_f = \ker \bar{\partial}_E = H^0(C, E),$$

We also need to know the cokernel of the operator; the key is that C has complex dimension 1, the E -valued Dolbeault complex is given by

$$0 \longrightarrow \Omega^{0,0}(C, E) \xrightarrow{\bar{\partial}_E} \Omega^{0,1}(C, E) \longrightarrow 0$$

The $(0, 1)$ -Dolbeault cohomology is just

$$H_{\bar{\partial}}^{0,1}(C, E) = \frac{\Omega^{0,1}(C, E)}{\text{im}(\bar{\partial}_E)} = \text{coker}(\bar{\partial}_E)$$

By Dolbeault theorem, $\text{coker}(\bar{\partial}_E) = H_{\bar{\partial}}^{0,1}(C, E) \cong H^1(C, E)$ ⁷, and thus, the Fredholm index of $\bar{\partial}_E$ is given by

$$\text{ind}_{\mathbb{C}}(D_f) = \text{ind}_{\mathbb{C}}(\bar{\partial}_E) = h^0(C, E) - h^1(C, E) = \chi(C, E)$$

Applies the Riemann-Roch theorem on the holomorphic vector bundles E ,

$$\chi(E) = (1 - g) \cdot \text{rank}(E) + \deg(E)$$

In our case, $\text{rank}_{\mathbb{C}}(f^*TX) = \dim_{\mathbb{C}} X = m$ and $\deg(f^*TX) = \langle c_1(TX), f_*[C] \rangle$, and thus we have the following proposition:

⁷For simplicity, we may use the notation that $H^1(C, E) := H^1(C, \mathcal{O}(E))$.

Proposition 6.6. For stable holomorphic map $f \in \overline{\mathcal{M}}_{g,n}(X, \beta; J)$ given by $f : (C, j) \rightarrow (X, J)$ with target space X of (complex) dimension m . Let $E := f^*TX$ be the pullback bundle on C , the index of linearized deformation operator is

$$\text{ind}_{\mathbb{C}}(D_f) = m(1 - g) + \int_{\beta} c_1(TX)$$

The real index is just

$$\text{ind}_{\mathbb{R}}(D_f) = 2 \text{ind}_{\mathbb{C}}(D_f) = 2m(1 - g) + 2 \int_{\beta} c_1(TX)$$

This is the contribution of the deformation of the J -holomorphic equation to the virtual dimension. However, it remains to analyze the domain deformation⁸ to obtain the virtual dimension.

Parametrization of domain deformation. In the main text, we may only provide the main theorem describe the deformation, while a more detailed illustration is in Appendix D, which can also be applied for nodal curves. Fix a smooth n -marked curve $(C; p_1, \dots, p_n)$ with genus g , let

$$D := p_1 + \dots + p_n$$

be a divisor on C . To understand the domain deformation, we need to consider three levels of the problem: deformation of complex structure, motion of marked points, and the infinitesimal automorphism. The moduli (deformation) of a smooth marked domain are given by

$$\mathcal{M}_{g,n} = \frac{\mathcal{J}^+(C) \times \text{Conf}_n(C)}{\text{Diff}^+(C)} \cong \mathcal{T}_{g,n}/\Gamma_{g,n}$$

The following theorem gives a complete description to the first-order domain deformation:

Theorem 6.7. For smooth marked curve $(C; p_1, \dots, p_n)$ and the given divisor $D = p_1 + \dots + p_n$, the infinitesimal automorphism is parametrized by $H^0(C, \mathcal{T}_C(-D))$, and the first-order deformation is parametrized by $H^1(C, \mathcal{T}_C(-D))$. Here, $\mathcal{T}_C(-D)$ is the section sheaf of holomorphic tangent bundle $\mathcal{T}_C := T^{1,0}C$ twisted by $-D$.

Thus, the contribution of the domain deformation is given by the Euler characteristics $\chi(C, \mathcal{T}_C(-D)) = h^0(C, \mathcal{T}_C(-D)) - h^1(C, \mathcal{T}_C(-D))$.

Virtual dimension. Once we have Theorem 6.7, we may compute the domain deformation contribution of the virtual dimension. It is not hard to compute the degree of the twisted line bundle $\mathcal{T}_C(-D)$ since for $\mathcal{T}_C(-D) \otimes \mathcal{O}_C(-D)$

$$\deg \mathcal{T}_C(-D) = \deg \mathcal{T}_C - \deg D = 2 - 2g - n$$

and the contribution of domain deformation can be computed by the Riemann-Roch theorem:

$$\chi(C, \mathcal{T}_C(-D)) = h^0(C, \mathcal{T}_C(-D)) - h^1(C, \mathcal{T}_C(-D)) = 3 - 3g - n$$

Recall that we are considering the case that the domain (C, p) is smooth and stable, $2 - 2g - n < 0$, and thus $\deg \mathcal{T}_C(-D) < 0$. This shows that $\mathcal{T}_C(-D)$ cannot have any holomorphic section, i.e., $H^0(C, \mathcal{T}_C(-D)) = 0$ and

$$h^1(C, \mathcal{T}_C(-D)) = 3g - 3 + n$$

Naively, we will expect the virtual dimension to be

$$\text{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{g,n}(X, \beta) = \text{ind}_{\mathbb{C}}(D_f) - \chi(C, \mathcal{T}_C(-D)) = \int_{\beta} c_1(TX) + (1 - g)(\dim_{\mathbb{C}} X - 3) + n$$

A more general discussion of the virtual dimension is in Appendix D.

⁸In smooth, stable case, the domain deformation is just $T_{(C,p)}\mathcal{M}_{g,n}$, and its contribution is expected to be $\dim_{\mathbb{C}} \mathcal{M}_{g,n} = 3g - 3 + n$.

6.4 Virtual Fundamental Class and GW Invariant

After we obtain the virtual dimension of moduli space, the natural question to ask is if there is a virtual homology class that can be used in the integration, which has degree exactly 2 times the virtual dimension. For simplicity, we still only consider a smooth domain. Note that in the previous section, we have identified the J -holomorphic operator $\bar{\partial}_J$ as a Fredholm section of the Banach vector bundle

$$s : \mathcal{B} = W^{k,p}(C, X) \rightarrow \mathcal{E}$$

with fiber $\mathcal{E}_f := W^{k-1,p}\Omega^{0,1}(C, f^*TX)$. The moduli space of the stable map is given by

$$\mathcal{M}_J(X, \beta; J) = s^{-1}(0_{\mathcal{E}})$$

where $0_{\mathcal{E}}$ is the zero section. More precisely, the section is defined by $s(u) := \bar{\partial}_J u$ and the J -holomorphic equation is just $s(u) = 0$. Near a solution u of the equation, we have the linearization

$$D_u : W^{k,p}(C, f^*TX) \rightarrow W^{k-1,p}\Omega^{0,1}(C, f^*TX)$$

Note that D_u is a Fredholm operator. Similar to the finite-dimensional case, we have the following definition:

Definition 6.8. $s \pitchfork 0_{\mathcal{E}}$ if and only if the linearization $D_f s$ is surjective.

In the case that $s \pitchfork 0_{\mathcal{E}}$, by infinite-dimensional implicit function theorem

$$\mathcal{M} = s^{-1}(0_{\mathcal{E}})$$

is a locally smooth manifold (or, at least orbifold), and

$$\dim \overline{\mathcal{M}} = \text{ind}(D_f s) = \text{vdim } \overline{\mathcal{M}}$$

In this case, we may simply take the fundamental class $[\overline{\mathcal{M}}] = [\overline{\mathcal{M}}]^{\text{vir}}$ and have no need for another virtual fundamental class.

In a general setting, things get more difficult. The global construction is hard in general, and requires significant technical arguments using Kuranishi structures. Thus, we may only consider the construction of a virtual class here locally near a stable map u_0 .

We choose the stable map

$$u_0 : (C, j_0; p_1, \dots, p_n) \rightarrow X$$

such that $u_0[C] = \beta$. If we allow the domain deformation, the unknown set will be $(j; \tilde{p}_1, \dots, \tilde{p}_n; u)$. To deal with this problem analytically, we need a local slice of deformation instead of the entire total space of deformation. In Theorem 6.7, we have already seen that the first-order deformation is parametrized by $H^1(C, \mathcal{T}_C(-D))$. Thus, locally near the moduli point $(C, p_i) \in \overline{\mathcal{M}}_{g,n}$, we may find a finite-dimensional parametrized pointed germ $(V, 0)$ with tangent space

$$T_0 V \cong H^1(C, \mathcal{T}_C(-D))$$

Also, as we mentioned before, we may take a family of smooth marked curves that $\pi : \mathcal{C} \rightarrow V$ with sections $\sigma_i : V \rightarrow \mathcal{C}$, $i = 1, \dots, n$ such that the central fiber is given by

$$(\mathcal{C}_0; \sigma_1(0), \dots, \sigma_n(0)) \cong (C; p_1, \dots, p_n)$$

and for each $a \in V$, the fiber is given by $(C_a, j_a; p_i(a))$ where a is the domain deformation parameter. Thus, locally, the unknown information can be written as a pair

$$(a, \xi) \in V \times W^{k,p}(C, u_0^*TX)$$

where $a \in V$ is finite-dimensional parameter and ξ is the map variation. With chosen $a \in V$ and the fixed stable map u_0 , we may defined the map $u_{a,\xi} : C_a \rightarrow X$

$$u_{a,\xi}(z) := \exp_{u_0(z)}(\xi(z)), \quad \xi \in W^{k,p}(C, u_0^*TX)$$

We may also define the local Banach section

$$\mathcal{F}(a, \xi) := \bar{\partial}_J(u_{a,\xi}; j_a)$$

which takes value in a Banach bundle with fiber

$$E_{a,\xi} \cong W^{k-1,p}(\Omega^{0,1}(C_a, u_{a,\xi}^*TX))$$

The zeros of the section $\mathcal{F}^{-1}(0)$ give a local moduli of stable maps near u_0 . We may consider the linearization of the section with the family of stable maps $u_{ta,t\xi}$ and the family of complex structures j_{ta} with $u_{0,0} = u_0$ and $j_0 = j$, and the linearization of the Banach setion $\mathcal{F}$ can be written as the following proposition:

Proposition 6.9. The linearization of $\mathcal{F}$ near $(0,0)$ gives a map

$$D\mathcal{F}_{(0,0)} : T_0V \oplus W^{k,p}(C, u_0^*TX) \rightarrow W^{k-1,p}(\Omega^{0,1}(C, u_0^*TX))$$

which is explicitly given by

$$D\mathcal{F}_{(0,0)}(a, \xi) = D_{u_0}\xi + \frac{1}{2}J \circ du_0 \circ \delta j(a)$$

where D_{u_0} is the linearization of $\bar{\partial}_J$ and $\delta j := dj_{ta}/dt|_{t=0} \in \Omega^{0,1}(TC)$ infinitesimal complex structure deformation corresponding to the parameter $a \in T_0V$.

Proof. We may directly compute the differential of $\mathcal{F}$ at $(0,0)$, by definition

$$\mathcal{F}(a, \xi) = \frac{1}{2} \left(du_{a,\xi} + J \circ du_{a,\xi} \circ j_a \right)$$

and thus, for any $(a, \xi) \in T_0V \oplus W^{k,p}(C, u_0^*TX)$, the linearization is given by

$$\begin{aligned} D\mathcal{F}_{(0,0)}(a, \xi) &= \frac{d}{dt} \Big|_{t=0} \mathcal{F}(ta, t\xi) = \frac{1}{2} \frac{d}{dt} \Big|_{t=0} \left(du_{ta,t\xi} + J \circ du_{ta,t\xi} \circ j_a \right) \\ &= \frac{1}{2} \frac{d}{dt} \Big|_{t=0} \left(du_{ta,t\xi} + J \circ du_{ta,t\xi} \circ j \right) + \frac{1}{2} \frac{d}{dt} \Big|_{t=0} \left(du_{ta,t\xi} + J \circ du_{ta,t\xi} \circ (j_a - j) \right) \\ &= D_{u_0}\xi + \frac{1}{2} \frac{d}{dt} \Big|_{t=0} \left(du_{ta,t\xi} + J \circ du_{ta,t\xi} \circ (j_a - j) \right) \end{aligned}$$

By the fact that when t is near 0, $j_{ta} - j = t\delta j(a) + o(t)$ and $du_{ta,t\xi} = du_0 + o(1)$. Thus, the second term in the equation above can be rewritten as

$$\frac{1}{2} \frac{d}{dt} \Big|_{t=0} \left(du_{ta,t\xi} + J \circ du_{ta,t\xi} \circ (j_a - j) \right) = \frac{1}{2} J \circ du_0 \circ \delta j(a)$$

This completes the proof of the proposition. $\square$

Note that in the previous section, we have shown that the variation $\delta j(a) \in \Omega^{0,1}(TC)$ is actually a sheaf cohomology class $[\{D_{\alpha\beta}j\}] \in H^1(C, \mathcal{T}_C(-D))$.

Also, since the map

$$a \mapsto \frac{1}{2} J \circ du_0 \circ \delta j(a)$$

is defined on a finite-dimensional domain T_0V , which implies

Proposition 6.10. The linearization $D\mathcal{F}_{(0,0)}$ is Fredholm.

To discuss the general case without transversality, we may define

$$K := \ker(D\mathcal{F}_{(0,0)}), \quad O := \operatorname{coker}(D\mathcal{F}_{(0,0)})$$

by the fact that $D\mathcal{F}_{(0,0)}$ is Fredholm, both K and O are finite-dimensional. Take the closed complementary subspace

$$X = K \oplus X', \quad Y = \operatorname{im}(D\mathcal{F}_{(0,0)}) \oplus O$$

and the restriction $L := D\mathcal{F}_{(0,0)}|_{X'} : X' \rightarrow \operatorname{im}(D\mathcal{F}_{(0,0)})$ is an isomorphism of Banach spaces. Let the unknown quantity be $x+y \in X$ such that $x \in K$ and $y \in X'$, the direct sum decomposition induce the natural projection:

$$\mathcal{F}(x+y) = \Pi_{\operatorname{im}}\mathcal{F}(x+y) + \Pi_O\mathcal{F}(x+y)$$

To solve the equation $\mathcal{F}(x+y) = 0$, we may first solve the image part:

$$\Pi_{\operatorname{im}}\mathcal{F}(x+y) = 0$$

since the linearization respect to y is just $L = D\mathcal{F}_{(0,0)}|_{X'}$ is isomorphism, by implicit function theorem, one can find a smooth function $\Phi : U_K \subseteq K \rightarrow X'$ such that $\Phi(0) = 0$ and

$$\Pi_{\operatorname{im}}\mathcal{F}(x + \Phi(x)) = 0$$

Then we deal with the obstruction part:

$$\mathfrak{o}(x) := \Pi_O\mathcal{F}(x + \Phi(x)) \in O$$

The space of solution is given by $\mathcal{F}^{-1}(0) \cong \mathfrak{o}^{-1}(0)$, and thus, we may only study the finite dimensional smooth map $\mathfrak{o} : K \rightarrow O$, and the expected dimension we have compute before is just $\operatorname{ind}(D\mathcal{F}_{(0,0)}) = \dim K - \dim O = \operatorname{vdim}$. The key on the construction of local virtual cycle is to treat the map $\mathfrak{o} : K \rightarrow O$ as a section of the trivial vector bundle $K \times O \rightarrow K$. This make the problem back the the Example 6.1, which, if $\mathfrak{o}$ does not transverse to $0 \in O$, we may take a a generic perturbation $\nu : K \rightarrow O$ such that $\mathfrak{o} + \nu \pitchfork 0$, the existence is given by the transversality theorem in finite dimensions. The perturbed zero set

$$Z_\nu := (\mathfrak{o} + \nu)^{-1}(0) \subseteq K$$

is a smooth submanifold of dimension $\dim K - \dim O$. Thus, we may define the virtual cycle locally as

$$[\mathcal{M}_{\operatorname{loc}}]^{\operatorname{vir}} := [Z_\nu]$$

We will give only a brief introduction to the existence of global virtual fundamental class, more detail can be seen in the expository article of Kuranishi struction in [Fuk+12].

Around each stable map u , by a finite-dimensional reduction, one can find a Kuranishi chart $(U_u, E_u, s_u, \Gamma_u, \psi_u)$ such that U_u is a finite dimensional manifold, $E_u \rightarrow U_u$ is a finite-rank obstruction bundle, $s_u : U_u \rightarrow E_u$ is a smooth Γ_u -invariant section, $\Gamma_u = \operatorname{Aut}(u)$ is a finite group, and

$$\psi_u : s_u^{-1}(0)/\Gamma_u \rightarrow \overline{\mathcal{M}}_{g,n}(X, \beta)$$

gives a local model of the moduli space. For each stable map u , finite-dimensional reduction yields a local chart for the moduli space. If these charts admit coordinate changes satisfying the cocycle condition, then they determine a Kuranishi structure (or, more generally, a virtual orbifold atlas). On such a structure, one can choose compatible transverse multisections $\mathfrak{s}_u$. Their zero sets are compatible on overlaps and glue together to form a compact, oriented, weighted branched orbifold of pure dimension vdim . Its rational fundamental class is independent of the chosen multisection and depends only on the underlying virtual atlas. We therefore define

$$[\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\operatorname{vir}} \in H_{2-\operatorname{vdim}_c}(\overline{\mathcal{M}}_{g,n}(X, \beta); \mathbb{Q}).$$

As shown above, there is a virtual fundamental class that corresponds to it and can be used for intersection theory and curve counting. At the end of this section, following the idea we have introduced at the beginning, we may define the Gromov-Witten invariant:

Definition 6.11 (Gromov-Witten Invariant). Given $\gamma_1, \dots, \gamma_n \in H^*(X, \mathbb{Q})$ and the virtual fundamental class $[\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\text{vir}} \in H_{2 \text{vdim}_{\mathbb{C}}}(\overline{\mathcal{M}}_{g,n}(X, \beta))$, the Gromov-Witten invariant is given by

$$\langle \gamma_1, \dots, \gamma_n \rangle_{g, \beta}^X := \int_{[\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\text{vir}}} \bigwedge_{i=1}^n \text{ev}_i^* \gamma_i$$

where for any stable map f , the evaluation map is defined by $\text{ev}_i : f \mapsto f(p_i)$.

It is a clear fact that the virtual fundamental class $[\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\text{vir}}$ is in pure dimension equal to the virtual dimension, and the Gromov-Witten Invariant is zero unless

$$\sum_{i=1}^n \deg \gamma_i = 2 \text{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{g,n}(X, \beta)$$

Note that we have forgetful morphism $\pi : \overline{\mathcal{M}}_{g,n}(X, \beta) \rightarrow \overline{\mathcal{M}}_{g,n}$, we can also defined the GW class as follows:

We define the Gromov-Witten invariant formally as follows:

$$\langle \gamma_1, \dots, \gamma_n \rangle_{g, \beta}^X := \pi_* \text{ev}^*(\gamma_1 \times \dots \times \gamma_n) \frown [\overline{\mathcal{M}}_{g,n}]$$

which make sense as $\overline{\mathcal{M}}_{g,n}$ is a smooth orbifold.

With the morphism above, we may also defined the GW class:

$$\begin{array}{ccc} \overline{\mathcal{M}}_{g,n}(X, \beta) & \xrightarrow{(\text{ev}_1, \dots, \text{ev}_n)} & X^n \\ \pi \downarrow & & \\ \overline{\mathcal{M}}_{g,n} & & \end{array}$$

Definition 6.12 (Gromov-Witten Class). The Gromov-Witten class with specified fundamental class $\beta \in H_2(X)$ is defined to be

$$\Omega_{g, \beta}^X(\gamma_1, \dots, \gamma_n) := \text{PD} \left[\pi_* \left(\bigwedge_{i=1}^n \text{ev}_i^*(\gamma_i) \frown [\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\text{vir}} \right) \right] \in H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q})$$

7 Gromov-Witten Theory as a CohFT

In this section, we will show that the GW class defined in the Definition 6.12 above give rise to a $\mathbb{Z}_2$ -graded CohFT over the graded $\mathbb{Q}$ -vector space $V := H^*(X, \mathbb{Q})$:

Theorem 7.1 (Gromov-Witten CohFT). Take $V := H^*(X, \mathbb{Q})$ with the pairing

$$\eta(a, b) := \int_X a \wedge b$$

and the unit element $\mathbf{1} := 1_X \in H^0(X, \mathbb{Q})$. The Gromov-Witten CohFT is given by the family of tensors $\Omega_{g,n}^X \in H^*(\overline{\mathcal{M}}_{g,n}, \mathbb{Q}) \otimes (V^*)^{\otimes n}$ given by $\forall \gamma_1, \dots, \gamma_n \in H^*(X, \mathbb{Q})$

$$\Omega_{g,n}^X(\gamma_1, \dots, \gamma_n) := \sum_{\beta \in H_2(X)} q^\beta \Omega_{g, \beta}^X(\gamma_1, \dots, \gamma_n)$$

In the definition above, the Gromov-Witten CohFT tensor is also called Novikov series, where the formal parameter q is called the Novikov variable, which is used to "track" the curve class $\beta \in H_2(X, \mathbb{Q})$ satisfies

$$q^{\beta_1} q^{\beta_2} = q^{\beta_1 + \beta_2} \quad \forall \beta_1, \beta_2 \in H_2(X, \mathbb{Q})$$

This may still looks a little odd, as we didn't really choose another group that consists q to make this notation really a group homomorphism. To make this precise, we can consider $\beta_1, \dots, \beta_n \in H_2(X)$ be a basis, and for any $\beta := \sum_i c_i \beta_i$ for $c_1, \dots, c_n \in \mathbb{Q}$, then we take the formal variables $\{q_i\}_{i=1}^n$ (that commutes to each other) and we defined the Novikov variable as

$$q^\beta := q_1^{c_1} q_2^{c_2} \cdots q_n^{c_n}$$

As q^β also multiply in similar way (i.e., $q^{\beta_1+\beta_2} = q^{\beta_1} q^{\beta_2}$). The benefits of writing q^β instead of $\prod_i q_i^{c_i}$ is that q^β is more well-defined, in other words, it does not depend on the choice of basis. Let $\Gamma := H_2(X, \mathbb{Z})/\text{tors}$, and the effective curve class $\Gamma_{\text{eff}} \subseteq \Gamma$ is a semigroup such that

1. $0 \in \Gamma_{\text{eff}}$
2. $\forall \beta_1, \beta_2 \in \Gamma_{\text{eff}} : \beta_1 + \beta_2 \in \Gamma_{\text{eff}}$

We define the Novikov ring as follows

Definition 7.2 (Novikov ring). The Novikoc ring on X is obtained by take the completion on the group algebra of effective curve classes

$$\Lambda_X := \widehat{\mathbb{Q}[\Gamma_{\text{eff}}]}$$

The completion is given by the following construction of a filtration: Taking $A_0 := \mathbb{Q}[\Gamma_{\text{eff}}]$ and fix a ample class $\omega \in H^2(X, \mathbb{R})$, let the valuation be

$$\nu_\omega \left(\sum_\beta a_\beta q^\beta \right) := \min \{ \langle \omega, \beta \rangle \mid a_\beta \neq 0 \}$$

and $F^\lambda A_0 := \{f \in A_0 \mid \nu_\omega(f) \geq \lambda\}$ gives the filtration.

Explicitly, we can treat the Novikov ring as the arbitrary formal series of elements in Γ_{eff} over $\mathbb{Q}$.

In the following passages, we shall first check that the definition above satisfies the $\mathbb{Z}_2$ -graded version of all three axioms (symmetry, gluing, and unit) in Section 5.

7.1 GW Theory as a CohFT

A1. The construction above obviously satisfies the $\mathbb{Z}_2$ -graded symmetry axiom since we have the graded commutativity of the cup product (or wedge) and the multilinearity of $\Omega_{g,n}$.

A2. We may check that the Gromov-Witten class satisfies the gluing (or splitting) axiom. The key idea is that when the curve as a domain splits to two (or more) components on the boundary piece in $\partial \overline{\mathcal{M}}_{g,n}$, the stable map also splits to two parts:

1. A component in $\overline{\mathcal{M}}_{g_1, n_1+1}(X, \beta_1)$.
2. Another component in $\overline{\mathcal{M}}_{g_2, n_2+1}(X, \beta_2)$

and $\beta = \beta_1 + \beta_2 \in H_2(X, \mathbb{Q})$. More importantly, the relation of these two components to the stable curve stratum is not simply the product (which forget the structure as a map) but the fiber product over X given by the following commutative diagram

$$\begin{array}{ccc} \mathcal{F}_{\beta_1, \beta_2} & \xrightarrow{\text{proj}_1} & \overline{\mathcal{M}}_{g_1, n_1+1}(X, \beta_1) \\ \text{proj}_2 \downarrow & & \downarrow \text{ev}_\bullet \\ \overline{\mathcal{M}}_{g_2, n_2+1}(X, \beta_2) & \xrightarrow{\text{ev}_\star} & X \end{array}$$

where $\text{ev}_\bullet$ and $\text{ev}_\star$ are the evaluation maps on the marked points that are used on gluing, and

$$\overline{\mathcal{M}}_{g_1, n_1+1} \times_X \overline{\mathcal{M}}_{g_2, n_2+1} := \mathcal{F}_{\beta_1, \beta_2} = \{(f_1, f_2) \mid \text{ev}_\bullet(f_1) = \text{ev}_\star(f_2)\}$$

More precisely, $\mathcal{F}_{\beta_1, \beta_2}$ is the gluing parameter space for stable map whose domains lie over boundary stratum determined by p . On cohomology level, the class on $\mathcal{F}_{\beta_1, \beta_2}$ is obtained by the pullback of the diagonal class $[\Delta_X] \in H^*(X \times X)$ from $X \times X$ (The Poincaré dual of Δ_X)⁹. Using the diagonal decomposition, with the basis $\{e_\mu\}$

$$[\Delta_X] = \sum_{\mu, \nu} \eta^{\mu\nu} e_\mu \boxtimes e_\nu$$

where $e_\mu \boxtimes e_\nu := (\text{proj}_1^X)^* e_\mu \wedge (\text{proj}_2^X)^* e_\nu$ is the external product, and the pullback of diagonal class along the following diagram, denote $\mathcal{K}_1 := \overline{\mathcal{M}}_{g_1, n_1+1}(X, \beta_1)$ and $\mathcal{K}_2 := \overline{\mathcal{M}}_{g_2, n_2+1}(X, \beta_2)$

$$\begin{array}{ccc} \mathcal{F}_{\beta_1, \beta_2} := \mathcal{K}_1 \times_X \mathcal{K}_2 & \hookrightarrow & \mathcal{K}_1 \times \mathcal{K}_2 \\ \pi_1 \times \pi_2 \downarrow & & \downarrow \text{ev} \\ X & \xrightarrow{\Delta} & X \times X \end{array}$$

where $\text{ev} := \text{ev}_\bullet \times \text{ev}_\star$ and $\mathcal{F}_{\beta_1, \beta_2} = (\text{ev}_\bullet \times \text{ev}_\star)^{-1}(\Delta_X) \subseteq \mathcal{K}_1 \times \mathcal{K}_2$, the pullback class is explicitly given by

$$\text{ev}^*[\Delta_X] = \sum_{\mu, \nu} \eta^{\mu\nu} (\text{proj}_1^* \text{ev}_\bullet^* e_\mu) \wedge (\text{proj}_1^* \text{ev}_\star^* e_\nu) = \sum_{\mu, \nu} \eta^{\mu\nu} \text{ev}_\bullet^* e_\mu \boxtimes \text{ev}_\star^* e_\nu$$

We may now introduce the intersection form, let the index set $I \sqcup J = \{1, \dots, n\}$ with $|I| = n_1$ and $|J| = n_2$, we define

$$A_1 := \bigwedge_{i \in I} \text{ev}_i^* \gamma_i \in H^*(\mathcal{K}_1, \mathbb{Q}), \quad A_2 := \bigwedge_{j \in J} \text{ev}_j^* \gamma_j \in H^*(\mathcal{K}_2, \mathbb{Q})$$

and the pullback class induced by the projection $\text{proj}_i : \mathcal{K}_1 \times \mathcal{K}_2 \rightarrow \mathcal{K}_i$, with $i = 1, 2$, is given by $\text{proj}_1^* A_1 \wedge \text{proj}_2^* A_2$. We still need the condition of fiber product, which is the reason why we introduce the diagonal class $\text{ev}^*[\Delta_X]$ and the Poincaré pairing of virtual class $[\mathcal{F}_{\beta_1, \beta_2}]^{\text{vir}} = [\mathcal{K}_1 \times_X \mathcal{K}_2]^{\text{vir}}$ satisfies

$$\int_{[\mathcal{F}_{\beta_1, \beta_2}]^{\text{vir}}} (-) = \int_{[\mathcal{K}_1]^{\text{vir}} \times [\mathcal{K}_2]^{\text{vir}}} (-) \wedge \text{ev}^*[\Delta_X]$$

So we may compute the invariant by computing the cup product between the intersection form and $\text{ev}^*[\Delta_X]$:

$$p_1^* A_1 \wedge p_2^* A_2 \wedge \text{ev}^*[\Delta_X] = \sum_{\mu, \nu} \eta^{\mu\nu} \text{proj}_1^* (A_1 \wedge \text{ev}_\bullet^* e_\mu) \wedge \text{proj}_2^* (A_2 \wedge \text{ev}_\star^* e_\nu)$$

We consider the pairing between the class above and the virtual fundamental class, in heuristic calculation, we consider

$$\begin{aligned} (p_1^* A_1 \wedge p_2^* A_2) \frown [\mathcal{F}_{\beta_1, \beta_2}]^{\text{vir}} &= (p_1^* A_1 \wedge p_2^* A_2 \wedge \text{ev}^*[\Delta_X]) \frown ([\mathcal{K}_1]^{\text{vir}} \times [\mathcal{K}_2]^{\text{vir}}) \\ &= \sum_{\mu, \nu} \eta^{\mu\nu} \left(\text{proj}_1^* (A_1 \wedge \text{ev}_\bullet^* e_\mu) \wedge \text{proj}_2^* (A_2 \wedge \text{ev}_\star^* e_\nu) \right) \frown ([\mathcal{K}_1]^{\text{vir}} \times [\mathcal{K}_2]^{\text{vir}}) \\ &= \sum_{\mu, \nu} \eta^{\mu\nu} \left(\text{proj}_1^* (A_1 \wedge \text{ev}_\bullet^* e_\mu) \frown [\mathcal{K}_1]^{\text{vir}} \right) \boxtimes \left(\text{proj}_2^* (A_2 \wedge \text{ev}_\star^* e_\nu) \frown [\mathcal{K}_2]^{\text{vir}} \right) \end{aligned}$$

If we pushforward this class using the forgetful map $\pi_i : \mathcal{K}_i \rightarrow \overline{\mathcal{M}}_{g_i, n_i+1}$, note that the definition of the GW class is defined to be

$$\Omega_{g_1, n_1+1, \beta_1}^X(\gamma_I, e_\mu) = (\pi_1)_* \left((A_1 \wedge \text{ev}_\bullet^*) \frown [\mathcal{K}_1]^{\text{vir}} \right)$$

⁹Rigorously, this is not true, a more accurate description is the Gysin pullback of the diagonal class.

and the pushforward class corresponding to the pairing above is just

$$(\pi_1 \times \pi_2)_* \left((p_1^* A_1 \wedge p_2^* A_2) \frown [\mathcal{F}_{\beta_1, \beta_2}]^{\text{vir}} \right) = \sum_{\mu, \nu} \eta^{\mu\nu} \Omega_{g_1, n_1+1, \beta_1}^X(\gamma_I, e_\mu) \boxtimes \Omega_{g_2, n_2+1, \beta_2}^X(\gamma_J, e_\nu)$$

By Künneth isomorphism $H^*(\overline{\mathcal{M}}_{g_1, n_1+1} \times \overline{\mathcal{M}}_{g_2, n_2+1}) \cong H^*(\overline{\mathcal{M}}_{g_1, n_1+1}) \otimes H^*(\overline{\mathcal{M}}_{g_2, n_2+1})$, after pushforward by forgetful maps and considering all possible cases of β_1 and β_2 (parametrized as $\mathcal{F}_{\beta_1, \beta_2}$), the equation above implies

$$\begin{aligned} r^* \Omega_{g, n, \beta}^X(\gamma_1, \dots, \gamma_2) &= \sum_{\beta_1 + \beta_2 = \beta} (\pi_1 \times \pi_2)_* \left((p_1^* A_1 \wedge p_2^* A_2) \frown [\mathcal{F}_{\beta_1, \beta_2}]^{\text{vir}} \right) \\ &= \sum_{\beta_1 + \beta_2 = \beta} \sum_{\mu, \nu} \eta^{\mu\nu} \Omega_{g_1, n_1+1, \beta_1}^X(\gamma_{i_1}, \dots, \gamma_{i_{n_1}}, e_\mu) \otimes \Omega_{g_2, n_2+1, \beta_2}^X(\gamma_{j_{n_1+1}}, \dots, \gamma_{j_{n_1+n_2}}, e_\nu) \end{aligned}$$

The nonseparated gluing case $r : \overline{\mathcal{M}}_{g, n+2} \rightarrow \overline{\mathcal{M}}_{g+1, n}$ is the same.

A3. Since in the definition of GW CohFT, we have defined the unit $\mathbf{1} = 1_X \in H^0(X, \mathbb{Q})$ to be the identity under cup product, we know that $\text{ev}_{n+1}^* 1_X = 1 \in H^*(\overline{\mathcal{M}}_{g, n+1}(X, \beta), \mathbb{Q})$ and

$$\bigwedge_{i=1}^{n+1} \text{ev}_i^* \gamma_i = \left(\bigwedge_{i=1}^n \text{ev}_i^* \gamma_i \right) \wedge \text{ev}_{n+1}^* 1_X = \bigwedge_{i=1}^n \text{ev}_i^* \gamma_i$$

In this case, we need to consider two forgetful maps on marked points $\tilde{p} : \overline{\mathcal{M}}_{g, n+1}(X, \beta) \rightarrow \overline{\mathcal{M}}_{g, n}(X, \beta)$ and $p : \overline{\mathcal{M}}_{g, n+1} \rightarrow \overline{\mathcal{M}}_{g, n}$, which is given by the fiber product $\mathcal{F} := \overline{\mathcal{M}}_{g, n}(X, \beta) \times_{\overline{\mathcal{M}}_{g, n}} \overline{\mathcal{M}}_{g, n+1} \subseteq \overline{\mathcal{M}}_{g, n+1}(X, \beta)$ such that the universal property gives the natural isomorphism $\varphi : \overline{\mathcal{M}}_{g, n+1}(X, \beta) \rightarrow \mathcal{F}$ such that

$$\text{proj}_1 \circ \varphi = \tilde{p} \quad \text{proj}_2 \circ \varphi = \pi_{n+1}$$

and the following diagram commutes

$$\begin{array}{ccccc} \overline{\mathcal{M}}_{g, n+1}(X, \beta) & \xrightarrow{\tilde{p}} & & & \overline{\mathcal{M}}_{g, n}(X, \beta) \\ & \searrow \exists! \varphi & & & \downarrow \pi_n \\ & & \overline{\mathcal{M}}_{g, n}(X, \beta) \times_{\overline{\mathcal{M}}_{g, n}} \overline{\mathcal{M}}_{g, n+1} & \xrightarrow{\text{proj}_1} & \overline{\mathcal{M}}_{g, n}(X, \beta) \\ & & \downarrow \text{proj}_2 & & \downarrow \pi_n \\ & & \overline{\mathcal{M}}_{g, n+1} & \xrightarrow{p} & \overline{\mathcal{M}}_{g, n} \\ & \searrow \pi_{n+1} & & & \\ & & & & \end{array}$$

We defined the evaluation map of $\overline{\mathcal{M}}_{g, n}(X, \beta)$ to be $\text{ev} := (\text{ev}_1, \dots, \text{ev}_n)$ and the evaluation map on $\overline{\mathcal{M}}_{g, n+1}$ be $\tilde{\text{ev}} := (\tilde{\text{ev}}_1, \dots, \tilde{\text{ev}}_{n+1})$. In the stable case $2 - 2g + n > 0$, since under the forgetful map, we only ignore the last marked point, for $1 \leq i \leq n$ the evaluation map satisfies

$$\tilde{\text{ev}}_i = \text{ev}_i \circ \tilde{p} = \text{ev}_i \circ \text{proj}_1 \circ \varphi$$

and the corresponding pullback is just $\tilde{\text{ev}}_i^* \gamma_i = \varphi^* \text{proj}_1^* \text{ev}_i^* \gamma_i$, and thus, the pullback of evaluation map on $\overline{\mathcal{M}}_{g, n+1}(X, \beta)$ is given by

$$\bigwedge_{i=1}^{n+1} \tilde{\text{ev}}_i^* v_i = \bigwedge_{i=1}^n \tilde{\text{ev}}_i^* v_i = \varphi^* \text{proj}_1^* \left(\bigwedge_{i=1}^{n+1} \text{ev}_i^* v_i \right)$$

where $v_i = \gamma_i$ for $1 \leq i \leq n$ and $v_{n+1} = 1_X$. We also need the compatibility condition of the virtual class and forgetful morphisms:

Lemma. *The natural morphism $\varphi : \overline{\mathcal{M}}_{g, n+1}(X, \beta) \rightarrow \mathcal{F}$ satisfies*

$$\varphi_* [\overline{\mathcal{M}}_{g, n+1}(X, \beta)] = \text{proj}_1^! [\overline{\mathcal{M}}_{g, n}(X, \beta)]$$

where $\text{proj}_1^!$ is the refined Gysin pullback corresponding to proj_1 . Since proj_1 is the pullback of the universal curve, the refined Gysin pullback is actually an ordinary flat pullback.

The proof is highly technical and will not be included in this article. To check the axiom, we first consider the stable case that $2g - 2 + n > 0$, then

$$\begin{aligned}\Omega_{g,n+1}^X(\gamma_1, \dots, \gamma_n, \mathbf{1}) &:= (\pi_{n+1})_* \left(\bigwedge_{i=1}^n \tilde{\text{ev}}_i^* \gamma_i \frown [\overline{\mathcal{M}}_{g,n+1}(X, \beta)]^{\text{vir}} \right) = (\text{proj}_2)_* \varphi_* \left(\bigwedge_{i=1}^n \tilde{\text{ev}}_i^* \gamma_i \frown [\overline{\mathcal{M}}_{g,n+1}(X, \beta)]^{\text{vir}} \right) \\ &= (\text{proj}_2)_* \varphi_* \left(\varphi^* \text{proj}_1^* \left(\bigwedge_{i=1}^{n+1} \text{ev}_i^* v_i \right) \frown [\overline{\mathcal{M}}_{g,n+1}(X, \beta)]^{\text{vir}} \right) = (\text{proj}_2)_* \left(\text{proj}_1^* \left(\bigwedge_{i=1}^{n+1} \text{ev}_i^* v_i \right) \frown \varphi_* [\overline{\mathcal{M}}_{g,n+1}(X, \beta)]^{\text{vir}} \right)\end{aligned}$$

and we apply the lemma

$$\begin{aligned}\Omega_{g,n+1}^X(\gamma_1, \dots, \gamma_n, \mathbf{1}) &= (\text{proj}_2)_* \left(\text{proj}_1^* \left(\bigwedge_{i=1}^{n+1} \text{ev}_i^* v_i \right) \frown \text{proj}_1^! [\overline{\mathcal{M}}_{g,n+1}(X, \beta)]^{\text{vir}} \right) \\ &= (\text{proj}_2)_* \text{proj}_1^* \left(\bigwedge_{i=1}^{n+1} \text{ev}_i^* v_i \frown [\overline{\mathcal{M}}_{g,n+1}(X, \beta)]^{\text{vir}} \right) = p^* (\pi_n)_* \left(\bigwedge_{i=1}^{n+1} \text{ev}_i^* v_i \frown [\overline{\mathcal{M}}_{g,n+1}(X, \beta)]^{\text{vir}} \right) \\ &= p^* \Omega_{g,n}^X(\gamma_1, \dots, \gamma_n)\end{aligned}$$

This shows that the GW class satisfies the unit axiom in the stable range. For the special case of $\Omega_{0,3,\beta}^X$, since $\overline{\mathcal{M}}_{0,3} = \{\text{pt}\}$, the output of the CohFT tensor is a number in $\mathbb{Q}$. Given curve class β , since $\text{ev}_3^* \mathbf{1} = 1$

$$\Omega_{0,3,\beta}^X(u, v, \mathbf{1}) = \int_{[\overline{\mathcal{M}}_{0,3}(X, \beta)]^{\text{vir}}} \text{ev}_1^* u \wedge \text{ev}_2^* v$$

For $\beta = 0$, every stable maps are constant and $\overline{\mathcal{M}}_{0,3}(X, 0) \cong X$. In this case, the virtual fundamental class is given by $[\overline{\mathcal{M}}_{0,3}(X, 0)]^{\text{vir}} = [X]$

$$\Omega_{0,3,\beta=0}^X(u, v, \mathbf{1}) = \int_{[X]} u \wedge v = \eta(u, v)$$

For the case that the fundamental curve class $\beta \neq 0$, we may check the forgetful map $\tilde{p} : \overline{\mathcal{M}}_{0,3}(X, \beta) \rightarrow \overline{\mathcal{M}}_{0,2}(X, \beta)$ and $\tilde{\text{ev}}_1, \tilde{\text{ev}}_2$ be the evaluation on $\overline{\mathcal{M}}_{0,2}(X, \beta)$. Then

$$\Omega_{0,3,\beta}^X(u, v, \mathbf{1}) = \int_{[\overline{\mathcal{M}}_{0,3}(X, \beta)]^{\text{vir}}} \tilde{p}^* (\text{ev}_1^* u \wedge \text{ev}_2^* v) = \int_{[\overline{\mathcal{M}}_{0,2}(X, \beta)]^{\text{vir}}} \text{ev}_1^* u \wedge \text{ev}_2^* v \wedge \tilde{p}_*(\mathbf{1})$$

Note that the Gysin pushforward (or integration along fiber) $\tilde{p}_* : H^k(\overline{\mathcal{M}}_{0,3}(X, \beta)) \rightarrow H^{k-2}(\overline{\mathcal{M}}_{0,2}(X, \beta))$ will send $\mathbf{1}$ to degree -2 , which means

$$\tilde{p}_*(\mathbf{1}) = 0 \in H^{-2}(\overline{\mathcal{M}}_{0,2}(X, \beta)) = 0$$

and thus, for $\beta \neq 0$, the GW class $\Omega_{0,3,\beta}^X(u, v, \mathbf{1}) = 0$. This completes the proof of the unit axiom.

The only remaining problem is the compatibility of the virtual fundamental class with boundary gluing and forgetful morphisms. This follows from the functoriality and pullback properties of the virtual fundamental class in [Beh97; BF97b; BM96b].

7.2 Gromov-Witten Potential and Quantum Cohomology

A natural way to think about the GW class is to study the generating function of $\langle \gamma_1, \dots, \gamma_n \rangle_{g,n,\beta}^X$. We need to first fix the following notations: Let X be a complex projective variety and $\gamma_0, \dots, \gamma_m \in H^{\text{even}}(X)$ be a basis of $H^*(X)$ such that

1. $\gamma_0 = 1 = \text{PD}[X] \in H^0(X)$
2. $\gamma_1, \dots, \gamma_r \in H^2(X)$ is a basis of $H^2(X)$

and since we treat GW theory as a ChoFT, we may consider $V := H^*(X, \mathbb{Q})$ as the domain of the CohFT with Poincaré pairing $\eta : V \otimes V \rightarrow \mathbb{Q}$.

Remark. For simplicity, we only consider the even degree cohomology $H^{\text{even}}(X)$, which allows us to ignore the $\mathbb{Z}_2$ -grading. And for any permutation $\sigma \in S_n$

$$\langle \gamma_1, \dots, \gamma_m \rangle_{g, \beta}^X = \langle \gamma_{\sigma(1)}, \dots, \gamma_{\sigma(m)} \rangle_{g, \beta}^X$$

Let $y_0, \dots, y_m$ be the formal variables corresponding to $\gamma_0, \dots, \gamma_m$ and define the following function: We denote the formal sum

$$t := \sum_{i=0}^m y_i \gamma_i \in V \otimes \mathbb{Q}[[y_0, \dots, y_m]]$$

and $\langle t^n \rangle_{g, \beta}^X = \langle t, \dots, t \rangle_{g, \beta}^X$.

Definition 7.3 (Gromov-Witten Potential). The genus g Gromov-Witten potential on X is defined by

$$\Phi_g^X(y_0, \dots, y_m, q) := \sum_{n \geq 0} \sum_{\beta \in H_2(X)} \frac{1}{n!} \langle t^n \rangle_{g, \beta}^X \in \Lambda_X[[y_0, \dots, y_m]]$$

where Λ_X is the Novikov ring.

Explicitly, one can write the GW potential as

$$\Phi_g^X(y_0, \dots, y_m, q) = \sum_{k_0, \dots, k_m \geq 0} \sum_{\beta \in H_2(X)} \langle \gamma_0^{k_0}, \dots, \gamma_m^{k_m} \rangle_{g, \beta}^X \prod_{i=1}^m \frac{y_i^{k_i}}{k_i!} q^\beta$$

where the notation $\gamma_i^{k_i}$ in the GW invariant means repeated entries instead of multiplicative exponents.

Proposition 7.4. For any $i_1, \dots, i_n \in \{0, \dots, m\}$, there is

$$\frac{\partial^r \Phi_g^X}{\partial y_{i_1} \cdots \partial y_{i_r}} = \sum_{n \geq 0} \sum_{\beta \in H_2(X)} \frac{1}{n!} \langle \gamma_{i_1}, \dots, \gamma_{i_r}, t^n \rangle_{g, n}^X q^\beta$$

When n is such that there is no stable map, the GW invariants are taken to be zero in the summation.

Proof. Start with the explicit formula

$$\Phi_g^X(y_0, \dots, y_m, q) = \sum_{k_0, \dots, k_m \geq 0} \sum_{\beta \in H_2(X)} \langle \gamma_0^{k_0}, \dots, \gamma_m^{k_m} \rangle_{g, \beta}^X \prod_{i=1}^m \frac{y_i^{k_i}}{k_i!} q^\beta$$

Note that the derivative

$$\frac{\partial}{\partial y_i} \left(\frac{y_i^{k+i}}{k_i!} \right) = \begin{cases} \frac{y_i^{k_i-1}}{(k_i-1)!}, & k_i \geq 1 \\ 0, & k_i = 0 \end{cases}$$

Thus, we redefine the index by let $l_i := k_i - 1$, and let $l_j = k_j$ for all $j \neq i$. This leads to

$$\begin{aligned} \frac{\partial \Phi_g^X}{\partial y_i} &= \sum_{\beta} \sum_{k_0, \dots, k_m \geq 0} \langle \gamma_0^{k_0}, \dots, \gamma_i^{k_i}, \dots, \gamma_m^{k_m} \rangle_{g, \beta}^X \frac{y_0^{k_0}}{k_0!} \cdots \frac{y_i^{k_i-1}}{(k_i-1)!} \cdots \frac{y_m^{k_m}}{k_m!} q^\beta \\ &= \sum_{\beta} \sum_{l_0, \dots, l_m \geq 0} \langle \gamma_i, \gamma_0^{l_0}, \dots, \gamma_m^{l_m} \rangle_{g, \beta}^X \prod_{j=0}^m \frac{y_j^{l_j}}{l_j!} q^\beta \end{aligned}$$

And since we only take the even cohomology, we don't need to consider any Koszul sign. Thus,

$$\frac{\partial \Phi_g^X}{\partial y_i} = \sum_{\beta} \sum_{n \geq 0} \frac{1}{n!} \langle \gamma_i, t^n \rangle_{g,\beta}^X q^\beta$$

By repeating this calculation for each variable, we can prove the proposition. $\square$

In particular, we may consider the topological (degree zero) part of the GW CohFT, we study the contribution of $\beta = 0$:

Proposition 7.5. For genus-zero GW invariants with the curve class $\beta = 0$, we have

$$\langle \gamma_1, \dots, \gamma_n \rangle_{0,\beta=0}^X = \begin{cases} \int_X \gamma_1 \wedge \gamma_2 \wedge \gamma_3, & n = 3 \\ 0, & \text{otherwise} \end{cases}$$

Proof. For $n \geq 3$ the stable map must be constant, thus

$$\overline{\mathcal{M}}_{0,n}(X, 0) \cong X \times \overline{\mathcal{M}}_{0,n}$$

The isomorphism sends a stable map to the image point $x \in X$ and its domain $(C; p_1, \dots, p_n)$.

Under this identification, with an abuse of notation, each evaluation map is given by the projection

$$\text{ev}_i = \text{proj}_X : X \times \overline{\mathcal{M}}_{0,n} \rightarrow X$$

and the invariant can be written as

$$\begin{aligned} \langle \gamma_1, \dots, \gamma_n \rangle_{0,\beta=0}^X &= \int_{X \times \overline{\mathcal{M}}_{0,n}} \bigwedge_{i=1}^n \text{proj}_X^* (\gamma_i) \\ &= \left(\int_X \gamma_1 \wedge \dots \wedge \gamma_n \right) \left(\int_{\overline{\mathcal{M}}_{0,n}} 1 \right) \end{aligned}$$

Where the second equal is due to Künnth formula. If $n = 0$, here

$$\langle \gamma_1, \gamma_2, \gamma_3 \rangle_{0,\beta=0}^X = \left(\int_X \gamma_1 \wedge \gamma_2 \wedge \gamma_3 \right) \left(\int_{\text{pt}} 1 \right) = \int_X \gamma_1 \wedge \gamma_2 \wedge \gamma_3$$

otherwise the pairing between $\overline{\mathcal{M}}_{0,n}$ and 1 id zero. This completes the proof. $\square$

Proposition 7.5 above also shows that the genus-zero GW priotentian is given by the following expression

$$\Phi_0^X(y_0, \dots, y_m, q) = \underbrace{\frac{1}{6} \int_X t^3}_{\text{classical part}} + \overbrace{\sum_{\beta \neq 0} \sum_{n \geq 0} \frac{1}{n!} \langle t^n \rangle_{0,\beta}^X q^\beta}^{\text{quantum part}}$$

where $t = y_0 \gamma_0 + \dots + y_m \gamma_m$ and the GW potential is been separated to classical and quantum parts. We may defined the third derivative

$$\Phi_{ijk}(y, q) := \frac{\partial \Phi_0^X}{\partial y_i \partial y_j \partial y_k}$$

Proposition 7.6. The third derivative that defined above is given by

$$\Phi_{ijk}(y, q) := \sum_{\beta \in H_2(X)} \sum_{n \geq 0} \frac{1}{n!} \langle \gamma_1, \gamma_2, \gamma_3, t^n \rangle_{0,\beta}^X q^\beta$$

The proof is simply by calculation. Note that the expression above can be also written as the classical part plus the quantum part

$$\Phi_{ijk}(y, q) = \int_X \gamma_i \wedge \gamma_j \wedge \gamma_k + \sum_{\beta \neq 0} \sum_{n \geq 0} \frac{1}{n!} \langle \gamma_1, \gamma_2, \gamma_3, t^n \rangle_{0, \beta}^X q^\beta$$

In Section 5, we have shown that the CohFT leads to a quantum product over the chosen vector space V . In this section, we may study the particular case of a GW CohFT on target space X , where the vector space we choose as the domain is the cohomology ring $V = H^*(X, \mathbb{Q})$. Recall that the quantum product associated to a CohFT is induced by the genus zero part of the CohFT, which is completely determined by the 3-points function $\Omega_{0,3}^X$.

Recall from section 5 and the beginning of this section that in Gromov-Witten CohFT we denote the Poincaré pairing

$$\eta_{ij} := \int_X \gamma_i \wedge \gamma_j, \quad (\eta^{ij}) = (\eta_{ij})^{-1}$$

The following theorem gives the fundamental equation of the genus-zero CohFT, the Witten-Dijkgraaf-Verlinde-Verlinde (WDVV) equation:

Proposition 7.7 (Witten-Dijkgraaf-Verlinde-Verlinde Equation). For any $\gamma_i, \gamma_j, \gamma_k, \gamma_l \in V$, we have

$$\sum_{\mu, \nu} \Phi_{ij\mu} \eta^{\mu\nu} \Phi_{\nu kl} = \sum_{\mu, \nu} \Phi_{jk\mu} \eta^{\mu\nu} \Phi_{\nu il}$$

Remark. For the $\mathbb{Z}_2$ -graded case, the WDVV equation needs to be modified due to the Koszul sign:

$$\sum_{\mu, \nu} \Phi_{ij\mu} \eta^{\mu\nu} \Phi_{\nu kl} = (-1)^{\deg \gamma_i (\deg \gamma_j + \deg \gamma_k)} \sum_{\mu, \nu} \Phi_{jk\mu} \eta^{\mu\nu} \Phi_{\nu il}$$

It will happen when we consider the state space to be $V = H^*(X, \mathbb{Q})$ other than $H^{\text{even}}(X, \mathbb{Q})$.

The key of the proof is to use $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ and the splitting axiom on boundary stratum $\partial \overline{\mathcal{M}}_{0,4} \cong \{0, 1, \infty\}$. The basic idea follows the proof of associativity in Theorem 5.3, and we will not rewrite it here. With the equation above, we can define the quantum product from the GW potential

Definition 7.8 (Big Quantum Product). The big quantum product $*_t$ on V is defined by

$$\eta(\gamma_i *_t \gamma_j, \gamma_k) = \Phi_{ijk}(y, q)$$

Equivalently, the big quantum product can be defined in coordinates

$$\gamma_i *_t \gamma_j := \sum_{k,l} \Phi_{ijk} \eta^{kl} \gamma_l$$

The nondegeneracy of η ensures the uniqueness of this coordinate representation.

Theorem 7.9. The big quantum product $*_t$ gives rise to a commutative (resp. graded commutative), associative, and unital Frobenius algebra structure on $V \otimes \Lambda_X[[y]]$ over $\Lambda_X[[y]]$, where $y := (y_0, \dots, y_n)$.

This theorem directly follows from Theorem 5.3. The definition above is the most general definition of the quantum product. However, recall from Section 5, the system is determined completely by the three-point function, so usually, we can just consider the small quantum product as the special case of the big quantum product at $y = 0$.

Definition 7.10 (Quantum Product). The quantum product, or small quantum product, is given by the specialization of the big quantum product at $y = 0$, i.e., $\forall a, b \in H^{\text{even}}(X, \mathbb{Q})$

$$a * b := (a *_t b)|_{y=0}$$

In local coordinates, this is just

$$\gamma_i * \gamma_j = \sum_{\beta \in H_2(X)} \langle a, b, c \rangle_{0,\beta}^X q^\beta$$

and the quantum cohomology ring is often defined as follows

Definition 7.11 (Quantum Cohomology Ring). The quantum cohomology ring of the target space X is given by $QH^*(X) := (H^*(X) \otimes \Lambda_X, *)$.

The quantum products in the following paragraph are all small quantum products.

We often see the claim that the quantum product is the deformation of the cup product; the following proposition is just the formal statement of this fact.

Proposition 7.12. The classical part of the quantum product is just the cup product, more precisely

$$a * b \equiv a \wedge b \pmod{\{q^\beta \mid \beta \neq 0\}}$$

Proof. By the definition of small quantum product, if we take $\beta = 0$ with the Novikov coefficient q^0 ,

$$\eta((a * b)|_{\beta=0}, c) = \int_X a \wedge b \wedge c = \eta(a \wedge b, c)$$

by the nondegeneracy of the Poincaré pairing η , this implies $(a * b)|_{\beta=0} = a \wedge b$. This completes the proof. $\square$

7.3 Some Examples of Quantum Cohomology Ring

In this section, we shall compute two examples of the quantum cohomology ring. Before we start the examples, we may recall the following notation convention.

For a smooth projective manifold X and $a, b, c \in V := H^*(X)$,

$$\eta(a * b, c) = \sum_{\beta \in H_2(X)} \langle a, b, c \rangle_{0,\beta}^X q^\beta$$

where $\eta : V \otimes V \rightarrow \mathbb{Q}$ is the Poincaré pairing. With a basis $\gamma_0, \dots, \gamma_m \in H^*(X)$ and the dual basis $\tilde{\gamma}^0, \dots, \tilde{\gamma}^m$ such that

$$\eta(\tilde{\gamma}^i, \gamma_j) = \delta_{ij} \quad \forall i, j \in \{0, \dots, m\}$$

We have a useful formula for the quantum product that

$$a * b = \sum_{\beta \in H_2(X)} q^\beta \sum_{i=0}^m \langle a, b, \gamma_i \rangle_{0,\beta}^X \tilde{\gamma}^i$$

Also, the virtual dimension gives rise to a degree constraint that

$$\deg_{\mathbb{R}} a + \deg_{\mathbb{R}} b + \deg_{\mathbb{R}} c = \dim_{\mathbb{R}} X + 2 \int_{\beta} c_1(TX)$$

For genus-zero three-point functions. In the following parts of this subsection, we are going to compute the quantum cohomology of $\mathbb{P}^1$ and $\mathbb{P}^1 \times \mathbb{P}^1$.

Quantum Cohomology of $\mathbb{P}^1$. Note that the cohomology ring of $\mathbb{P}^1$ is given by taking $\gamma_0 = 1 \in H^0(\mathbb{P}^1, \mathbb{Q})$ and $\gamma_1 = x \in H^2(X, \mathbb{Q})$ where x is the point class, i.e.,

$$\int_{\mathbb{P}^1} x = 1$$

and the cohomology ring is given by $H^*(\mathbb{P}^1, \mathbb{Q}) \cong \mathbb{Q} \cdot 1 \oplus \mathbb{Q} \cdot x$. The Poincaré pairing is given by $\eta(1, x) = \eta(x, 1) = 1$, $\eta(1, 1) = \eta(x, x) = 0$, and the dual basis is just $1^\vee = x$, $x^\vee = 1$. The effective curve class in $\mathbb{P}^1$ is just $d[l]$, for $d \geq 0$, and $[l]$ is the class of first order maps and the first Chern class $c_1(T\mathbb{P}^1) = 2x$. The dimension constraint requires

$$\deg a + \deg b + \deg c = 2 + 4d$$

Thus, possible cases include $d = 0, 1$ since for $d \geq 2$, the expected dimension $2 + 4d \geq 10$, which is impossible for $\deg 1 = 0$ and $\deg x = 2$ in the three-point function. We may list all possible GW invariants

1. For $d = 0$, the only case is

$$\langle a, b, c \rangle_{0,0} = \int_{\mathbb{P}^1} a \wedge b \wedge c$$

The only possibility is $\langle 1, 1, x \rangle_{0,0}^X = 1$ and its permutation.

2. For $d = 1$, the real dimension is 6, so the only possibility is

$$\langle x, x, x \rangle_{0,[l]}^X = 1$$

since the degree 1 map $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ is nothing but an automorphism, and three distinct points determine a unique Möbius transformation.

To compute the quantum cohomology of $\mathbb{P}^1$, note that 1 has been chosen as the unit of quantum product. So we only need to know the quantum product on x . Let $q^d := q^{d[l]}$, apply the equation given at the beginning of this subsection,

$$x * x = \sum_{d \geq 0} q^d \left(\langle x, x, 1 \rangle_{0,d[l]}^X x + \langle x, x, x \rangle_{0,d[l]}^X 1 \right) = q$$

This shown that the quantum cohomology ring of $\mathbb{P}^1$ is $QH^*(\mathbb{P}^1) \cong \mathbb{Q}[x, q]/(x^2 - q)$.

Actually, we have the fact that for n -dimensional complex projective space $\mathbb{P}^n$, the quantum cohomology is given by $QH^*(\mathbb{P}^n) \cong \mathbb{Q}[x, q]/(x^{n+1} - q)$.

Quantum Cohomology of $\mathbb{P}^1 \times \mathbb{P}^1$. Let $X := \mathbb{P}^1 \times \mathbb{P}^2$ and the projection $\text{proj}_1, \text{proj}_2 : \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^1$. Let $a_i := \text{proj}_i^*(x)$ where $x \in H^2(\mathbb{P}^1)$ is the point class on $\mathbb{P}^1$, and both a_1 and a_2 are divisor class. Let $p := a_1 \wedge a_2 \in H^4(\mathbb{P}^1 \times \mathbb{P}^1)$ be the point class on $\mathbb{P}^1 \times \mathbb{P}^1$. Thus,

$$H^*(\mathbb{P}^1 \times \mathbb{P}^1, \mathbb{Q}) \cong \mathbb{Q} \cdot 1 \oplus \mathbb{Q} \cdot a_1 \oplus \mathbb{Q} \cdot a_2 \oplus \mathbb{Q} \cdot p$$

The Poincaré pairing $\eta(1, p) = \eta(a_2, a_1) = \eta(a_1, a_2) = \eta(p, 1) = 1$ and others are all zero. Let $F_1 = [\mathbb{P}^1 \times \{\text{pt}\}]$ and $F_2 = [\{\text{pt}\} \times \mathbb{P}^1]$, then any effective curve class can be written as

$$\beta = d_1 F_1 + d_2 F_2$$

with the corresponding Novikov variable $q^\beta = q_1^{d_1} q_2^{d_2}$. The intersection number satisfies

$$\int_{a_i} F_j = \delta_{ij}$$

and $c_1(TX) = 2a_1 + 2a_2$, the virtual dimension gives

$$\deg a + \deg b + \deg c = 4 + 4(d_1 + d_2)$$

To make the GW invariant exist, we need $d_1 + d_2 \leq 2$. Here are all the nonzero GW invariants:

1. $(d_1, d_2) = (0, 0)$. Then there are two possibilities

$$\langle 1, 1, p \rangle_{0,(0,0)}^X = \langle 1, a_1, a_2 \rangle_{0,(0,0)}^X = 1$$

and their permutation.

2. $(d_1, d_2) = (1, 0)$ and $(0, 1)$, These two cases are parallel. In this case, the possible case are $\langle a_1, a_1, p \rangle_{0,(1,0)}^X$ and $\langle a_2, a_2, p \rangle_{0,(0,1)}^X$. Furthermore, the curve class F_1 (resp. F_2) only relate with a_1 (resp. a_2), thus, possible GW invariants are

$$\langle a_1, a_1, p \rangle_{0,(1,0)}^X = \langle a_2, a_2, p \rangle_{0,(0,1)}^X = 1$$

3. $(d_1, d_2) = (1, 1)$. In this case, the expected dimension is $4 + 4 + 4 = 12$, which means we have only one possible choice that $\langle p, p, p \rangle_{0,(1,1)}^X = 1$.
4. The case that $(d_1, d_2) = (2, 0)$ or $(0, 2)$ does not contribute to the quantum product, since the stable map with image in $\mathbb{P}^1 \times \{x_0\}$ (resp. $\{\tilde{x}_0\} \times \mathbb{P}^1$) cannot pass three generic point, since their image under projection π_2 (resp. π_1) is constant.

Apply the formula of quantum product, we can compute the product over the basis

$$a_1 * a_1 = \sum_{d_1, d_2 \geq 0} q_1^{d_1} q_2^{d_2} \langle a_1, a_1, 1 \rangle_{0,(1,0)}^X p = q_1$$

and parallely, $B * B = q_2$. Also,

$$p * a_1 = \sum_{d_1, d_2 \geq 0} q_1^{d_1} q_2^{d_2} \langle a_1, a_1, p \rangle_{0,(1,0)}^X a_2 = q_1 a_2$$

and parallely, $p * a_2 = q_2 a_1$. Finally, we may consider $a_1 * a_2$ and $p * p$

$$a_1 * a_2 = \sum_{d_1, d_2 \geq 0} q_1^{d_1} q_2^{d_2} \langle a_1, a_2, 1 \rangle_{0,(0,0)}^X = p, \quad p * p = \sum_{d_1, d_2 \geq 0} q_1^{d_1} q_2^{d_2} \langle p, p, p \rangle_{0,(1,1)}^X 1 = q_1 q_2$$

In summary, $QH^*(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Q}[a_1, a_2, q_1, q_2]/(a_1^2 - q_1, a_2^2 - q_2)$.

Actually, from these two examples, we can find a general lower bound for the quantum cohomology of a product space

Proposition 7.13. Let X, Y be smooth projective manifold, then $QH^*(X \times Y) \cong QH^*(X) \otimes QH^*(Y)$.

This fact actually follows from the fact that

$$\langle a_1 \boxtimes a_2, b_1 \boxtimes b_2, c_1 \boxtimes c_2 \rangle_{0,(\beta_1, \beta_2)}^{X \times Y} = \langle a_1, b_1, c_1 \rangle_{0, \beta_1}^X \langle a_2, b_2, c_2 \rangle_{0, \beta_2}^Y$$

where $\boxtimes$ is the external product defined by cup product after pullback via projection. The proof of the proposition above simply follows from the definition, and we omitted the details of the proof here.

7.4 Frobenius Manifolds

In previous subsections, we showed that the genus-zero Gromov–Witten potential determines a family of products $*_t$ on $H^*(X, \mathbb{Q}) \otimes \Lambda_X[[y_0, \dots, y_m]]$, whose structure constants are the third derivatives of the genus-zero potential. The WDVV equation implies that this product is associative, while the fundamental class axiom gives the unit. This is precisely the starting point of the Frobenius manifold formalism: roughly speaking, a Frobenius manifold is a manifold whose tangent spaces are equipped with a smoothly varying Frobenius algebra structure, compatible with a flat metric and encoded by a potential function. Since the Gromov–Witten potential is a formal power series, the resulting structure is, strictly speaking, a formal Frobenius manifold germ.

Let M be a smooth manifold over a field $\mathbb{k}$ of characteristic 0, or more generally a formal manifold. Denote by TM its tangent bundle. We begin with the notion of a Frobenius algebra on each tangent space. Recall that a Frobenius algebra structure over the tangent space $T_p M$ consists of

1. A commutative, associative multiplication $*_p : T_p M \times T_p M \rightarrow T_p M$.
2. A distinguished unit element $e_p \in T_p M$.
3. A nondegenerate symmetric bilinear form $g_p : T_p M \otimes T_p M \rightarrow \mathbb{k}$ such that the following identity holds

$$g_p(X_p *_p Y_p, Z_p) = g_p(X_p, Y_p *_p Z_p), \quad \forall X_p, Y_p, Z_p \in T_p M$$

This is exactly the commutative Frobenius algebra (Definition 3.1) structure we have discussed in Section 3 and 5.

To globalize this notion, one asks for these structures to vary compatibly over the manifold.

Definition 7.14 (Frobenius Manifold). A Frobenius manifold consists the following data:

1. A symmetric nondegenerate flat metric $g \in \Gamma(\text{Sym}^2(TM))$.
2. A commutative, associative $\mathcal{O}_M$ -bilinear multiplication $*$: $TM \otimes TM \rightarrow TM$.
3. A global unit vector field $e \in \Gamma(TM)$.

such that the following conditions holds:

1. The unit vector field is flat under the Levi-Civita connection $\nabla e = 0$.
2. The symmetric 3-tensor defined by $A(X, Y, Z) := g(X * Y, Z)$ is corresponding to a local potential function Φ such that for flat coordinate system $t^0, \dots, t^m$

$$A_{ijk} = \partial_i \partial_j \partial_k \Phi$$

Such function is called a Frobenius potential.

Remark. We have to remark again about the $\mathbb{Z}_2$ -graded case, which we need to up grade the manifold M to a supermanifold. More detail can be find in the book [Man99].

The existence and uniqueness of the multiplication $*$ for given symmetric 3-tensor is follows from the nondegeneracy of g , which means there is canonical isomorphism

$$\flat_g : TM \xrightarrow{\sim} T^*M, \quad V \mapsto \flat_g(V) := g(V, -)$$

and it is clear that the multiplication is $\mathcal{O}_M$ -bilinear.

The nondegeneracy of g ensures that we can reconstruct the multiplication $*$ from the 3-tensor $A \in \Gamma(\text{Sym}^3(TM))$ with the chosen flat coordinate $t^0, \dots, t^m$:

Proposition 7.15. The product on Frobenius manifold is given by

$$\partial_i * \partial_j = \sum_k A_{ij}{}^k \partial_k$$

Furthermore, the 3-tensor must satisfies the WDVV equation

$$\sum_{\mu, \nu} A_{ij\mu} g^{\mu\nu} A_{\nu k}{}^l \partial_l = \sum_{\mu, \nu} A_{jk\mu} g^{\mu\nu} A_{\nu i}{}^l \partial_l$$

which is equivalent to the the associativity $(\partial_i * \partial_j) * \partial_k = \partial_i * (\partial_j * \partial_k)$.

Now we return to the GW theory, which we still take $V := H^{\text{even}}(X)$ for simplicty, with basis $\gamma_0 = \mathbf{1}, \gamma_1, \dots, \gamma_m$. We now show that this is precisely a Frobenius manifold structure, in the formal sense.

Theorem 7.16. Let X be a target space for which the genus-zero Gromov–Witten potential is defined. Then the formal spectrum

$$M_X := \mathrm{Spf}(\Lambda_X[[y_0, \dots, y_m]])$$

carries a canonical structure of a formal Frobenius manifold, where:

1. The metric is the constant Poincaré pairing

$$\eta(\partial_i, \partial_j) := \eta_{ij} = \int_X \gamma_i \wedge \gamma_j$$

2. The Frobenius potential is given by the GW potential Φ_0^X , and the multiplication is given by the big quantum product

$$\partial_i * \partial_j := \sum_{k,l} A_{ijk} \eta^{kl} \partial_l$$

3. The unit vector field is $e := \partial_0$ corresponding to the class $\mathbf{1} \in H^0(X)$

The proof of the theorem is simply check the requirement in the definition of Frobenius manifold. The previous theorem implies that every point t of the formal manifold determines a Frobenius algebra structure on the tangent space $T_t M_X$

Proposition 7.17. At the origin $t = 0$, the tangent Frobenius algebra of the Gromov–Witten Frobenius manifold is the small quantum cohomology ring $QH^*(X)$.

Since the genus-zero potential defines the multiplication and WDVV expresses its associativity, it is natural to repackage the entire structure into a zero-curvature condition. This is precisely Dubrovin’s viewpoint. Let $(M, \eta, *, e)$ be a Frobenius manifold, and let ∇ be the Levi–Civita connection of the flat metric η . For each vector field X , define an endomorphism

$$A_X \in \mathrm{End}(TM), \quad A_X(Y) = X * Y$$

Equivalently, the endomorphism can be treat as a TM -valued 1-form on M . The Dubrovin connection pencil is the 1-parameter family of connections

$$\nabla_X^{(\lambda)} Y := \nabla_X Y + X * Y \quad \lambda \in \mathbb{C}$$

i.e., a connection defined by $\nabla_X^{(\lambda)} := \nabla_X + \lambda A_X$. Under the flat coordinate $t^1, \dots, t^n$, if

$$\partial_i * \partial_j = A_{ij}^k \partial_k$$

Then for $A_i := A_{\partial_i}$ we have $(A_i)_j^k = A_{ij}^k$, and $\nabla_i^{(\lambda)} = \partial_i + \lambda A_i$. The importance of this construction is that the Frobenius manifold axioms may be reformulated as the flatness of the above connection for all values of the parameter λ .

Proposition 7.18. Let $(M, \eta, *, e)$ is a Frobenius manifold if and only if the Dubrovin connection pencil

$$\nabla^{(\lambda)} = \nabla + \lambda A$$

is flat for every $\lambda \in \mathbb{C}$.

Proof. ($\Rightarrow$) Since η is flat, we may work in flat coordinates $t^1, \dots, t^n$, so that $\nabla_i = \partial_i$. The curvature of $\nabla^{(\lambda)}$ is

$$R_{ij}^{(\lambda)} = \lambda(\partial_i A_j - \partial_j A_i) + \lambda^2[A_i, A_j]$$

We claim that both coefficients vanish. First, since the multiplication is encoded by a local potential Φ , one has

$$A_{ijk} = \eta(\partial_i * \partial_j, \partial_k) = \partial_i \partial_j \partial_k \Phi$$

Since η_{ij} is constant under flat coordinate, this implies

$$\partial_i A_{jl}^k = \partial_j A_{il}^k$$

Hence, $\partial_i A_j = \partial_j A_i$. Second, associativity of the multiplication is equivalent to $A_i A_j = A_j A_i$, that is $[A_i, A_j] = 0$. Therefore, $R_{ij=0}^{(\lambda)} \forall i, j, \lambda$, so $\nabla^{(\lambda)}$ is flat for any $\lambda \in \mathbb{C}$.

($\Leftarrow$) From the expression of the curvature, we know that

$$\partial_i A_j = \partial_j A_i \quad \text{and} \quad [A_i, A_j] = 0$$

The second equation is just the commutativity. We claim that the first equation is equivalent to the associativity of the product. For the first equation, write

$$\partial_i * \partial_j = A_{ij}^k \partial_k = \eta^{kl} A_{ijl} \partial_k$$

Because η is constant in flat coordinates and Frobenius compatibility holds, the tensor A_{ijk} is totally symmetric. The identity

$$\partial_i A_j = \partial_j A_i$$

is equivalent to $\partial_i A_{jkl} = \partial_j A_{ikl}$. By the Poincaré lemma, applied repeatedly, there exists locally a function Φ such that

$$A_{ijk} = \partial_i \partial_j \partial_k \Phi$$

Thus the multiplication is locally determined by a Frobenius potential, and the remaining Frobenius manifold axioms are already part of the assumptions. Hence M is a Frobenius manifold. $\square$

Therefore, once the flat metric, the flat unit, and the Frobenius symmetry with respect to the metric are fixed, Dubrovin's zero-curvature condition is equivalent to the potentiality and associativity of the multiplication. In this sense, the pencil $\nabla^{(\lambda)}$ provides an equivalent differential-geometric reformulation of the Frobenius manifold axioms.

A Automorphisms of Complete Hyperbolic Surface

In this appendix, we may prove the statement that the automorphism group of a complete hyperbolic surface is finite, which we used in the proof of Theorem 4.5.

Theorem A.1. Suppose Σ is a compact Riemann surface with $\chi(\Sigma) < 0$ or the genus $g \geq 2$. Then the automorphism $|\text{Aut}(\Sigma)| < \infty$.

Proof. Let Σ be a compact Riemann surface with $\chi(\Sigma) < 0$, equivalently of genus $g \geq 2$. By the Uniformization Theorem, Σ admits a unique complete hyperbolic metric of constant curvature -1 in its conformal class. Concretely, there exists a discrete, torsion-free, cocompact Fuchsian group $\Gamma \leq \text{PSL}_2(\mathbb{R})$ such that

$$\Sigma \cong \mathbb{H}/\Gamma$$

as a Riemann surface, where $\mathbb{H}$ is the upper half-plane with its standard hyperbolic metric

$$ds^2 = \frac{dx^2 + dy^2}{y^2}$$

In particular, the hyperbolic area of Σ is finite and given by Gauss–Bonnet:

$$\text{Area}(\Sigma) = \int_{\Sigma} dA = -2\pi \chi(\Sigma) = 2\pi(2g - 2) > 0$$

Let $f \in \text{Aut}(\Sigma)$ be a biholomorphism. We claim f is an isometry of the hyperbolic metric on Σ . Indeed, since f is conformal, the pullback metric f^*g (where g is the hyperbolic metric on Σ) is pointwise conformal to g . Moreover, curvature is invariant under pullback, hence f^*g also has constant curvature -1 . Therefore, f^*g is another hyperbolic metric in the same conformal class as g . By uniqueness of the hyperbolic metric in a given conformal class (a consequence of uniformization), we must have $f^*g = g$, i.e., f is an isometry. Thus, we have an injective homomorphism

$$\text{Aut}(\Sigma) \hookrightarrow \text{Isom}^+(\Sigma, g)$$

where Isom^+ denotes orientation-preserving isometries. Hence it suffices to show $\text{Isom}^+(\Sigma, g)$ is finite

Fix a point $p \in \Sigma$. Let $\text{inj}(\Sigma) > 0$ denote the injectivity radius of (Σ, g) , defined by

$$\text{inj}(\Sigma) = \inf_{x \in \Sigma} \text{inj}_x(\Sigma)$$

where $\text{inj}_x(\Sigma)$ is the supremum of radius r for which the exponential map at x is a diffeomorphism on the open ball of radius r in $T_x \Sigma$. Since Σ is compact, $\text{inj}(\Sigma) > 0$.

Choose r with $0 < r < \text{inj}(\Sigma)$. Then for every $x \in \Sigma$, the hyperbolic ball $B(x, r)$ is embedded and geodesically convex, and in particular two distinct points $x \neq y$ satisfy

$$B(x, r) \cap B(y, r) \neq \emptyset \implies d(x, y) < 2r$$

Now let $\phi \in \text{Isom}^+(\Sigma, g)$ be nontrivial. We claim:

$$d(p, \phi(p)) \geq 2r$$

Suppose for contradiction that $d(p, \phi(p)) < 2r$. Then by contrapositive, we have $B(p, r) \cap B(\phi(p), r) \neq \emptyset$. Let q be a point in the intersection. Since ϕ is an isometry, $\phi(B(p, r)) = B(\phi(p), r)$. Thus $q \in B(p, r)$ and also $q = \phi(q')$ for some $q' \in B(p, r)$, i.e. ϕ maps $B(p, r)$ into itself and has a point q in common with its image.

Consider the restriction $\phi|_{B(p, r)}$. Because $B(p, r)$ is embedded and geodesically convex (as $r < \text{inj}(\Sigma)$), it is isometric to a hyperbolic disk. An orientation-preserving isometry of a hyperbolic disk is uniquely determined by its value and differential at a single point, and in particular, if it fixes a point and has differential the identity there, it is the identity. We will show that ϕ must fix a point in $B(p, r)$.

Indeed, in a simply connected, complete, negatively curved surface, the squared distance function to a point is strictly convex along geodesics; the same holds on a convex ball in Σ via the universal cover. Define $F : B(p, r) \rightarrow \mathbb{R}$ by

$$F(x) = d(x, \phi(x))^2$$

Since $B(p, r)$ is compact with boundary, F attains a minimum at some $x_0 \in \overline{B(p, r)}$. If the minimum occurs on the boundary, we can shrink slightly to a closed ball $\overline{B(p, r')}$ with $r' < r$ still satisfying $d(p, \phi(p)) < 2r'$, so that the minimum occurs in the interior; thus assume $x_0 \in B(p, r)$.

At an interior minimizer, standard first-variation arguments imply the geodesic segment from x_0 to $\phi(x_0)$ is orthogonal to its image under ϕ , which forces $\phi(x_0) = x_0$ in negative curvature. (Equivalently: in negative curvature, the displacement function $x \mapsto d(x, \phi(x))$ is strictly convex along geodesics unless ϕ has a fixed point; an interior minimum implies a fixed point.) Hence ϕ fixes some $x_0 \in B(p, r)$.

Now, an orientation-preserving isometry of a hyperbolic surface that fixes a point lifts to an orientation-preserving isometry of $\mathbb{H}$ fixing a lift of that point. Such an isometry is elliptic. But Γ is torsion-free, so Σ has no nontrivial elliptic deck transformations, and consequently, the only orientation-preserving isometry of Σ with a fixed point is the identity. Therefore $\phi = \text{id}$, contradicting nontriviality. This proves that $d(p, \phi(p)) \geq 2r$.

Let $G = \text{Isom}^+(\Sigma, g)$. For each $\phi \in G$, consider the ball $B(\phi(p), r)$. If $\phi \neq \psi$, then by $d(p, \phi(p)) \geq 2r$ applied to $\psi^{-1}\phi \neq \text{id}$ we have

$$d(\phi(p), \psi(p)) = d(p, \psi^{-1}\phi(p)) \geq 2r$$

so the balls $B(\phi(p), r)$ and $B(\psi(p), r)$ are disjoint. Thus, we obtain a collection of pairwise disjoint embedded balls

$$\{B(\phi(p), r) \mid \phi \in G\}$$

inside Σ .

Taking areas and using the Gauss-Bonnet theorem, we get

$$|G| \cdot \text{Area}(B_{\mathbb{H}}(r)) \leq \text{Area}(\Sigma) = -2\pi\chi(\Sigma)$$

where $\text{Area}(B_{\mathbb{H}}(r))$ denotes the area of a hyperbolic disk of radius r in $\mathbb{H}$, given by

$$\text{Area}(B_{\mathbb{H}}(r)) = 2\pi(\cosh r - 1) > 0$$

Hence

$$|G| \leq \frac{-2\pi\chi(\Sigma)}{2\pi(\cosh r - 1)} < \infty,$$

so G is finite. Since $\text{Aut}(\Sigma) \hookrightarrow G$, it follows that $|\text{Aut}(\Sigma)| < \infty$.

This completes the proof. $\square$

B C^∞ Topology on Moduli Space

In this appendix, we explain the natural C^∞ topology on the space of complex structures on a fixed oriented smooth surface, and the quotient topology induced on the corresponding Teichmüller space and moduli space.

Let S be a closed connected oriented smooth surface. We write

$$\mathcal{J}^+(S) := \{j \in \Gamma^\infty(\text{End}(TS)) \mid j^2 = -\text{id}_{TS}, \text{ and } (v, jv) \text{ is positively oriented for every } v \neq 0\}$$

Lemma. *Every element $j \in \mathcal{J}^+(S)$ is integrable. Consequently, $\mathcal{J}^+(S)$ may be identified with the set of complex structures on the underlying oriented smooth surface S .*

Proof. Fix $j \in \mathcal{J}^+(S)$. Consider the $(0,1)$ -subbundle $T_j^{0,1}S \subset T_{\mathbb{C}}S$. Since $\dim_{\mathbb{R}} S = 2$, the complex rank of $T_j^{0,1}S$ is 1. Thus, on any sufficiently small open set $U \subseteq S$, there exists a nowhere-vanishing local section X of $T_j^{0,1}S|_U$, and every local section of $T_j^{0,1}S|_U$ is of the form fX for some smooth complex-valued function f on U .

Now let fX and gX be two such local sections. A direct computation gives

$$[fX, gX] = (fX(g) - gX(f))X$$

which is again a local section of $T_j^{0,1}S|_U$. Hence $T_j^{0,1}S$ is involutive. By the Newlander–Nirenberg theorem, j is integrable. Therefore, every $j \in \mathcal{J}^+(S)$ determines a genuine complex structure on S . $\square$

Proposition B.1. The space $\Gamma^\infty(\text{End}(TS))$ carries a natural Fréchet C^∞ topology, and $\mathcal{J}^+(S)$ is naturally a topological subspace.

Proof. Choose a finite smooth atlas on S , a connection ∇ on TS , and a bundle metric on $\text{End}(TS)$. For each integer $k \geq 0$, define a seminorm on $\Gamma^\infty(\text{End}(TS))$ by

$$\|A\|_{C^k} := \max_{x \in S} \sum_{m=0}^k |\nabla^m A(x)|$$

These seminorms determine the standard Fréchet C^∞ topology. Since $\mathcal{J}^+(S) \subset \Gamma^\infty(\text{End}(TS))$ is just a subset cut out by pointwise algebraic conditions, it inherits the subspace topology. $\square$

Let $\text{Diff}^+(S)$ be the group of orientation-preserving diffeomorphisms of S , and let $\text{Diff}_0(S)$ be its identity component. For $\phi \in \text{Diff}^+(S)$ and $j \in \mathcal{J}^+(S)$, define

$$\phi^*j := d\phi^{-1} \circ j \circ d\phi$$

Lemma. *The pullback action*

$$\mathrm{Diff}^+(S) \times \mathcal{J}^+(S) \longrightarrow \mathcal{J}^+(S), \quad (\phi, j) \longmapsto \phi^*j$$

is continuous with respect to the C^∞ topology.

Proof. Endow $\mathrm{Diff}^+(S)$ with its standard C^∞ topology. In a local coordinate chart, the matrix of ϕ^*j is given by

$$D\phi^{-1} \cdot j \cdot D\phi$$

Thus, the local coefficients of ϕ^*j are obtained from the local coefficients of j , the first derivatives of ϕ , and the inverse of $D\phi$ by smooth algebraic operations. Since matrix inversion is smooth on $\mathrm{GL}^+(2, \mathbb{R})$, and since differentiation, multiplication, and composition are continuous in the Whitney C^∞ topology, it follows that $(\phi, j) \mapsto \phi^*j$ is continuous. $\square$

We now define

$$\mathcal{T}(S) := \mathcal{J}^+(S)/\mathrm{Diff}_0(S), \quad \mathcal{M}(S) := \mathcal{J}^+(S)/\mathrm{Diff}^+(S)$$

Theorem B.2. The spaces $\mathcal{T}(S)$ and $\mathcal{M}(S)$ carry natural quotient topologies induced from the C^∞ topology on $\mathcal{J}^+(S)$. Equivalently, a subset $U \subseteq \mathcal{M}(S)$ is open if and only if its full preimage under the quotient map

$$q : \mathcal{J}^+(S) \rightarrow \mathcal{M}(S)$$

is open in $\mathcal{J}^+(S)$, and similarly for $\mathcal{T}(S)$. In this sense, the topology on the moduli space is precisely the topology of smooth variation of complex structures modulo reparametrization.

Proof. By the previous lemma, $\mathrm{Diff}^+(S)$ acts continuously on $\mathcal{J}^+(S)$, and the subgroup $\mathrm{Diff}_0(S)$ acts continuously as well. Therefore, the orbit spaces

$$\mathcal{J}^+(S)/\mathrm{Diff}^+(S) \quad \text{and} \quad \mathcal{J}^+(S)/\mathrm{Diff}_0(S)$$

are well-defined topological quotient spaces. By definition of the quotient topology, a subset of the quotient is open exactly when its preimage in $\mathcal{J}^+(S)$ is open.

This is the natural topology for deformation theory on a fixed smooth surface: if $j_\nu \rightarrow j$ in C^∞ , then their classes converge in the quotient; more generally, if $\phi_\nu \in \mathrm{Diff}^+(S)$ and $\phi_\nu^*j_\nu \rightarrow j$ in C^∞ , then the classes $[j_\nu]$ and $[\phi_\nu^*j_\nu]$ define the same points of moduli space and hence have the same limit there. Thus, the quotient topology records exactly smooth variation modulo reparametrization. $\square$

Proposition B.3. The topology above describes deformations of complex structures on a fixed smooth surface. It does not include degenerations to nodal curves.

Proof. The essential point is that the ambient parameter space is

$$\mathcal{J}^+(S) \subset \Gamma^\infty(\mathrm{End}(TS))$$

so every point of $\mathcal{J}^+(S)$ is, by definition, a *smooth* bundle endomorphism of the *fixed* tangent bundle TS over the *fixed* smooth surface S , satisfying

$$j^2 = -\mathrm{id}_{TS}$$

together with the positivity condition. Thus, before taking any quotient, the space $\mathcal{J}^+(S)$ parametrizes only smooth complex structures on one and the same differentiable surface S .

Now suppose that

$$j_\nu \longrightarrow j \quad \text{in } C^\infty$$

By definition of the C^∞ topology, this means that in every local coordinate chart, all local coefficient functions of j_ν and all of their derivatives converge uniformly to those of j on compact subsets. In particular, the limit j is again a smooth section of $\text{End}(TS)$. Since the relation

$$j_\nu^2 = -\text{id}_{TS}$$

is pointwise algebraic, it passes to the limit and yields

$$j^2 = -\text{id}_{TS}$$

Likewise, the positivity condition is preserved under such convergence, so the limit still belongs to $\mathcal{J}^+(S)$. Hence, every convergent sequence in $\mathcal{J}^+(S)$ has a limit which is again a smooth complex structure on the same smooth surface S .

Passing to the quotient by $\text{Diff}^+(S)$ does not enlarge the class of objects under consideration. Indeed, an orientation-preserving diffeomorphism

$$\phi : S \rightarrow S$$

acts on $\mathcal{J}^+(S)$ by pullback,

$$\phi^* j = d\phi^{-1} \circ j \circ d\phi$$

which merely changes the parametrization of the same smooth surface. Thus, two points of $\mathcal{J}^+(S)$ lying in the same $\text{Diff}^+(S)$ -orbit represent biholomorphically equivalent complex structures on the same underlying differentiable manifold. Therefore the quotient

$$\mathcal{M}(S) = \mathcal{J}^+(S)/\text{Diff}^+(S)$$

still records only smooth complex structures on the fixed topological type of S .

More concretely, by definition of the quotient topology, if

$$[j_\nu] \longrightarrow [j] \quad \text{in } \mathcal{M}(S)$$

then one may choose representatives so that there exist diffeomorphisms $\phi_\nu \in \text{Diff}^+(S)$ with

$$\phi_\nu^* j_\nu \longrightarrow j \quad \text{in } C^\infty$$

for some $j \in \mathcal{J}^+(S)$. Thus, the limit object is again a smooth complex structure on S ; only the parametrization of the approximating sequence is allowed to vary.

A nodal degeneration is of an entirely different nature. In such a degeneration, the limiting object is no longer a smooth Riemann surface, but a singular complex curve with at least one point having a local analytic model $xy = 0$. Equivalently, one may view it as the central fiber of a one-parameter family locally modeled by $xy = t$, whose fibers are smooth for $t \neq 0$ but singular when $t = 0$. The limiting object therefore fails to be a smooth surface, and at the node, there is no smooth rank-2 tangent bundle on which an element of $\mathcal{J}^+(S)$ could be defined. Consequently, such a singular curve cannot be represented by any point of $\mathcal{J}^+(S)$, nor can it occur as a limit in the C^∞ topology on $\mathcal{J}^+(S)$.

Therefore, the topology on $\mathcal{M}(S)$ induced from the C^∞ topology on $\mathcal{J}^+(S)$ describes only smooth deformations of complex structures on a fixed smooth surface, modulo reparametrization. It does not detect nodal degenerations. To incorporate such limits, one must enlarge the moduli problem itself, for instance by passing to the Deligne–Mumford compactification, where stable nodal curves appear as additional boundary points. $\square$

C Contractibility of $\mathcal{J}^+(S)$

In this appendix, we prove that the space $\mathcal{J}^+(S)$ of positive complex structures on a closed oriented smooth surface is contractible.

Let $\text{Met}(S)$ denote the space of smooth Riemannian metrics on S , endowed with the C^∞ topology inherited from the Fréchet space $\Gamma^\infty(S^2 T^* S)$.

Lemma. *The space $\text{Met}(S)$ is contractible.*

Proof. If $g_0, g_1 \in \text{Met}(S)$ and $t \in [0, 1]$, then

$$g_t := (1 - t)g_0 + tg_1$$

is again a smooth positive definite symmetric bilinear form on TS , hence a Riemannian metric. Thus $\text{Met}(S)$ is a convex subset of the Fréchet space $\Gamma^\infty(S^2T^*S)$, and therefore it is contractible. $\square$

Lemma. *For every metric $g \in \text{Met}(S)$, there exists a unique element $j_g \in \mathcal{J}^+(S)$ such that, for each $x \in S$, the endomorphism $j_g(x)$ is the rotation by $+\pi/2$ with respect to the inner product g_x and the given orientation on T_xS .*

Proof. Fix $x \in S$. Since (T_xS, g_x) is an oriented Euclidean plane, there exists a unique orientation-preserving g_x -orthogonal endomorphism J_x satisfying

$$J_x^2 = -\text{id}_{T_xS}$$

Define $j_g(x) := J_x$. In a local positively oriented g -orthonormal frame, the matrix of j_g is constantly

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

so j_g depends smoothly on x . It follows that $j_g \in \Gamma^\infty(\text{End}(TS))$, satisfies $j_g^2 = -\text{id}_{TS}$, and is positively oriented by construction. Hence $j_g \in \mathcal{J}^+(S)$. $\square$

Therefore, we obtain a map

$$\pi : \text{Met}(S) \rightarrow \mathcal{J}^+(S), \quad \pi(g) := j_g$$

Fix once and for all a smooth positive area form ρ on S .

Lemma. *For each $j \in \mathcal{J}^+(S)$, the formula*

$$g_j(v, w) := \rho(v, jw)$$

defines a Riemannian metric on S .

Proof. We first show that g_j is symmetric. Since $j^2 = -\text{id}$, the map $j_x : T_xS \rightarrow T_xS$ is a complex structure on the oriented plane T_xS , hence orientation-preserving. Therefore $\det(j_x) = 1$, and thus

$$\rho(jv, jw) = \rho(v, w)$$

for all tangent vectors v, w . Replacing w by jw , we obtain

$$\rho(jv, -w) = \rho(v, jw)$$

that is,

$$\rho(jv, w) = -\rho(v, jw)$$

Hence

$$g_j(w, v) = \rho(w, jv) = -\rho(jv, w) = \rho(v, jw) = g_j(v, w)$$

so g_j is symmetric.

Next, let $v \neq 0$. Since $j \in \mathcal{J}^+(S)$, the pair (v, jv) is positively oriented, and ρ is a positive area form. Therefore

$$g_j(v, v) = \rho(v, jv) > 0$$

So g_j is positive definite. Smoothness is immediate from the smoothness of ρ and j . Hence g_j is a Riemannian metric. $\square$

Thus, we obtain a second map

$$s : \mathcal{J}^+(S) \rightarrow \text{Met}(S), \quad s(j) := g_j$$

Lemma. *The map s is a continuous section of π , namely*

$$\pi \circ s = \text{id}_{\mathcal{J}^+(S)}$$

Proof. Let $j \in \mathcal{J}^+(S)$ and put $g := s(j) = g_j$. We claim that the positive complex structure determined by g is precisely j . Indeed, for any tangent vector v ,

$$g(v, jv) = \rho(v, j^2v) = \rho(v, -v) = 0$$

so jv is g -orthogonal to v . Moreover,

$$g(jv, jv) = \rho(jv, j^2v) = \rho(jv, -v) = \rho(v, jv) = g(v, v)$$

so j preserves the g -norm. Since j also rotates each nonzero vector in the positive direction, it is exactly the positive quarter-turn determined by g . By uniqueness in the previous lemma,

$$\pi(s(j)) = j$$

Therefore $\pi \circ s = \text{id}_{\mathcal{J}^+(S)}$. □

Theorem C.1. The space $\mathcal{J}^+(S)$ is contractible.

Proof. By the first lemma, $\text{Met}(S)$ is contractible. We shall show that $\mathcal{J}^+(S)$ is a deformation retract of $\text{Met}(S)$.

Consider the maps

$$\pi : \text{Met}(S) \rightarrow \mathcal{J}^+(S), \quad s : \mathcal{J}^+(S) \rightarrow \text{Met}(S)$$

constructed above. Since $\pi \circ s = \text{id}_{\mathcal{J}^+(S)}$, it is enough to deform $\text{Met}(S)$ onto the image of s .

Define

$$H : [0, 1] \times \text{Met}(S) \rightarrow \text{Met}(S), \quad H(t, g) := (1 - t)g + t s(\pi(g))$$

Because $\text{Met}(S)$ is convex, $H(t, g)$ is again a Riemannian metric for every (t, g) . Moreover,

$$H(0, g) = g, \quad H(1, g) = s(\pi(g))$$

If $g = s(j)$ lies in the image of s , then

$$H(t, g) = (1 - t)s(j) + t s(\pi(s(j))) = (1 - t)s(j) + t s(j) = s(j)$$

since $\pi \circ s = \text{id}_{\mathcal{J}^+(S)}$. Hence H is a deformation retraction of $\text{Met}(S)$ onto $s(\mathcal{J}^+(S))$.

Since $\text{Met}(S)$ is contractible, so is $s(\mathcal{J}^+(S))$. Finally, because s is a continuous section of π , it is a homeomorphism from $\mathcal{J}^+(S)$ onto its image. Therefore, $\mathcal{J}^+(S)$ itself is contractible. □

Proposition C.2. The homotopical complexity of Teichmüller space and moduli space does not come from $\mathcal{J}^+(S)$ itself, but from quotienting by the diffeomorphism group.

Proof. The theorem shows that the total space $\mathcal{J}^+(S)$ of positive complex structures has trivial homotopy type. Therefore, any nontrivial topology in

$$\mathcal{T}(S) = \mathcal{J}^+(S)/\text{Diff}_0(S) \quad \text{or} \quad \mathcal{M}(S) = \mathcal{J}^+(S)/\text{Diff}^+(S)$$

must arise from the action of the reparametrization group and the topology of the quotient, rather than from the parameter space $\mathcal{J}^+(S)$ itself. □

D An Algebraic Approach to Domain Deformation

For simplicity, let $A := \mathbb{C}[\epsilon]/(\epsilon^2)$ be the dual number. As we have mentioned in the main text, a first-order deformation of a pointed curve is a flat family $\pi : \mathcal{C} \rightarrow \text{Spec } A$ with sections $s_i : \text{Spec } A \rightarrow \mathcal{C}$ such that the special fiber can be identified as the original curve:

$$(\mathcal{C}_0, s_1(0), \dots, s_n(0)) \cong (C, p_1, \dots, p_n)$$

and we consider the divisor $D := p_1 + \dots + p_n$.

Smooth curve. We shall illustrate the following theorem of the deformation of smooth domain:

Theorem. For smooth marked curve $(C; p_1, \dots, p_n)$ and the given divisor $D = p_1 + \dots + p_n$, the infinitesimal automorphism is parametrized by $H^0(C, \mathcal{T}_C(-D))$, and the first-order deformation is parametrized by $H^1(C, \mathcal{T}_C(-D))$. Here, $\mathcal{T}_C(-D)$ is the section sheaf of holomorphic tangent bundle $\mathcal{T}_C := T^{1,0}C$ twisted by $-D$.

Our discussion will be based on the deformation of the complex structure on the curve C . Let $A := \mathbb{C}[\epsilon]/(\epsilon^2)$, a first order deformation is flat morphism

$$\pi : \mathcal{C} \rightarrow \text{Spec } A$$

such that $\mathcal{C}$ is an A -scheme representing the deformation family of the curve. The forgetful map $A := \mathbb{C}[\epsilon]/(\epsilon^2) \rightarrow \mathbb{C}$ leads to a map $\text{Spec } \mathbb{C} \rightarrow \text{Spec } A$, and we require the following fiber product to satisfy the following isomorphism of special fiber

$$\mathcal{C}_0 := \mathcal{C} \times_{\text{Spec } A} \text{Spec } \mathbb{C} \cong C$$

with a series of sections $s_1, \dots, s_n : \text{Spec } A \rightarrow \mathcal{C}$ that points out the marked points $p_1, \dots, p_n$.

Since C is a smooth curve, in a good cover¹⁰ $\mathcal{U} := \{(U_\alpha, z_\alpha : U_\alpha \xrightarrow{\sim} V_\alpha \subseteq \mathbb{C})\}_{\alpha \in I}$ of holomorphic charts, we always have a local trivialization for some open set $U_\alpha \subseteq C$ such that

$$\mathcal{U}_\alpha \times_{\text{Spec } A} \text{Spec } \mathbb{C} \cong U_\alpha \subseteq C$$

and with the chosen trivialization, $\mathcal{U}_\alpha \cong U_\alpha \times \text{Spec } A$ ¹¹. This trivialization tells us that the nontrivial deformation only exists in global level, this already give a hint that we need to use Čech cohomology to parametrize the deformation. Mor precisely, the local deformation is always trivial in the following sense in consider of the structural ring (structural sheaf) of a ringed space:

Proposition D.1. The ring of (holomorphic) sections on $\mathcal{U}_\alpha$ satisfies $\mathcal{O}_\mathcal{C}(\mathcal{U}_\alpha) \cong \mathcal{O}_C(U_\alpha) \otimes_\mathbb{C} A$.

Proof. We start from the local trivialization

$$\phi_\alpha : \mathcal{U}_\alpha \xrightarrow{\sim} U_\alpha \times \text{Spec } A$$

Since $\text{Spec } A = \{(\epsilon)\}$ is a single point, the underlying topological space is the same as U_α , and the true difference is in the structural sheaf. Defined the sheaf $\mathcal{A}_\alpha$ on U_α as

$$\mathcal{A}_\alpha(V) := \mathcal{O}_C(V) \otimes_\mathbb{C} A$$

where $V \subseteq U_\alpha$ is open. The restriction map is defined by

$$\rho_{VW}^{\mathcal{A}_\alpha} = \rho_{VW}^{\mathcal{O}_C} \otimes \text{id}_A$$

¹⁰Leray cover.

¹¹The idea of thickening is to keep the underlying topological space U_α unchanged and the structure sheaf change from $\mathcal{O}_{U_\alpha}$ to $\mathcal{O}_{U_\alpha} \otimes_\mathbb{C} A$, i.e., as a ringed space, $\mathcal{U}_\alpha = (U_\alpha, \mathcal{O}_{U_\alpha} \otimes_\mathbb{C} A)$.

Since $A = \mathbb{C}[\epsilon]/(\epsilon^2)$ is a finite dimensional $\mathbb{C}$ -vector space, the functor $(-)\otimes_{\mathbb{C}} A$ is exact and preserves arbitrary product. Thus, $(U_\alpha, \mathcal{A}_\alpha)$ is the ringed space decomposition of $U_\alpha \times \text{Spec } A$. In other words, by taking $V = U_\alpha$

$$\mathcal{O}_{U_\alpha \times \text{Spec } A}(U_\alpha) = \mathcal{O}_C(U_\alpha) \otimes_{\mathbb{C}} A$$

The isomorphism of ringed spaces ϕ_α induces the isomorphism of the section ring

$$\phi_\alpha^\#(U_\alpha) : \mathcal{O}_{U_\alpha \times \text{Spec } A}(U_\alpha) \xrightarrow{\sim} \mathcal{O}_C(\mathcal{U}_\alpha)$$

which is just $\mathcal{O}_C(\mathcal{U}_\alpha) \cong \mathcal{O}_C(U_\alpha) \otimes_{\mathbb{C}} A$. □

Thus, on a trivial local chart, the deformation family sends the holomorphic functions to a first-order jet by isomorphism

$$\mathcal{O}_C(\mathcal{U}_\alpha) \cong \mathcal{O}_C(U_\alpha) \otimes_{\mathbb{C}} A \cong \mathcal{O}_C(U_\alpha) \oplus \epsilon \mathcal{O}_C(U_\alpha)$$

which has multiplication that $(f + \epsilon g)(h + \epsilon k) = fh + \epsilon(fk + gh)$. For $p \in U_\alpha$, we may take the stalk at p using any open set U that contains p and the restriction map, and we can also obtain the isomorphism of the local ring

$$\mathcal{O}_{\mathcal{U}_\alpha, p} = \varinjlim_{p \in V \subseteq U_\alpha} \mathcal{O}_C(\mathcal{U}_\alpha) \cong \mathcal{O}_{C, p} \otimes_{\mathbb{C}} A$$

With the discussion above of the local representation, for good cover $\mathfrak{U} = \{(U_\alpha, z_\alpha)\}_{\alpha \in I}$, we may consider the transition function $f_{\alpha\beta}$ on the overlap $U_{\alpha\beta} = U_\alpha \cap U_\beta$ such that

$$z_\alpha = f_{\alpha\beta} \circ z_\beta : U_{\alpha\beta} \rightarrow V_\alpha \quad \forall \alpha, \beta \in I$$

Under the deformation, we may allow it to contain the first-order perturbation:

$$\tilde{z}_\alpha = f_{\alpha\beta} \circ \tilde{z}_\beta + \epsilon \eta_{\alpha\beta} \circ \tilde{z}_\beta$$

where $\eta_{\alpha\beta}$ is a holomorphic function. Note that the perturbation has the nature of being a vector field or derivation. Let $U \subseteq C$ be an open set, then its "first order thickening" is the corresponding open set $\mathcal{U} \cong U \times \text{Spec } A$ in $\mathcal{C}$, the deformation of the complex structure locally is given by an A -scheme automorphism that is identity on the special fiber $\mathcal{C}_0$, i.e., an automorphism $g : \mathcal{U} \rightarrow \mathcal{U}$ such that $g \equiv \text{id} \pmod{\epsilon}$

Proposition D.2. There is a one-to-one correspondence between the automorphism we defined above and the holomorphic derivation $D \in \text{Der}_{\mathbb{C}}(\mathcal{O}_U, \mathcal{O}_U)$. Equivalently, the automorphism that is identity on the special fiber $\mathcal{C}_0 \cong C$ can be viewed as a holomorphic vector field $\theta \in \Gamma(U, T^{1,0}C)$

Proof. By the condition that $g \equiv \text{id} \pmod{\epsilon}$, for any $f \in \mathcal{O}_U$ (we consider f as $f \otimes 1 \in \mathcal{O}_{\mathcal{U}}$), we have the induced map

$$g^\#(f) = f \pmod{\epsilon}$$

There must be unique $\mathbb{C}$ -linear map $D : \mathcal{O}_U \rightarrow \mathcal{O}_U$ such that

$$g^\#(f) = f + \epsilon D(f)$$

Since $\epsilon^2 = 0$, the multiplicative $g^\#(fg) = g^\#(g)g^\#(g)$ leads to

$$fg + \epsilon D(fg) = fg + \epsilon(D(f)g + fD(g))$$

and thus, $D(fg) = D(f)g + fD(g)$, i.e., the automorphism g induce a derivation D .

Conversely, if we take derivation $D \in \text{Der}_{\mathbb{C}}(\mathcal{O}_U, \mathcal{O}_U)$, we may consider

$$g := \text{id} + \epsilon D$$

and by Leibniz rule, it is an A -automorphism with inverse $(\text{id} + \epsilon D)(\text{id} - \epsilon D) = \text{id} - \epsilon^2 D^2 = \text{id}$, and thus, we have an one-to-one correspondence

$$\{g \in \text{Aut}(\mathcal{U}) \text{ reduce to id}\} \longleftrightarrow \text{Der}_{\mathbb{C}}(\mathcal{O}_U, \mathcal{O}_U)$$

and complex manifolds have an isomorphism $\text{Der}_{\mathbb{C}}(\mathcal{O}_U, \mathcal{O}_U) \cong \Gamma(U, T^{1,0}C)$, which completes the proof of the proposition that the automorphism reduces to identity, can be viewed as a holomorphic vector field. □

And we further claim the local coordinate form of the corresponding vector field:

Proposition D.3. Let z be the local coordinate over the open set $U \subseteq C$, then such a derivation D can be written uniquely as

$$D = \eta(z) \frac{\partial}{\partial z}$$

where $\eta(z) := D(z)$ is holomorphic function.

Proof. Let $\eta(z) := D(z)$ for any holomorphic function $f = \sum_{m \geq 0} a_m z^m$, we may take

$$D(f) := \sum_{m \geq 0} a_m D(z^m)$$

By the Leibniz rule, we may compute each term by

$$D(z^m) = m z^{m-1} D(z) = m z^{m-1} \eta(z)$$

Thus,

$$D = \eta(z) \frac{\partial}{\partial z} =: \theta$$

and the uniqueness is trivial. $\square$

Corollary. Under local chart (U, z) , for any holomorphic function f , the first-order deformation can be written as $g^\sharp(f)(z) = f(z) + \epsilon \eta(z) f'(z)$.

If U contains a marked point such that $z(p) = 0$, and g preserves p , then the corresponding vector field satisfies $\eta(0) = 0$, i.e., vanishes at the marked point. This shows that the infinitesimal automorphism is parametrized by $H^0(C, \mathcal{T}_C(-D))$, which is the first part of Theorem 6.7. Thus,

$$\text{Marked point } p \text{ is a fixed point of } g \iff \text{the corresponding vector field } \theta \in \Gamma(C, \mathcal{T}_C(-p))$$

Then, we may go back to global deformation $(\mathcal{C}, s_i)$. We already have the adapted trivialization on each U_α

$$\psi_\alpha : \mathcal{U}_\alpha \xrightarrow{\sim} U_{\alpha,A} \cong U_\alpha \times \text{Spec } A$$

In the overlapping region $U_{\alpha\beta} := U_\alpha \cap U_\beta$, the difference of two trivialization gives an A -automorphism

$$g_{\alpha\beta} := \psi_\alpha \circ \psi_\beta^{-1} : U_{\alpha\beta,A} \rightarrow U_{\alpha\beta,A}$$

Note that this difference of isomorphism needs to be identity modulo ϵ , since the adapted trivialization requires

$$\psi_\alpha|_{\epsilon=0} : \mathcal{U}_\alpha \times_{\text{Spec } A} \text{Spec } \mathbb{C} \rightarrow U_{\alpha,A} \times_{\text{Spec } A} \text{Spec } \mathbb{C} \cong U_\alpha$$

and we may treat this as just the identity id_{U_α} . By Proposition D.2, we may find the unique derivation (or vector field) $D_{\alpha\beta} \in \Gamma(U_{\alpha\beta}, \mathcal{T}_C)$ such that $g_{\alpha\beta}^\sharp = \text{id} + \epsilon D_{\alpha\beta}$, and we also need to require the vector field to be adapted to the marked points p_i , the automorphism $g_{\alpha\beta}$ should preserves $p_i \times \text{Spec } A$, and thus, we have a Čech 1-cochain that twisted by the marked points:

$$\{D_{\alpha\beta}\} \in \check{C}^1(\mathfrak{U}, \mathcal{T}_C(-D))$$

Then, in triple overlap $U_{\alpha\beta\gamma} := U_\alpha \cap U_\beta \cap U_\gamma$, there is a cocycle condition $g_{\alpha\beta} \circ g_{\beta\gamma} = g_{\alpha\gamma}$ and on a structural sheaf, due to the fact that the induced pullback satisfies

$$g_{\beta\gamma}^\sharp \circ g_{\alpha\beta}^\sharp = g_{\alpha\gamma}^\sharp$$

To check if $\{D_{\alpha\beta}\}$ is a 1-cocycle, we expand the pullback with the derivation as

$$(\text{id} + \epsilon D_{\beta\gamma})(\text{id} + \epsilon D_{\alpha\beta}) = \text{id} + \epsilon(D_{\alpha\beta} + D_{\beta\gamma}) = \text{id} + \epsilon D_{\alpha\gamma}$$

Thus, $D_{\alpha\beta} + D_{\beta\gamma} = D_{\alpha\gamma}$. Equivalently, we have the 1-cocycle condition $D_{\alpha\beta} + D_{\beta\gamma} + D_{\gamma\alpha} = 0$. Thus, $\{D_{\alpha\beta}\} \in \check{Z}^1(\mathfrak{U}, \mathcal{T}_C(-D))$ is a Čech 1-cocycle.

We can rewrite the cocycle condition above in coordinates. With $z_\alpha = f_{\alpha\beta} \circ z_\beta$, recall that the pullback of the automorphism is given by $g_{\alpha\beta}^\#(z_\alpha) = f_{\alpha\beta}(z_\beta) + \epsilon \eta_{\alpha\beta}(z_\beta)$, and the local perturbation corresponds to the vector field given by

$$\theta_{\alpha\beta} := \eta_{\alpha\beta}(z_\beta) \frac{\partial}{\partial z_\alpha} \in \Gamma(U_{\alpha\beta}, \mathcal{T}_C(-D))$$

On the triple intersection $U_{\alpha\beta\gamma}$, by substitution and use the fact that $\epsilon^2 = 0$, we may write that

$$\begin{aligned} z_\beta &= f_{\alpha\beta}(f_{\beta\gamma}(z_\gamma) + \epsilon \eta_{\beta\gamma}(z_\gamma)) + \epsilon \eta_{\alpha\beta}(f_{\beta\gamma}(z_\gamma) + \epsilon \eta_{\beta\gamma}(z_\gamma)) \\ &= f_{\alpha\beta}(f_{\beta\gamma}(z_\gamma)) + \epsilon f'_{\alpha\beta}(f_{\beta\gamma}(z_\gamma)) \eta_{\beta\gamma}(z_\gamma) + \epsilon \eta_{\alpha\beta}(f_{\beta\gamma}(z_\gamma)) \end{aligned}$$

The Čech cocycle condition requires also $f_{\alpha\beta} \circ f_{\beta\gamma} = f_{\alpha\gamma}$, apply this condition, we can get

$$z_\beta = f_{\alpha\gamma}(z_\gamma) + \epsilon (\eta_{\alpha\beta} \circ f_{\beta\gamma}(z_\gamma) + \eta_{\beta\gamma}(z_\gamma)(f'_{\alpha\beta} \circ f_{\beta\gamma})(z_\gamma))$$

Use the first-order term of ϵ , the equation above implies

$$\eta_{\alpha\gamma}(z_\gamma) = \eta_{\alpha\beta}(f_{\beta\gamma}(z_\gamma)) + \eta_{\beta\gamma}(z_\gamma) f'_{\alpha\beta}(f_{\beta\gamma}(z_\gamma))$$

This exactly shows that $\eta_{\alpha\beta}$ is not a scalar 1-cocycle; instead, it transforms as a vector. In a coordinate-free form, it is exactly

$$\theta_{\alpha\beta} + \theta_{\beta\gamma} + \theta_{\gamma\alpha} = 0$$

To show that every deformation is a cohomology class in $H^1(C, \mathcal{T}_C(-D))$, we should also prove that the change of trivialization will only result in the difference of a Čech coboundary. Suppose on each $U_\alpha \in \mathfrak{U}$, the trivialization becomes

$$\psi_{;\alpha} := h_\alpha \circ \psi_\alpha : \mathcal{U}_\alpha \xrightarrow{\sim} U_{\alpha,A}$$

where h_α is an automorphism that reduces to identity under modulo ϵ , and we also need to require it to keep the marked sections s_i . By Proposition D.2, we can write

$$h_\alpha^\# = \text{id} + \epsilon \xi_\alpha, \quad \xi_\alpha \in \Gamma(U_\alpha, \mathcal{T}_C(-D))$$

The new transition map will be $g'_{\alpha\beta} = h_\alpha \circ g_{\alpha\beta} \circ h_\beta^{-1}$, and by the expression of $h_\alpha^\#$, one can show that

$$g'_{\alpha\beta}^\# = \text{id} + \epsilon(D_{\alpha\beta} + \xi_\alpha - \xi_\beta)$$

Thus, the new cocycle is just $D_{\alpha\beta} + \xi_\alpha - \xi_\beta$, which exactly is the original one plus a Čech coboundary. Since we have chosen a good cover, we can safely claim that the deformation corresponds to a first sheaf cohomology class

$$[\{D_{\alpha\beta}\}] \in H^1(C, \mathcal{T}_C(-D))$$

Conversely, it is not hard to see that given a Čech 1-cocycle, we may construct an automorphism $g_{\alpha\beta} : U_{\alpha\beta,A} \rightarrow U_{\alpha\beta,A}$ with pullback $g_{\alpha\beta}^\# = \text{id} + \epsilon D_{\alpha\beta}$. Since such $g_{\alpha\beta}$ satisfies the cocycle condition, it can be glued to a ringed space $\mathcal{C}$, and locally

$$\mathcal{O}_{\mathcal{C}}|_{U_\alpha} \cong \mathcal{O}_{U_\alpha} \otimes A$$

and one may check that the gluing is well-defined on sections of marked points. Thus, the equivalent class of first-order domain deformation is parametrized using $H^1(C, \mathcal{T}_C(-D))$, and this is exactly the second half of Theorem 6.7.

Then, we may discuss the general case that $(C, p_1, \dots, p_n)$ is a pointed nodal curve.

General case. For the deformation of (possibly nodal) curves, we need to consider the smoothing directions of nodes. Let $\mathbb{k}$ be an algebraically closed field (we may just work over $\mathbb{C}$), $(C, \mathbf{p}) := (C, p_1, \dots, p_n)$ is a path-connected nodal curve with distinct marked points p_i on the smooth locus. Let $D := p_1 + \dots + p_n$ be a divisor, N be the set of singularities (nodes) on C . Let the normalization $\nu : \tilde{C} \rightarrow C$ and

$$Q := \nu^{-1}(N) = \sum_{q \in N} (q' + q''), \quad \tilde{D} := \nu^{-1}(D)$$

We also defined the tangent sheaf and the twisted tangent sheaf as

$$\mathcal{T}_C := \mathcal{H}om(\Omega_C, \mathcal{O}_C), \quad \mathcal{T}_C(-D) := \mathcal{H}om(\Omega_C(D), \mathcal{O}_C)$$

Note that $\mathcal{H}\mathbb{I}$ here is the sheaf of morphisms, and $\mathcal{H}om(-, -) = H^0(\mathcal{H}om(-, -))$. Then we have the following theorem:

Theorem D.4. (1) The infinitesimal automorphism is parametrized by

$$\text{InfAut}(C, \mathbf{p}) \cong H^0(C, \mathcal{T}_C(-D))$$

and with the natural isomorphism $\mathcal{T}_C(-D) \cong \nu_* \mathcal{T}_{\tilde{C}}(-\tilde{D} - Q)$. Thus,

$$\text{InfAut}(C, \mathbf{p}) \cong H^0(\tilde{C}, \mathcal{T}_{\tilde{C}}(-\tilde{D} - Q))$$

(2) The first order deformation of the pointed nodal curve is given by

$$T^1(C, \mathbf{p}) \cong \text{Ext}_C^1(\Omega_C(D), \mathcal{O}_C)$$

Now we may prove the theorem.

Recall that we take $A := \mathbb{C}[\epsilon]/(\epsilon^2)$, a infinitesimal automorphism of $(C, \mathbf{p})$ is a A -morphism

$$\Phi^\sharp : C_A := C \times \text{Spec } A \rightarrow C_A$$

such that $\Phi^\sharp \equiv \text{id} \pmod{\epsilon}$ and $\Phi^\sharp$ preserves all sections p_i . On the structure sheaf, $\Phi^\sharp$ has the form

$$\Phi^\sharp(f) = f + \epsilon \delta(f), \quad \delta \in \text{Der}_{\mathbb{C}}(\mathcal{O}_C, \mathcal{O}_C)$$

Then we need to consider the condition that $\Phi^\sharp$ preserves the marked points. Let $I_D := \mathcal{O}_C(-D) \subseteq \mathcal{O}_C$ be the ideal sheaf of the divisor D , the condition that $\Phi^\sharp$ preserves every p_i is just

$$\delta(I_D) \subseteq I_D$$

Thus, we have

$$\text{InfAut}(C, \mathbf{p}) = H^0(\mathcal{O}_C, \mathcal{O}_C(-D))$$

By the universal property of Kähler differential,

$$\text{Der}_k(\mathcal{O}_C, \mathcal{O}_C(-D)) \cong \mathcal{H}om(\mathcal{O}_C, \mathcal{O}_C(-D)) \cong \mathcal{H}om(\mathcal{O}_C(D), \mathcal{O}_C) = \mathcal{T}_C(-D)$$

Thus, the infinitesimal automorphism is given by

$$\text{InfAut}(C, \mathbf{p}) \cong H^0(\mathcal{O}_C, \mathcal{T}_C(-D))$$

This proves the first part of the theorem.

Now we take a node $q \in C$, the local ring is given by

$$R := \hat{\mathcal{O}}_{C,q} \cong \mathbb{C}[[x, y]]/(xy)$$

since all marked points are given to be smooth points, locally near q , we have $\Omega_C(D)_q \cong \Omega_{C,q}$. By the standard presentation of Kähler differential,

$$\Omega_R \cong \frac{R \, dx \oplus R \, dy}{R \cdot (y \, dx + x \, dy)}$$

which means we have a free resolution given by

$$0 \longrightarrow R \xrightarrow{\alpha} R \, dx \oplus R \, dy \longrightarrow \Omega_R \longrightarrow 0$$

where $\alpha(1) = y \, dx + x \, dy$. Apply $\text{Hom}_R(-, R)$ to the resolution above, we can get

$$0 \longrightarrow \text{Hom}_R(\Omega_R, R) \longrightarrow R \, \partial_x \oplus R \, \partial_y \xrightarrow{\alpha^\vee} R \longrightarrow \text{Ext}_R^1(\Omega_R, R) \longrightarrow 0$$

where $\alpha^\vee(a, b) = ay + bx$, by the exactness of the sequence above, one can know that

$$\text{Hom}_R(\Omega_R, R) \cong \ker(\alpha^\vee) = (x)\partial_x \oplus (y)\partial_y$$

Since $\text{coker}(\alpha^\vee) = R/(x, y)$, apply the exactness again, we can obtain

$$\text{Ext}_R^1(\Omega_R, R) \cong R/(x, y) \cong \mathbb{C}$$

Thus, each node contributes a first-order deformation parameter.

Since on a smooth point, $\Omega_C(D)$ is locally free of rank 1, on the smooth locus, we have $\mathcal{E}xt^1(\Omega_C(D), \mathcal{O}_C) = 0$. Thus, the sheaf $\mathcal{E}xt^1$ only supported on nodes, and

$$\mathcal{E}xt^1(\Omega_C(D), \mathcal{O}_C) = \bigoplus_{q \in N} k_q$$

as a skyscraper sheaf. To identify the tangent sheaf in terms of the normalization, we compare the stalks at a node. The normalization of $R = \mathbb{C}[[x, y]]/(xy)$ is $\tilde{R} \cong \mathbb{C}[[x]] \oplus \mathbb{C}[[y]]$, corresponding to the two preimages $q', q'' \in \tilde{C}$ of the node q . On the normalization, the stalk of the twisted tangent sheaf at $q' \sqcup q''$ is

$$\mathcal{T}_{\tilde{C}}(-Q)_{q' \sqcup q''} \cong x\mathbb{C}[[x]] \, \partial_x \oplus y\mathbb{C}[[y]] \, \partial_y.$$

After pushing forward along ν , this is identified with

$$(x)\partial_x \oplus (y)\partial_y = \text{Hom}_R(\Omega_R, R) = \mathcal{T}_{C,q}$$

Away from the nodes, ν is an isomorphism, hence we obtain a natural isomorphism of sheaves

$$\mathcal{T}_C \cong \nu_* \mathcal{T}_{\tilde{C}}(-Q).$$

Twisting by the marked divisor gives

$$\mathcal{T}_C(-D) \cong \nu_* \mathcal{T}_{\tilde{C}}(-\tilde{D} - Q)$$

Since ν is finite, we have $R^i \nu_* = 0$ for all $i > 0$. Therefore

$$H^i(C, \mathcal{T}_C(-D)) \cong H^i(\tilde{C}, \mathcal{T}_{\tilde{C}}(-\tilde{D} - Q)), \quad i = 0, 1$$

In particular,

$$\text{InfAut}(C, \mathbf{p}) \cong H^0(C, \mathcal{T}_C(-D)) \cong H^0(\tilde{C}, \mathcal{T}_{\tilde{C}}(-\tilde{D} - Q))$$

Now we turn to the first-order deformation. By the standard classification of square-zero extensions compatible with the marked sections, the set of isomorphism classes of first-order deformations of the pointed nodal curve $(C, \mathbf{p})$ is naturally identified with

$$T^1(C, \mathbf{p}) \cong \text{Ext}_C^1(\Omega_C(D), \mathcal{O}_C)$$

To make the smoothing directions explicit, we apply the local-to-global Ext spectral sequence

$$E_2^{p,q} = H^p(C, \mathcal{E}xt^q(\Omega_C(D), \mathcal{O}_C)) \implies \text{Ext}_C^{p+q}(\Omega_C(D), \mathcal{O}_C)$$

Since C is one-dimensional, we have $H^2(C, -) = 0$.

Moreover, $\Omega_C(D)$ has projective dimension at most 1: it is locally free of rank 1 on the smooth locus, and at each node it admits the length-one free resolution computed above. Hence

$$\mathcal{E}xt^q(\Omega_C(D), \mathcal{O}_C) = 0, \quad q \geq 2$$

Therefore the spectral sequence degenerates in total degree 1 and yields a short exact sequence

$$0 \longrightarrow H^1(C, \mathcal{H}om(\Omega_C(D), \mathcal{O}_C)) \longrightarrow \text{Ext}_C^1(\Omega_C(D), \mathcal{O}_C) \longrightarrow H^0(C, \mathcal{E}xt^1(\Omega_C(D), \mathcal{O}_C)) \longrightarrow 0$$

Using $\mathcal{H}om(\Omega_C(D), \mathcal{O}_C) = \mathcal{T}_C(-D)$ and $\mathcal{E}xt^1(\Omega_C(D), \mathcal{O}_C) = \bigoplus_{q \in N} k_q$, we obtain the short exact sequence

$$0 \longrightarrow H^1(C, \mathcal{T}_C(-D)) \longrightarrow T^1(C, \mathbf{p}) \longrightarrow \bigoplus_{q \in N} k_q \longrightarrow 0$$

This exact sequence gives the precise decomposition of the first-order deformation space: the term

$$H^1(C, \mathcal{T}_C(-D))$$

describes the locally trivial deformation of the normalization together with the marked points and the preimages of the nodes, while each factor k_q gives the one-dimensional smoothing direction of the node q .

Indeed, in the local model $R = \mathbb{C}[[x, y]]/(xy)$, a first-order smoothing of the node is given by

$$\mathbb{C}[\epsilon, x, y]/(xy - \epsilon a), \quad a \in \mathbb{C}$$

Thus the parameter a gives exactly one independent first-order smoothing direction. Equivalently, each node contributes one copy of the ground field to the right-hand term above. In a coordinate-free form, this smoothing line can also be written as

$$k_q \cong T_{q'} \tilde{C} \otimes T_{q''} \tilde{C}$$

where $q', q'' \in \tilde{C}$ are the two preimages of q .

Finally, if C is smooth, then $N = \emptyset$, so the short exact sequence reduces to

$$T^1(C, \mathbf{p}) \cong H^1(C, \mathcal{T}_C(-D))$$

which recovers the smooth case. This completes the proof of the theorem.

References

- [Abr96] Lowell Abrams. “Two-Dimensional Topological Quantum Field Theories and Frobenius Algebras”. In: *Journal of Knot Theory and Its Ramifications* 5.5 (1996), pp. 569–587.
- [Ati88] Michael F. Atiyah. “Topological Quantum Field Theories”. In: *Publications Mathématiques de l’IHÉS* 68 (1988), pp. 175–186.
- [Beh97] Kai Behrend. “Gromov-Witten invariants in algebraic geometry”. In: *Invent. Math.* 127.3 (1997), pp. 601–617. DOI: [10.1007/s002220050132](https://doi.org/10.1007/s002220050132).
- [BF97a] Kai Behrend and Barbara Fantechi. “The Intrinsic Normal Cone”. In: *Inventiones Mathematicae* 128 (1997), pp. 45–88.
- [BF97b] Kai Behrend and Barbara Fantechi. “The intrinsic normal cone”. In: *Invent. Math.* 128.1 (1997), pp. 45–88. DOI: [10.1007/s002220050136](https://doi.org/10.1007/s002220050136).

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- [BM96a] Kai Behrend and Yuri Manin. “Stacks of Stable Maps and Gromov–Witten Invariants”. In: *Duke Mathematical Journal* 85 (1996).
 - [BM96b] Kai Behrend and Yuri Manin. “Stacks of stable maps and Gromov–Witten invariants”. In: *Duke Math. J.* 85.1 (1996), pp. 1–60. DOI: [10.1215/S0012-7094-96-08501-4](https://doi.org/10.1215/S0012-7094-96-08501-4).
 - [CG07] Tom Coates and Alexander Givental. “Quantum Riemann–Roch, Lefschetz and Serre”. In: *Annals of Mathematics* 165 (2007), pp. 15–53.
 - [DM69] Pierre Deligne and David Mumford. “The Irreducibility of the Space of Curves of Given Genus”. In: *Publications Mathématiques de l’IHÉS* 36 (1969), pp. 75–109.
 - [Dub96] Boris Dubrovin. “Geometry of 2D Topological Field Theories”. In: *Integrable Systems and Quantum Groups*. Vol. 1620. Lecture Notes in Mathematics. Springer, 1996.
 - [Fre12] Daniel S. Freed. *Bordism: Old and New*. 2012. eprint: <https://people.math.harvard.edu/~dafr/M392C-2012/index.html>.
 - [FO99] Kenji Fukaya and Kaoru Ono. “Arnold Conjecture and Gromov–Witten Invariant”. In: *Topology* 38 (1999).
 - [Fuk+12] Kenji Fukaya et al. “Technical details on Kuranishi structure and virtual fundamental chain”. In: (2012). arXiv: [1209.4410](https://arxiv.org/abs/1209.4410) [math.SG]. URL: <https://arxiv.org/abs/1209.4410>.
 - [FP97] William Fulton and Rahul Pandharipande. “Notes on Stable Maps and Quantum Cohomology”. In: *Algebraic Geometry—Santa Cruz 1995*. Vol. 62. Proceedings of Symposia in Pure Mathematics. American Mathematical Society, 1997.
 - [Giv01] Alexander Givental. “Gromov–Witten Invariants and Quantization of Quadratic Hamiltonians”. In: *Moscow Mathematical Journal* 1 (2001).
 - [HM98] Joe Harris and Ian Morrison. *Moduli of Curves*. Vol. 187. Graduate Texts in Mathematics. Springer, 1998.
 - [Koc04] Joachim Kock. *Frobenius Algebras and 2D Topological Quantum Field Theories*. Vol. 59. London Mathematical Society Student Texts. Cambridge University Press, 2004.
 - [Kon95] Maxim Kontsevich. “Enumeration of Rational Curves via Torus Actions”. In: *The Moduli Space of Curves*. Ed. by Robbert Dijkgraaf, Carel Faber, and Gerard van der Geer. Vol. 129. Progress in Mathematics. Birkhäuser, 1995.
 - [KM94] Maxim Kontsevich and Yuri Manin. “Gromov–Witten Classes, Quantum Cohomology, and Enumerative Geometry”. In: *Communications in Mathematical Physics* 164 (1994), pp. 525–562.
 - [Lee03] J.M. Lee. *Introduction to Smooth Manifolds*. Graduate Texts in Mathematics. Springer, 2003. ISBN: 9780387954486. URL: <https://link.springer.com/book/10.1007/978-1-4419-9982-5>.
 - [LT98] Jun Li and Gang Tian. “Virtual Moduli Cycles and Gromov–Witten Invariants of Algebraic Varieties”. In: *Journal of the American Mathematical Society* 11 (1998), pp. 119–174.
 - [Man99] Yuri Manin. *Frobenius Manifolds, Quantum Cohomology, and Moduli Spaces*. Vol. 47. American Mathematical Society Colloquium Publications. American Mathematical Society, 1999.
 - [RT95] Yongbin Ruan and Gang Tian. “A Mathematical Theory of Quantum Cohomology”. In: *Journal of Differential Geometry* 42 (1995), pp. 259–367.
 - [Tel12] Constantin Teleman. “The Structure of 2D Semi-Simple Field Theories”. In: *Inventiones Mathematicae* 188 (2012).
 - [Wit88] Edward Witten. “Topological Sigma Models”. In: *Communications in Mathematical Physics* 118 (1988), pp. 411–449.